Human review packet: PG24NoisyMatchingMarkets

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: PG24NoisyMatchingMarkets
Paper id: PG24NoisyMatchingMarkets
Source version: arXiv:2402.16771 source archive SHA256 5779a3e63007b043c1f4db4200b7add8455929a2730c5261bfdee0e5102a59ef
Source record: audit/paper_statement_map.json
Rows: 19
Generated: 2026-09-08
Source URL: official source

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper PG24NoisyMatchingMarkets --serve
```

Open http://127.0.0.1:8765/ in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.FiniteRanking.valueTieKey

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → (α → ℝ) → α → Lex (ℝ × α)

value:
fun {α : Type u_1} (value : α → ℝ) (a : α) => toLex (value a, a)
```

AppliedModelingLib.FiniteRanking.valueTieKeySet

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [DecidableEq α] → Finset α → (α → ℝ) → Finset (Lex (ℝ × α))

value:
fun {α : Type u_1} [DecidableEq α] (s : Finset α) (value : α → ℝ) =>
  Finset.image (AppliedModelingLib.FiniteRanking.valueTieKey value) s
```

Direct retained dependencies

- AppliedModelingLib.FiniteRanking.valueTieKey

AppliedModelingLib.FiniteRanking.rankKeyByValue

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} →
  [LinearOrder α] → [DecidableEq α] → (s : Finset α) → (α → ℝ) → {k : ℕ} → s.card = k → Fin k → Lex (ℝ × α)

value:
fun {α : Type u_1} [LinearOrder α] [DecidableEq α] (s : Finset α) (value : α → ℝ) {k : ℕ} (hcard : s.card = k)
    (a : Fin k) =>
  ((AppliedModelingLib.FiniteRanking.valueTieKeySet s value).orderEmbOfFin
      (AppliedModelingLib.FiniteRanking.rankKeyByValue._proof_1 s value hcard))
    a
```

Direct retained dependencies

- AppliedModelingLib.FiniteRanking.valueTieKeySet

AppliedModelingLib.FiniteRanking.rankValueByValue

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [LinearOrder α] → [DecidableEq α] → (s : Finset α) → (α → ℝ) → {k : ℕ} → s.card = k → Fin k → ℝ

value:
fun {α : Type u_1} [LinearOrder α] [DecidableEq α] (s : Finset α) (value : α → ℝ) {k : ℕ} (hcard : s.card = k)
    (a : Fin k) =>
  (ofLex (AppliedModelingLib.FiniteRanking.rankKeyByValue s value hcard a)).1
```

Direct retained dependencies

- AppliedModelingLib.FiniteRanking.rankKeyByValue

AppliedModelingLib.Matching.CutoffMarket

Declaration location: reusable library

Lean declaration body

```
field Cutoff:
{Student : Type u} → {College : Type v} → AppliedModelingLib.Matching.CutoffMarket Student College → Type w

field Matching:
{Student : Type u} → {College : Type v} → AppliedModelingLib.Matching.CutoffMarket Student College → Type x

field demandAt:
{Student : Type u} →
  {College : Type v} →
    (self : AppliedModelingLib.Matching.CutoffMarket Student College) → self.Cutoff → Student → Option College

field aggregateDemand:
{Student : Type u} →
  {College : Type v} → (self : AppliedModelingLib.Matching.CutoffMarket Student College) → self.Cutoff → College → ℝ

field capacity:
{Student : Type u} → {College : Type v} → AppliedModelingLib.Matching.CutoffMarket Student College → College → ℝ

field isStable:
{Student : Type u} →
  {College : Type v} → (self : AppliedModelingLib.Matching.CutoffMarket Student College) → self.Matching → Prop

field marketClearing:
{Student : Type u} →
  {College : Type v} → (self : AppliedModelingLib.Matching.CutoffMarket Student College) → self.Cutoff → Prop

field representedByCutoff:
{Student : Type u} →
  {College : Type v} →
    (self : AppliedModelingLib.Matching.CutoffMarket Student College) → self.Matching → self.Cutoff → Prop
```

AppliedModelingLib.Matching.CutoffMarket.MarketClearing

Declaration location: reusable library

Lean declaration body

```
type:
{Student : Type u} →
  {College : Type v} → (M : AppliedModelingLib.Matching.CutoffMarket Student College) → M.Cutoff → Prop

value:
fun {Student : Type u} {College : Type v} (M : AppliedModelingLib.Matching.CutoffMarket Student College)
    (a : M.Cutoff) =>
  M.marketClearing a
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket

AppliedModelingLib.Matching.CutoffMarket.RepresentedByCutoff

Declaration location: reusable library

Lean declaration body

```
type:
{Student : Type u} →
  {College : Type v} → (M : AppliedModelingLib.Matching.CutoffMarket Student College) → M.Matching → M.Cutoff → Prop

value:
fun {Student : Type u} {College : Type v} (M : AppliedModelingLib.Matching.CutoffMarket Student College)
    (a : M.Matching) (a_1 : M.Cutoff) =>
  M.representedByCutoff a a_1
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket

AppliedModelingLib.Matching.CutoffMarket.Stable

Declaration location: reusable library

Lean declaration body

```
type:
{Student : Type u} →
  {College : Type v} → (M : AppliedModelingLib.Matching.CutoffMarket Student College) → M.Matching → Prop

value:
fun {Student : Type u} {College : Type v} (M : AppliedModelingLib.Matching.CutoffMarket Student College)
    (a : M.Matching) =>
  M.isStable a
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket

AppliedModelingLib.Matching.MarketClearingCapacityInterface

Declaration location: reusable library

Lean declaration body

```
field marketClearing_aggregateDemand_eq_capacity:
∀ {Student : Type u} {College : Type v} {M : AppliedModelingLib.Matching.CutoffMarket Student College},
  AppliedModelingLib.Matching.MarketClearingCapacityInterface M →
    ∀ (P : M.Cutoff), M.MarketClearing P → ∀ (c : College), M.aggregateDemand P c = M.capacity c
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket
- AppliedModelingLib.Matching.CutoffMarket.MarketClearing

AppliedModelingLib.Matching.SupplyDemandInterface

Declaration location: reusable library

Lean declaration body

```
field stable_iff_exists_marketClearing_cutoff:
∀ {Student : Type u} {College : Type v} {M : AppliedModelingLib.Matching.CutoffMarket Student College},
  AppliedModelingLib.Matching.SupplyDemandInterface M →
    ∀ (μ : M.Matching), M.Stable μ ↔ ∃ (P : M.Cutoff), M.MarketClearing P ∧ M.RepresentedByCutoff μ P

field marketClearing_induces_stable:
∀ {Student : Type u} {College : Type v} {M : AppliedModelingLib.Matching.CutoffMarket Student College},
  AppliedModelingLib.Matching.SupplyDemandInterface M →
    ∀ (P : M.Cutoff), M.MarketClearing P → ∃ (μ : M.Matching), M.Stable μ ∧ M.RepresentedByCutoff μ P
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket
- AppliedModelingLib.Matching.CutoffMarket.MarketClearing
- AppliedModelingLib.Matching.CutoffMarket.RepresentedByCutoff
- AppliedModelingLib.Matching.CutoffMarket.Stable

AppliedModelingLib.Matching.activeCapacity

Declaration location: reusable library

Lean declaration body

```
type:
{College : Type u} → Finset College → (College → ℝ) → ℝ

value:
fun {College : Type u} (active : Finset College) (capacity : College → ℝ) => ∑ c ∈ active, capacity c
```

AppliedModelingLib.Matching.chosenInActive

Declaration location: reusable library

Lean declaration body

```
type:
{Ω : Type u_1} → {College : Type u} → (Ω → Option College) → Finset College → Ω → Prop

value:
fun {Ω : Type u_1} {College : Type u} (choice : Ω → Option College) (active : Finset College) (ω : Ω) =>
  ∃ c ∈ active, choice ω = some c
```

AppliedModelingLib.Matching.cutoffCrossed

Declaration location: reusable library

Lean declaration body

```
type:
{College : Type u} → (College → ℝ) → (College → ℝ) → Prop

value:
fun {College : Type u} (score cutoff : College → ℝ) => ∃ (c : College), cutoff c < score c
```

AppliedModelingLib.Matching.cutoffCrossedOn

Declaration location: reusable library

Lean declaration body

```
type:
{College : Type u} → Finset College → (College → ℝ) → (College → ℝ) → Prop

value:
fun {College : Type u} (active : Finset College) (score cutoff : College → ℝ) => ∃ c ∈ active, cutoff c < score c
```

AppliedModelingLib.Matching.noisyScore

Declaration location: reusable library

Lean declaration body

```
type:
{College : Type u} → ℝ → (College → ℝ) → College → ℝ

value:
fun {College : Type u} (v : ℝ) (noise : College → ℝ) (a : College) => v + noise a
```

AppliedModelingLib.Matching.cutoffCrossingProbability

Declaration location: reusable library

Lean declaration body

```
type:
{College : Type u} →
  [inst : MeasurableSpace (College → ℝ)] → MeasureTheory.Measure (College → ℝ) → Finset College → ℝ → (College → ℝ) → ℝ

value:
fun {College : Type u} [MeasurableSpace (College → ℝ)] (μ : MeasureTheory.Measure (College → ℝ))
    (active : Finset College) (v : ℝ) (cutoff : College → ℝ) =>
  μ.real
    {noise : College → ℝ |
      AppliedModelingLib.Matching.cutoffCrossedOn active (AppliedModelingLib.Matching.noisyScore v noise) cutoff}
```

Direct retained dependencies

- AppliedModelingLib.Matching.cutoffCrossedOn
- AppliedModelingLib.Matching.noisyScore

AppliedModelingLib.Probability.ascendingOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → (Fin n → ℝ) → Fin n → ℝ

value:
fun {n : ℕ} (sample : Fin n → ℝ) (rank : Fin n) => sample ((Tuple.sort sample) rank)
```

AppliedModelingLib.Probability.topSampleRank

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → [NeZero n] → Fin n

value:
fun {n : ℕ} [NeZero n] => ⟨0, AppliedModelingLib.Probability.topSampleRank._proof_1⟩
```

AppliedModelingLib.Probability.upperOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → (Fin n → ℝ) → Fin n → ℝ

value:
fun {n : ℕ} (sample : Fin n → ℝ) (rankFromTop : Fin n) =>
  AppliedModelingLib.Probability.ascendingOrderStatistic sample rankFromTop.rev
```

Direct retained dependencies

- AppliedModelingLib.Probability.ascendingOrderStatistic

AppliedModelingLib.Probability.expectedUpperOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → MeasureTheory.Measure (Fin n → ℝ) → Fin n → ℝ

value:
fun {n : ℕ} (μ : MeasureTheory.Measure (Fin n → ℝ)) (rankFromTop : Fin n) =>
  ∫ (sample : Fin n → ℝ), AppliedModelingLib.Probability.upperOrderStatistic sample rankFromTop ∂μ
```

Direct retained dependencies

- AppliedModelingLib.Probability.upperOrderStatistic

AppliedModelingLib.Probability.expectedTopOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → [NeZero n] → MeasureTheory.Measure (Fin n → ℝ) → ℝ

value:
fun {n : ℕ} [NeZero n] (μ : MeasureTheory.Measure (Fin n → ℝ)) =>
  AppliedModelingLib.Probability.expectedUpperOrderStatistic μ AppliedModelingLib.Probability.topSampleRank
```

Direct retained dependencies

- AppliedModelingLib.Probability.expectedUpperOrderStatistic
- AppliedModelingLib.Probability.topSampleRank

AppliedModelingLib.Probability.expectedTopOrderStatisticSeq

Declaration location: reusable library

Lean declaration body

```
type:
((n : ℕ) → MeasureTheory.Measure (Fin (n + 1) → ℝ)) → ℕ → ℝ

value:
fun (sampleLaw : (n : ℕ) → MeasureTheory.Measure (Fin (n + 1) → ℝ)) (n : ℕ) =>
  AppliedModelingLib.Probability.expectedTopOrderStatistic (sampleLaw n)
```

Direct retained dependencies

- AppliedModelingLib.Probability.expectedTopOrderStatistic

AppliedModelingLib.Probability.upperTailMass

Declaration location: reusable library

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → ℝ → ℝ

value:
fun (μ : MeasureTheory.Measure ℝ) (x : ℝ) => μ.real (Set.Ioi x)
```

AppliedModelingLib.measureProb

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [inst : MeasurableSpace α] → MeasureTheory.Measure α → (α → Prop) → ℝ

value:
fun {α : Type u_1} [MeasurableSpace α] (μ : MeasureTheory.Measure α) (p : α → Prop) => (μ {a : α | p a}).toReal
```

AppliedModelingLib.Matching.eventMass

Declaration location: reusable library

Lean declaration body

```
type:
{Ω : Type u_1} → [inst : MeasurableSpace Ω] → MeasureTheory.Measure Ω → (Ω → Prop) → ℝ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] (μ : MeasureTheory.Measure Ω) (event : Ω → Prop) =>
  AppliedModelingLib.measureProb μ event
```

Direct retained dependencies

- AppliedModelingLib.measureProb

AppliedModelingLib.Matching.choiceMass

Declaration location: reusable library

Lean declaration body

```
type:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] → {College : Type u} → MeasureTheory.Measure Ω → (Ω → Option College) → Finset College → ℝ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] {College : Type u} (μ : MeasureTheory.Measure Ω) (choice : Ω → Option College)
    (active : Finset College) =>
  AppliedModelingLib.Matching.eventMass μ (AppliedModelingLib.Matching.chosenInActive choice active)
```

Direct retained dependencies

- AppliedModelingLib.Matching.chosenInActive
- AppliedModelingLib.Matching.eventMass

PG24NoisyMatchingMarkets.CoalitionLargeSubset

Declaration location: paper-local

Lean declaration body

```
field subset:
∀ {College : Type v} {coalition largeSubset : Finset College} {epsilon : ℝ},
  PG24NoisyMatchingMarkets.CoalitionLargeSubset coalition largeSubset epsilon → largeSubset ⊆ coalition

field coalition_card_pos:
∀ {College : Type v} {coalition largeSubset : Finset College} {epsilon : ℝ},
  PG24NoisyMatchingMarkets.CoalitionLargeSubset coalition largeSubset epsilon → 0 < ↑coalition.card

field large_ratio:
∀ {College : Type v} {coalition largeSubset : Finset College} {epsilon : ℝ},
  PG24NoisyMatchingMarkets.CoalitionLargeSubset coalition largeSubset epsilon →
    1 - epsilon < ↑largeSubset.card / ↑coalition.card
```

PG24NoisyMatchingMarkets.LongTailedSurvival

Declaration location: paper-local

Lean declaration body

```
type:
(ℝ → ℝ) → Prop

value:
fun (survival : ℝ → ℝ) =>
  ∀ (d : ℝ), 0 < d → Filter.Tendsto (fun (x : ℝ) => survival (x + d) / survival x) Filter.atTop (nhds 1)
```

PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.localCoordinates

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome →
              Outcome → ℝ × (Fin (C + 1) → ℝ)

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [MeasurableSpace StudentType]
    {Outcome : Type u} [MeasurableSpace Outcome]
    (sampling : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome)
    (a : Outcome) =>
  (sampling.commonValue (sampling.sourceStudent a), sampling.coalitionNoise a)
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling

PG24NoisyMatchingMarkets.PG24HolderIntervalRegular

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → Prop

value:
fun (eta : MeasureTheory.Measure ℝ) =>
  ∃ (holderConstant : ℝ) (exponent : ℝ),
    0 < exponent ∧
      0 ≤ holderConstant ∧
        ∀ (x delta : ℝ), 0 < delta → eta.real (Set.Ioo x (x + delta)) ≤ holderConstant * delta.rpow exponent
```

PG24NoisyMatchingMarkets.PG24SourceCutoffMarket

Declaration location: paper-local

Lean declaration body

```
field demandAt:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} →
      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff →
        Cutoff → Outcome → Option GlobalCollege

field aggregateDemand:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} →
      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff → Cutoff → GlobalCollege → ℝ

field capacity:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} → PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff → GlobalCollege → ℝ

field isStable:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} →
      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff →
        (Outcome → Option GlobalCollege) → Prop

field marketClearing:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} → PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff → Cutoff → Prop

field representedByCutoff:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} →
      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff →
        (Outcome → Option GlobalCollege) → Cutoff → Prop
```

PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket

Declaration location: paper-local

Lean declaration body

```
type:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    {Cutoff : Type w} →
      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff →
        AppliedModelingLib.Matching.CutoffMarket Outcome GlobalCollege

value:
fun {Outcome : Type u} {GlobalCollege : Type v} {Cutoff : Type w}
    (sourceMarket : PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff) =>
  { Cutoff := Cutoff, Matching := Outcome → Option GlobalCollege, demandAt := sourceMarket.demandAt,
    aggregateDemand := sourceMarket.aggregateDemand, capacity := sourceMarket.capacity,
    isStable := sourceMarket.isStable, marketClearing := sourceMarket.marketClearing,
    representedByCutoff := sourceMarket.representedByCutoff }
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket

PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance

Declaration location: paper-local

Lean declaration body

```
field sourceMarket:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff

field supplyDemand:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
```

```
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  AppliedModelingLib.Matching.SupplyDemandInterface self.sourceMarket.toCutoffMarket

field clearingCapacity:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  AppliedModelingLib.Matching.MarketClearingCapacityInterface self.sourceMarket.toCutoffMarket

field sampling:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome

field selectedMatching:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Outcome → Option GlobalCollege

field selectedStable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  self.sourceMarket.isStable self.selectedMatching

field selectedCutoff:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Cutoff

field selectedCutoff_marketClearing:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  self.sourceMarket.marketClearing self.selectedCutoff

field selectedCutoff_represents:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  self.sourceMarket.representedByCutoff self.selectedMatching self.selectedCutoff

field selectedMatching_eq_demandAt:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (outcome : Outcome), self.selectedMatching outcome = self.sourceMarket.demandAt self.selectedCutoff outcome

field selected_choiceMass_eq_aggregateDemand:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
```

```
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  AppliedModelingLib.Matching.choiceMass self.sampling.outcomeLaw self.selectedMatching Finset.univ =
    ∑ c : GlobalCollege, self.sourceMarket.aggregateDemand self.selectedCutoff c

field totalCapacity_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  ∑ c : GlobalCollege, self.sourceMarket.capacity c = totalSupply

field cutoffCoordinates:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Cutoff → GlobalCollege → ℝ

field globalScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Cutoff → Outcome → GlobalCollege → ℝ

field coalitionEmbedding:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Fin (C + 1) ↪ GlobalCollege

field coalition_score_ae:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  ∀ᵐ (outcome : Outcome) ∂self.sampling.outcomeLaw,
    ∀ (c : Fin (C + 1)),
      self.globalScore self.selectedCutoff outcome (self.coalitionEmbedding c) =
        (self.sampling.localCoordinates outcome).1 + (self.sampling.localCoordinates outcome).2 c

field demand_none_iff_no_global_cutoff_crossing:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (outcome : Outcome),
  self.sourceMarket.demandAt self.selectedCutoff outcome = none ↔
    ¬AppliedModelingLib.Matching.cutoffCrossed (self.globalScore self.selectedCutoff outcome)
        (self.cutoffCoordinates self.selectedCutoff)
```

Direct retained dependencies

- AppliedModelingLib.Matching.MarketClearingCapacityInterface
- AppliedModelingLib.Matching.SupplyDemandInterface
- AppliedModelingLib.Matching.choiceMass
- AppliedModelingLib.Matching.cutoffCrossed
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.localCoordinates
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket

PG24NoisyMatchingMarkets.betaMaxConcentratingVariance

Declaration location: paper-local

Lean declaration body

```
type:
(ℕ → ℝ) → ℝ → Prop

value:
fun (maxVariance : ℕ → ℝ) (β : ℝ) =>
  0 < β ∧ ∃ (K : ℝ) (N : ℕ), 0 ≤ K ∧ ∀ (n : ℕ), N ≤ n → maxVariance n ≤ K * (↑(n + 1)).rpow (-β)
```

PG24NoisyMatchingMarkets.capacityRegular

Declaration location: paper-local

Lean declaration body

```
type:
{College : Type v} → (College → ℝ) → ℝ → ℕ → Prop

value:
fun {College : Type v} (capacity : College → ℝ) (alpha : ℝ) (C : ℕ) => ∀ (c : College), capacity c < alpha / ↑C
```

PG24NoisyMatchingMarkets.coalitionCutoffRestriction

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {GlobalCollege : Type v} →
    {Student : Type u} →
      {M : AppliedModelingLib.Matching.CutoffMarket Student GlobalCollege} →
        (Fin (C + 1) ↪ GlobalCollege) → (M.Cutoff → GlobalCollege → ℝ) → M.Cutoff → Fin (C + 1) → ℝ

value:
fun {C : ℕ} {GlobalCollege : Type v} {Student : Type u}
    {M : AppliedModelingLib.Matching.CutoffMarket Student GlobalCollege}
    (coalitionEmbedding : Fin (C + 1) ↪ GlobalCollege) (globalCutoff : M.Cutoff → GlobalCollege → ℝ) (P : M.Cutoff)
    (a : Fin (C + 1)) =>
  globalCutoff P (coalitionEmbedding a)
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket

PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.localCutoff

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      Fin (C + 1) → ℝ

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (inst_3 :
```

```
    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (a : Fin (C + 1)) =>
  PG24NoisyMatchingMarkets.coalitionCutoffRestriction inst_3.coalitionEmbedding inst_3.cutoffCoordinates
    inst_3.selectedCutoff a
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket
- PG24NoisyMatchingMarkets.coalitionCutoffRestriction

PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Declaration location: paper-local

Lean declaration body

```
type:
{College : Type v} →
  [inst : MeasurableSpace (College → ℝ)] → MeasureTheory.Measure (College → ℝ) → Finset College → ℝ → (College → ℝ) → ℝ

value:
fun {College : Type v} [MeasurableSpace (College → ℝ)] (noiseLaw : MeasureTheory.Measure (College → ℝ))
    (active : Finset College) (v : ℝ) (cutoff : College → ℝ) =>
  AppliedModelingLib.Matching.cutoffCrossingProbability noiseLaw active v cutoff
```

Direct retained dependencies

- AppliedModelingLib.Matching.cutoffCrossingProbability

PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.pMu

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff →
                      ℝ → Finset (Fin (C + 1)) → ℝ

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (inst_3 :
      PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply StudentType Outcome
        GlobalCollege Cutoff)
    (value : ℝ) (active : Finset (Fin (C + 1))) =>
  PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
    active value inst_3.localCutoff
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.localCutoff
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

PG24NoisyMatchingMarkets.epsilonFloorSplitIndex

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ → ℕ

value:
fun (epsilon : ℝ) (C : ℕ) => ⌊epsilon * ↑(C + 1)⌋₊
```

PG24NoisyMatchingMarkets.indexSuffixLarge

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → (C : ℕ) → Finset (Fin (C + 1))

value:
fun (k C : ℕ) => {c : Fin (C + 1) | k ≤ ↑c}
```

PG24NoisyMatchingMarkets.source_assumption_beta_max_variance_bound

Declaration location: paper-local

Lean declaration body

```
type:
((n : ℕ) → MeasureTheory.Measure (Fin (n + 1) → ℝ)) → (ℕ → ℝ) → Prop

value:
fun (sampleLaw : (n : ℕ) → MeasureTheory.Measure (Fin (n + 1) → ℝ)) (maxVariance : ℕ → ℝ) =>
  (∀ᶠ (n : ℕ) in Filter.atTop,
      MeasureTheory.MemLp
        (fun (sample : Fin (n + 1) → ℝ) =>
          AppliedModelingLib.Probability.upperOrderStatistic sample AppliedModelingLib.Probability.topSampleRank)
        2 (sampleLaw n)) ∧
    ∀ᶠ (n : ℕ) in Filter.atTop,
      ProbabilityTheory.variance
          (fun (sample : Fin (n + 1) → ℝ) =>
            AppliedModelingLib.Probability.upperOrderStatistic sample AppliedModelingLib.Probability.topSampleRank)
          (sampleLaw n) ≤
        maxVariance n
```

Direct retained dependencies

- AppliedModelingLib.Probability.topSampleRank
- AppliedModelingLib.Probability.upperOrderStatistic

PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → (ℕ → ℝ) → Prop

value:
fun (noiseLaw : MeasureTheory.Measure ℝ) (maxVariance : ℕ → ℝ) =>
  PG24NoisyMatchingMarkets.source_assumption_beta_max_variance_bound
    (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) maxVariance
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.source_assumption_beta_max_variance_bound

PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → Prop

value:
fun (noiseLaw : MeasureTheory.Measure ℝ) => MeasureTheory.MemLp (fun (x : ℝ) => x) 2 noiseLaw
```

PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock

Declaration location: paper-local

Lean declaration body

```
type:
{College : Type u} → [DecidableEq College] → Finset College → (College → ℝ) → ℝ → Finset College

value:
fun {College : Type u} [DecidableEq College] (active : Finset College) (cutoff : College → ℝ) (pivot : ℝ) =>
  {c ∈ active | pivot ≤ cutoff c}
```

PG24NoisyMatchingMarkets.theorem1TailDenom

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (β γ : ℝ) => 3 * β * γ + 2 * β + 5 * γ + 6
```

PG24NoisyMatchingMarkets.theorem1TailK

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (β γ : ℝ) => 2 * β * γ / PG24NoisyMatchingMarkets.theorem1TailDenom β γ
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailDenom

PG24NoisyMatchingMarkets.theorem1TailPhi1

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (β γ : ℝ) => -(2 * β) / PG24NoisyMatchingMarkets.theorem1TailDenom β γ
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailDenom

PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ

value:
fun (C : ℕ) (beta gamma : ℝ) => (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma)
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailPhi1

PG24NoisyMatchingMarkets.theorem1TailPhi2

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℝ

value:
fun (β γ : ℝ) => (5 * γ + 6) / PG24NoisyMatchingMarkets.theorem1TailDenom β γ
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailDenom

PG24NoisyMatchingMarkets.theorem1DenseGroupSize

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℕ

value:
fun (C : ℕ) (beta gamma : ℝ) => max 1 ⌈(↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)⌉₊
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailPhi2

PG24NoisyMatchingMarkets.theorem1DenseGroupIndex

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℕ

value:
fun (C : ℕ) (beta gamma : ℝ) => PG24NoisyMatchingMarkets.theorem1DenseGroupSize C beta gamma - 1
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1DenseGroupSize

PG24NoisyMatchingMarkets.theorem1DenseGroupCenter

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure ℝ → ℕ → ℝ → ℝ → ℝ

value:
fun (noiseLaw : MeasureTheory.Measure ℝ) (C : ℕ) (beta gamma : ℝ) =>
  AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
    (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw)
    (PG24NoisyMatchingMarkets.theorem1DenseGroupIndex C beta gamma)
```

Direct retained dependencies

- AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
- PG24NoisyMatchingMarkets.theorem1DenseGroupIndex

PG24NoisyMatchingMarkets.theorem1TailPhi3

Declaration location: paper-local

Lean declaration body

type:
ℝ → ℝ → ℝ

value:
```
fun (β γ : ℝ) => 1 - 2 * β * γ / PG24NoisyMatchingMarkets.theorem1TailDenom β γ
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailDenom

PG24NoisyMatchingMarkets.theorem1TailPhi4

Declaration location: paper-local

Lean declaration body

type:
ℝ → ℝ → ℝ

value:
```
fun (β γ : ℝ) => -β / 2 + β * γ / PG24NoisyMatchingMarkets.theorem1TailDenom β γ
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem1TailDenom

PG24NoisyMatchingMarkets.theorem2_smallFirmSourceBound

Declaration location: paper-local

Lean declaration body

type:
ℝ → ℝ → ℝ → ℝ → ℝ → Prop

value:
```
fun (totalSupply alpha epsilon sigma p : ℝ) =>
  p ≤ alpha * √epsilon / (1 - totalSupply - epsilon - (1 - Real.exp (-(2 * epsilon * sigma))))
```

PG24NoisyMatchingMarkets.theorem3DenseWindowCount

Declaration location: paper-local

Lean declaration body

type:
ℕ → ℝ → ℕ

value:
```
fun (C : ℕ) (exponent : ℝ) => ⌈(↑C).rpow exponent⌉₊
```

PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank

Declaration location: paper-local

Lean declaration body

type:
ℕ → ℝ → ℕ

value:
```
fun (C : ℕ) (exponent : ℝ) => ⌊(↑C).rpow exponent⌋₊
```

PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount

Declaration location: paper-local

Lean declaration body

type:
ℕ → ℝ → ℝ → ℕ

value:
```
fun (C : ℕ) (denseExponent prefixExponent : ℝ) =>
  PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C prefixExponent /
    PG24NoisyMatchingMarkets.theorem3DenseWindowCount C denseExponent
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem3DenseWindowCount
- PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank

PG24NoisyMatchingMarkets.theorem3RankedCutoff

Declaration location: paper-local

Lean declaration body

```
type:
(C : ℕ) → (Fin (C + 1) → ℝ) → Fin (C + 1) → ℝ

value:
fun (C : ℕ) (cutoff : Fin (C + 1) → ℝ) (a : Fin (C + 1)) =>
  AppliedModelingLib.FiniteRanking.rankValueByValue Finset.univ cutoff
    (PG24NoisyMatchingMarkets.theorem3_ranked_cutoff_univ_card C) a
```

Direct retained dependencies

- AppliedModelingLib.FiniteRanking.rankValueByValue

PG24NoisyMatchingMarkets.theorem3RankedCutoffNat

Declaration location: paper-local

Lean declaration body

```
type:
(C : ℕ) → (Fin (C + 1) → ℝ) → ℕ → ℝ

value:
fun (C : ℕ) (cutoff : Fin (C + 1) → ℝ) (rank : ℕ) =>
  PG24NoisyMatchingMarkets.theorem3RankedCutoff C cutoff
    ⟨min rank C, PG24NoisyMatchingMarkets.theorem3RankedCutoffNat._proof_1 C rank⟩
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem3RankedCutoff

PG24NoisyMatchingMarkets.theorem4AffordableColleges

Declaration location: paper-local

Lean declaration body

```
type:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    [Fintype GlobalCollege] →
      {Cutoff : Type w} →
        (Cutoff → GlobalCollege → ℝ) → (Cutoff → Outcome → GlobalCollege → ℝ) → Cutoff → Outcome → Finset GlobalCollege

value:
fun {Outcome : Type u} {GlobalCollege : Type v} [Fintype GlobalCollege] {Cutoff : Type w}
    (cutoffCoordinates : Cutoff → GlobalCollege → ℝ) (globalScore : Cutoff → Outcome → GlobalCollege → ℝ) (P : Cutoff)
    (outcome : Outcome) =>
  {college : GlobalCollege | cutoffCoordinates P college < globalScore P outcome college}
```

PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

Declaration location: paper-local

Lean declaration body

```
type:
{Outcome : Type u} →
  {GlobalCollege : Type v} →
    [Fintype GlobalCollege] →
      {Cutoff : Type w} →
        (Outcome → GlobalCollege → ℕ) →
          (Cutoff → GlobalCollege → ℝ) →
            (Cutoff → Outcome → GlobalCollege → ℝ) → Cutoff → Outcome → Option GlobalCollege

value:
fun {Outcome : Type u} {GlobalCollege : Type v} [Fintype GlobalCollege] {Cutoff : Type w}
    (preferenceRank : Outcome → GlobalCollege → ℕ) (cutoffCoordinates : Cutoff → GlobalCollege → ℝ)
    (globalScore : Cutoff → Outcome → GlobalCollege → ℝ) (P : Cutoff) (outcome : Outcome) =>
  List.argmin (preferenceRank outcome)
    (PG24NoisyMatchingMarkets.theorem4AffordableColleges cutoffCoordinates globalScore P outcome).toList
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem4AffordableColleges

PG24NoisyMatchingMarkets.PG24FinitePreferredDemand

Declaration location: paper-local

Lean declaration body

```
field outcomeLaw:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff →
            MeasureTheory.Measure Outcome

field preferenceRank:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff → Outcome → GlobalCollege → ℕ

field preferenceRank_injective:
∀ {Outcome : Type u} [inst : MeasurableSpace Outcome] {GlobalCollege : Type v} [inst_1 : Fintype GlobalCollege]
  {Cutoff : Type w} (self : PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff)
  (outcome : Outcome), Function.Injective (self.preferenceRank outcome)

field cutoffCoordinates:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff → Cutoff → GlobalCollege → ℝ

field globalScore:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff →
            Cutoff → Outcome → GlobalCollege → ℝ

field singletonDemandMeasurable:
∀ {Outcome : Type u} [inst : MeasurableSpace Outcome] {GlobalCollege : Type v} [inst_1 : Fintype GlobalCollege]
  {Cutoff : Type w} (self : PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff)
  (P : Cutoff) (college : GlobalCollege),
  MeasurableSet
    {outcome : Outcome |
      PG24NoisyMatchingMarkets.theorem4DemandFromPreferences self.preferenceRank self.cutoffCoordinates self.globalScore
          P outcome =
        some college}
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt

Declaration location: paper-local

Lean declaration body

```
type:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff →
            Cutoff → Outcome → Option GlobalCollege

value:
fun {Outcome : Type u} [MeasurableSpace Outcome] {GlobalCollege : Type v} [Fintype GlobalCollege] {Cutoff : Type w}
    (model : PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff) (a : Cutoff)
    (a_1 : Outcome) =>
  PG24NoisyMatchingMarkets.theorem4DemandFromPreferences model.preferenceRank model.cutoffCoordinates model.globalScore
    a a_1
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand

Declaration location: paper-local

Lean declaration body

```
type:
{Outcome : Type u} →
  [inst : MeasurableSpace Outcome] →
    {GlobalCollege : Type v} →
      [inst_1 : Fintype GlobalCollege] →
        {Cutoff : Type w} →
          PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff → Cutoff → GlobalCollege → ℝ

value:
fun {Outcome : Type u} [MeasurableSpace Outcome] {GlobalCollege : Type v} [Fintype GlobalCollege] {Cutoff : Type w}
    (model : PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff) (a : Cutoff)
    (a_1 : GlobalCollege) =>
  AppliedModelingLib.Matching.eventMass model.outcomeLaw fun (outcome : Outcome) => model.demandAt a outcome = some a_1
```

Direct retained dependencies

- AppliedModelingLib.Matching.eventMass
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt

PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.marketClearing

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              Cutoff → Prop

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Cutoff : Type v}
    (market : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
    (P : Cutoff) =>
  ∀ (college : Fin (C + 1)), market.demand.aggregateDemand P college = market.capacity college
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket

PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.isStable

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              (StudentType × (Fin (C + 1) → ℝ) → Option (Fin (C + 1))) → Prop

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Cutoff : Type v}
    (market : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
    (matching : StudentType × (Fin (C + 1) → ℝ) → Option (Fin (C + 1))) =>
  ∃ (P : Cutoff), market.marketClearing P ∧ matching = market.demand.demandAt P
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.marketClearing

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData

Declaration location: paper-local

Lean declaration body

```
field demand:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff

field sampling:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome

field sampling_outcomeLaw_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  self.sampling.outcomeLaw = self.demand.outcomeLaw

field capacity:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      GlobalCollege → ℝ

field totalCapacity_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  ∑ college : GlobalCollege, self.capacity college = totalSupply

field selectedCutoff:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      Cutoff

field selectedCutoff_clearing:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
```

```
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (college : GlobalCollege), self.demand.aggregateDemand self.selectedCutoff college = self.capacity college

field approximateScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      Outcome → GlobalCollege → ℝ

field demand_globalScore_eq_approximateScore:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (P : Cutoff) (outcome : Outcome) (college : GlobalCollege),
  self.demand.globalScore P outcome college = self.approximateScore outcome college

field coalitionEmbedding:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      Fin (C + 1) ↪ GlobalCollege

field coalition_score_ae:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  ∀ᵐ (outcome : Outcome) ∂self.sampling.outcomeLaw,
    ∀ (college : Fin (C + 1)),
      self.approximateScore outcome (self.coalitionEmbedding college) =
        (self.sampling.localCoordinates outcome).1 + (self.sampling.localCoordinates outcome).2 college
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.localCoordinates
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand

PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance

Declaration location: paper-local

Lean declaration body

```
field studentLaw:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType
                Cutoff →
              MeasureTheory.Measure StudentType

field studentLaw_isProbability:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff),
  MeasureTheory.IsProbabilityMeasure self.studentLaw

field value:
```

```
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType
                Cutoff →
              StudentType → ℝ
```

```
field value_measurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff),
  Measurable self.value
```

```
field value_marginal:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff),
  MeasureTheory.Measure.map self.value self.studentLaw = eta
```

```
field literal:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType
                Cutoff →
              PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType
                (StudentType × (Fin (C + 1) → ℝ)) (Fin (C + 1)) Cutoff
```

```
field sampling_outcomeLaw_eq_iid:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff),
  self.literal.sampling.outcomeLaw = self.studentLaw.prod (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
```

```
field coalitionEmbedding_id:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (c : Fin (C + 1)), self.literal.coalitionEmbedding c = c
```

```
field basic_score:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (P : Cutoff) (outcome : StudentType × (Fin (C + 1) → ℝ)) (college : Fin (C + 1)),
  self.literal.demand.globalScore P outcome college = self.value outcome.1 + outcome.2 college
```

```
field capacity_regular:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff),
  PG24NoisyMatchingMarkets.capacityRegular self.literal.capacity alpha (C + 1)
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.capacityRegular

PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance.selectedCutoffVector

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType
                Cutoff →
              Fin (C + 1) → ℝ
```

```
value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Cutoff : Type v}
```

```
  (inst_1 :
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (a : Fin (C + 1)) =>
 inst_1.literal.demand.cutoffCoordinates inst_1.literal.selectedCutoff a
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.marketClearing

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
 {noiseLaw eta : MeasureTheory.Measure ℝ} →
  {totalSupply : ℝ} →
   {StudentType : Type u} →
    [inst : MeasurableSpace StudentType] →
     {Outcome : Type v} →
      [inst_1 : MeasurableSpace Outcome] →
       {GlobalCollege : Type w} →
        [inst_2 : Fintype GlobalCollege] →
         {Cutoff : Type x} →
          PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
            GlobalCollege Cutoff →
           Cutoff → Prop

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
  [Fintype GlobalCollege] {Cutoff : Type x}
  (data :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
     Cutoff)
  (P : Cutoff) =>
 ∀ (college : GlobalCollege), data.demand.aggregateDemand P college = data.capacity college
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.isStable

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
 {noiseLaw eta : MeasureTheory.Measure ℝ} →
  {totalSupply : ℝ} →
   {StudentType : Type u} →
    [inst : MeasurableSpace StudentType] →
     {Outcome : Type v} →
      [inst_1 : MeasurableSpace Outcome] →
       {GlobalCollege : Type w} →
        [inst_2 : Fintype GlobalCollege] →
         {Cutoff : Type x} →
          PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
            GlobalCollege Cutoff →
           (Outcome → Option GlobalCollege) → Prop

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
  [Fintype GlobalCollege] {Cutoff : Type x}
  (data :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
     Cutoff)
  (matching : Outcome → Option GlobalCollege) =>
 ∃ (P : Cutoff), data.marketClearing P ∧ matching = data.demand.demandAt P
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.marketClearing

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selectedMatching

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      Outcome → Option GlobalCollege

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
        Cutoff)
    (a : Outcome) =>
  data.demand.demandAt data.selectedCutoff a
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.sourceMarket

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24SourceCutoffMarket Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
        Cutoff) =>
  { demandAt := data.demand.demandAt, aggregateDemand := data.demand.aggregateDemand, capacity := data.capacity,
    isStable := data.isStable, marketClearing := data.marketClearing,
    representedByCutoff := fun (matching : Outcome → Option GlobalCollege) (P : Cutoff) =>
      matching = data.demand.demandAt P }
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData

- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.isStable
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.marketClearing
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.clearingCapacity

Declaration location: paper-local

Lean declaration body

```
type:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (data :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  AppliedModelingLib.Matching.MarketClearingCapacityInterface data.sourceMarket.toCutoffMarket

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
        Cutoff) =>
  {
    marketClearing_aggregateDemand_eq_capacity :=
      fun (P : data.sourceMarket.toCutoffMarket.Cutoff) (hclearing : data.sourceMarket.toCutoffMarket.MarketClearing P)
        (college : GlobalCollege) =>
      hclearing college }
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket
- AppliedModelingLib.Matching.CutoffMarket.MarketClearing
- AppliedModelingLib.Matching.MarketClearingCapacityInterface
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.sourceMarket
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.supplyDemand

Declaration location: paper-local

Lean declaration body

```
type:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (data :
    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  AppliedModelingLib.Matching.SupplyDemandInterface data.sourceMarket.toCutoffMarket

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
        Cutoff) =>
  {
    stable_iff_exists_marketClearing_cutoff :=
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.stable_iff_exists_marketClearing_cutoff data,
    marketClearing_induces_stable :=
      fun (P : data.sourceMarket.toCutoffMarket.Cutoff)
        (hclearing : data.sourceMarket.toCutoffMarket.MarketClearing P) =>
      Exists.intro (data.demand.demandAt P) ⟨Exists.intro P ⟨hclearing, rfl⟩, Eq.refl (data.demand.demandAt P)⟩ }
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket
- AppliedModelingLib.Matching.CutoffMarket.MarketClearing
- AppliedModelingLib.Matching.CutoffMarket.RepresentedByCutoff
- AppliedModelingLib.Matching.CutoffMarket.Stable
- AppliedModelingLib.Matching.SupplyDemandInterface
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.marketClearing
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.sourceMarket
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket

PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome
                        GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance C noiseLaw eta totalSupply
                        StudentType Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
        Cutoff) =>
  { sourceMarket := data.sourceMarket,
    supplyDemand := PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.supplyDemand data,
    clearingCapacity := PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.clearingCapacity data,
    sampling := data.sampling, selectedMatching := data.selectedMatching,
    selectedStable := PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selectedMatching_stable data,
    selectedCutoff := data.selectedCutoff,
    selectedCutoff_marketClearing :=
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selectedCutoff_marketClearing data,
    selectedCutoff_represents :=
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selectedMatching_represents_selectedCutoff data,
    selectedMatching_eq_demandAt :=
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance._proof_1 data,
    selected_choiceMass_eq_aggregateDemand :=
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selected_choiceMass_eq_aggregateDemand data,
    totalCapacity_eq := data.totalCapacity_eq, cutoffCoordinates := data.demand.cutoffCoordinates,
    globalScore := fun (x : Cutoff) => data.approximateScore, coalitionEmbedding := data.coalitionEmbedding,
    coalition_score_ae := data.coalition_score_ae,
    demand_none_iff_no_global_cutoff_crossing := fun (outcome : Outcome) =>
      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.demand_none_iff_no_global_cutoff_crossing data outcome }
```

Direct retained dependencies

- AppliedModelingLib.Matching.CutoffMarket.MarketClearing
- AppliedModelingLib.Matching.CutoffMarket.RepresentedByCutoff
- AppliedModelingLib.Matching.CutoffMarket.Stable
- AppliedModelingLib.Matching.choiceMass
- AppliedModelingLib.Matching.cutoffCrossed
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand

- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.clearingCapacity
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.selectedMatching
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.sourceMarket
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.supplyDemand
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket
- PG24NoisyMatchingMarkets.PG24SourceCutoffMarket.toCutoffMarket

PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData

Declaration location: paper-local

Lean declaration body

```
field sampling:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome

field rankOfStudent:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      StudentType → GlobalCollege → ℕ

field rankOfStudent_injective:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (student : StudentType), Function.Injective (self.rankOfStudent student)

field cutoffCoordinates:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff → GlobalCollege → ℝ

field globalScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff → Outcome → GlobalCollege → ℝ

field singletonDemandMeasurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
```

```
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (P : Cutoff) (college : GlobalCollege),
  MeasurableSet
    {outcome : Outcome |
      PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
          (fun (outcome : Outcome) (college : GlobalCollege) =>
            self.rankOfStudent (self.sampling.sourceStudent outcome) college)
          self.cutoffCoordinates self.globalScore P outcome =
        some college}

field capacity:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      GlobalCollege → ℝ

field totalCapacity_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  ∑ college : GlobalCollege, self.capacity college = totalSupply

field selectedCutoff:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff

field selectedCutoff_clearing:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (college : GlobalCollege),
  (AppliedModelingLib.Matching.eventMass self.sampling.outcomeLaw fun (outcome : Outcome) =>
      PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
          (fun (outcome : Outcome) (college : GlobalCollege) =>
            self.rankOfStudent (self.sampling.sourceStudent outcome) college)
          self.cutoffCoordinates self.globalScore self.selectedCutoff outcome =
        some college) =
    self.capacity college

field approximateScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Outcome → GlobalCollege → ℝ

field globalScore_eq_approximateScore:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff)
  (P : Cutoff) (outcome : Outcome) (college : GlobalCollege),
  self.globalScore P outcome college = self.approximateScore outcome college

field coalitionEmbedding:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
```

```
          [inst_2 : Fintype GlobalCollege] →
            {Cutoff : Type x} →
              PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                  Outcome GlobalCollege Cutoff →
                Fin (C + 1) ↪ GlobalCollege

field coalition_score_ae:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
      GlobalCollege Cutoff),
  ∀ᵐ (outcome : Outcome) ∂self.sampling.outcomeLaw,
    ∀ (college : Fin (C + 1)),
      self.approximateScore outcome (self.coalitionEmbedding college) =
        (self.sampling.localCoordinates outcome).1 + (self.sampling.localCoordinates outcome).2 college
```

Direct retained dependencies

- AppliedModelingLib.Matching.eventMass
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.localCoordinates
- PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.demand

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24FinitePreferredDemand Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
        GlobalCollege Cutoff) =>
  { outcomeLaw := data.sampling.outcomeLaw,
    preferenceRank := fun (outcome : Outcome) (college : GlobalCollege) =>
      data.rankOfStudent (data.sampling.sourceStudent outcome) college,
    preferenceRank_injective := PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.demand._proof_1 data,
    cutoffCoordinates := data.cutoffCoordinates, globalScore := data.globalScore,
    singletonDemandMeasurable := data.singletonDemandMeasurable }
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData

PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.ofTrueValueSource

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
```

```
            [inst_2 : Fintype GlobalCollege] →
              {Cutoff : Type x} →
                (sampling :
                    PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome) →
                  (rankOfStudent : StudentType → GlobalCollege → ℕ) →
                    (∀ (student : StudentType), Function.Injective (rankOfStudent student)) →
                      (cutoffCoordinates : Cutoff → GlobalCollege → ℝ) →
                        (globalScore : Cutoff → Outcome → GlobalCollege → ℝ) →
                          (∀ (P : Cutoff) (college : GlobalCollege),
                              MeasurableSet
                                {outcome : Outcome |
                                  PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
                                      (fun (outcome : Outcome) (college : GlobalCollege) =>
                                        rankOfStudent (sampling.sourceStudent outcome) college)
                                      cutoffCoordinates globalScore P outcome =
                                    some college}) →
                            (capacity : GlobalCollege → ℝ) →
                              ∑ college : GlobalCollege, capacity college = totalSupply →
                                (selectedCutoff : Cutoff) →
                                  (∀ (college : GlobalCollege),
                                      (AppliedModelingLib.Matching.eventMass sampling.outcomeLaw
                                          fun (outcome : Outcome) =>
                                          PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
                                              (fun (outcome : Outcome) (college : GlobalCollege) =>
                                                rankOfStudent (sampling.sourceStudent outcome) college)
                                              cutoffCoordinates globalScore selectedCutoff outcome =
                                            some college) =
                                        capacity college) →
                                    (approximateScore : Outcome → GlobalCollege → ℝ) →
                                      (∀ (P : Cutoff) (outcome : Outcome) (college : GlobalCollege),
                                          globalScore P outcome college = approximateScore outcome college) →
                                        (coalitionEmbedding : Fin (C + 1) ↪ GlobalCollege) →
                                          (trueValue : StudentType → GlobalCollege → ℝ) →
                                            (∀ᵐ (student : StudentType) ∂sampling.studentLaw,
                                                ∀ (college : Fin (C + 1)),
                                                  trueValue student (coalitionEmbedding college) =
                                                    sampling.commonValue student) →
                                              (∀ᵐ (outcome : Outcome) ∂sampling.outcomeLaw,
                                                  ∀ (college : Fin (C + 1)),
                                                    approximateScore outcome (coalitionEmbedding college) =
                                                      trueValue (sampling.sourceStudent outcome)
                                                          (coalitionEmbedding college) +
                                                        sampling.coalitionNoise outcome college) →
                                                PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw
                                                  eta totalSupply StudentType Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (sampling : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome)
    (rankOfStudent : StudentType → GlobalCollege → ℕ)
    (rankOfStudent_injective : ∀ (student : StudentType), Function.Injective (rankOfStudent student))
    (cutoffCoordinates : Cutoff → GlobalCollege → ℝ) (globalScore : Cutoff → Outcome → GlobalCollege → ℝ)
    (singletonDemandMeasurable :
      ∀ (P : Cutoff) (college : GlobalCollege),
        MeasurableSet
          {outcome : Outcome |
            PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
                (fun (outcome : Outcome) (college : GlobalCollege) =>
                  rankOfStudent (sampling.sourceStudent outcome) college)
                cutoffCoordinates globalScore P outcome =
              some college})
    (capacity : GlobalCollege → ℝ) (totalCapacity_eq : ∑ college : GlobalCollege, capacity college = totalSupply)
    (selectedCutoff : Cutoff)
    (selectedCutoff_clearing :
      ∀ (college : GlobalCollege),
        (AppliedModelingLib.Matching.eventMass sampling.outcomeLaw fun (outcome : Outcome) =>
            PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
                (fun (outcome : Outcome) (college : GlobalCollege) =>
                  rankOfStudent (sampling.sourceStudent outcome) college)
                cutoffCoordinates globalScore selectedCutoff outcome =
              some college) =
          capacity college)
    (approximateScore : Outcome → GlobalCollege → ℝ)
    (globalScore_eq_approximateScore :
      ∀ (P : Cutoff) (outcome : Outcome) (college : GlobalCollege),
        globalScore P outcome college = approximateScore outcome college)
    (coalitionEmbedding : Fin (C + 1) ↪ GlobalCollege) (trueValue : StudentType → GlobalCollege → ℝ)
    (hcoalition_trueValue :
      ∀ᵐ (student : StudentType) ∂sampling.studentLaw,
        ∀ (college : Fin (C + 1)), trueValue student (coalitionEmbedding college) = sampling.commonValue student)
    (hscore_trueValue :
      ∀ᵐ (outcome : Outcome) ∂sampling.outcomeLaw,
        ∀ (college : Fin (C + 1)),
          approximateScore outcome (coalitionEmbedding college) =
            trueValue (sampling.sourceStudent outcome) (coalitionEmbedding college) +
              sampling.coalitionNoise outcome college) =>
  { sampling := sampling, rankOfStudent := rankOfStudent, rankOfStudent_injective := rankOfStudent_injective,
    cutoffCoordinates := cutoffCoordinates, globalScore := globalScore,
    singletonDemandMeasurable := singletonDemandMeasurable, capacity := capacity, totalCapacity_eq := totalCapacity_eq,
    selectedCutoff := selectedCutoff, selectedCutoff_clearing := selectedCutoff_clearing,
    approximateScore := approximateScore, globalScore_eq_approximateScore := globalScore_eq_approximateScore,
    coalitionEmbedding := coalitionEmbedding,
    coalition_score_ae :=
      PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.coalition_score_ae_of_trueValue sampling coalitionEmbedding
        trueValue approximateScore hcoalition_trueValue hscore_trueValue }
```

Direct retained dependencies

- AppliedModelingLib.Matching.eventMass
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling.localCoordinates
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData
- PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.toLiteralSourceStableData

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
        GlobalCollege Cutoff) =>
  { demand := data.demand, sampling := data.sampling,
    sampling_outcomeLaw_eq :=
      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.toLiteralSourceStableData._proof_1 data,
    capacity := data.capacity, totalCapacity_eq := data.totalCapacity_eq, selectedCutoff := data.selectedCutoff,
    selectedCutoff_clearing :=
      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.toLiteralSourceStableData._proof_2 data,
    approximateScore := data.approximateScore,
    demand_globalScore_eq_approximateScore := data.globalScore_eq_approximateScore,
    coalitionEmbedding := data.coalitionEmbedding, coalition_score_ae := data.coalition_score_ae }
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.aggregateDemand
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.demand

PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData

Declaration location: paper-local

Lean declaration body

```
field sampling:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome

field rankOfStudent:
{C : ℕ} →
```

```
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      StudentType → GlobalCollege → ℕ

field rankOfStudent_injective:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (student : StudentType), Function.Injective (self.rankOfStudent student)

field cutoffCoordinates:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff → GlobalCollege → ℝ

field globalScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff → Outcome → GlobalCollege → ℝ

field singletonDemandMeasurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (P : Cutoff) (college : GlobalCollege),
  MeasurableSet
    {outcome : Outcome |
      PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
          (fun (outcome : Outcome) (college : GlobalCollege) =>
            self.rankOfStudent (self.sampling.sourceStudent outcome) college)
          self.cutoffCoordinates self.globalScore P outcome =
        some college}

field capacity:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      GlobalCollege → ℝ

field totalCapacity_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  ∑ college : GlobalCollege, self.capacity college = totalSupply

field selectedCutoff:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
```

```
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Cutoff

field selectedCutoff_clearing:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (college : GlobalCollege),
  (AppliedModelingLib.Matching.eventMass self.sampling.outcomeLaw fun (outcome : Outcome) =>
      PG24NoisyMatchingMarkets.theorem4DemandFromPreferences
          (fun (outcome : Outcome) (college : GlobalCollege) =>
            self.rankOfStudent (self.sampling.sourceStudent outcome) college)
          self.cutoffCoordinates self.globalScore self.selectedCutoff outcome =
        some college) =
    self.capacity college

field approximateScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Outcome → GlobalCollege → ℝ

field globalScore_eq_approximateScore:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff)
  (P : Cutoff) (outcome : Outcome) (college : GlobalCollege),
  self.globalScore P outcome college = self.approximateScore outcome college

field coalitionEmbedding:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      Fin (C + 1) ↪ GlobalCollege

field trueValue:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      StudentType → GlobalCollege → ℝ

field coalition_trueValue_ae:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  ∀ᵐ (student : StudentType) ∂self.sampling.studentLaw,
    ∀ (college : Fin (C + 1)),
      self.trueValue student (self.coalitionEmbedding college) = self.sampling.commonValue student

field score_eq_trueValue_plus_noise_ae:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Outcome : Type v} [inst_1 : MeasurableSpace Outcome] {GlobalCollege : Type w}
  [inst_2 : Fintype GlobalCollege] {Cutoff : Type x}
  (self :
    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome GlobalCollege
      Cutoff),
  ∀ᵐ (outcome : Outcome) ∂self.sampling.outcomeLaw,
    ∀ (college : Fin (C + 1)),
      self.approximateScore outcome (self.coalitionEmbedding college) =
        self.trueValue (self.sampling.sourceStudent outcome) (self.coalitionEmbedding college) +
```

```
self.sampling.coalitionNoise outcome college
```

Direct retained dependencies

- AppliedModelingLib.Matching.eventMass
- PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
- PG24NoisyMatchingMarkets.theorem4DemandFromPreferences

PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData.toStudentTypedSourceStableData

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff →
                      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                        Outcome GlobalCollege Cutoff

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome
        GlobalCollege Cutoff) =>
  PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.ofTrueValueSource data.sampling data.rankOfStudent
    data.rankOfStudent_injective data.cutoffCoordinates data.globalScore data.singletonDemandMeasurable data.capacity
    data.totalCapacity_eq data.selectedCutoff data.selectedCutoff_clearing data.approximateScore
    data.globalScore_eq_approximateScore data.coalitionEmbedding data.trueValue data.coalition_trueValue_ae
    data.score_eq_trueValue_plus_noise_ae
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.ofTrueValueSource
- PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData

PG24NoisyMatchingMarkets.theorem4EmbeddedCoalition

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → {GlobalCollege : Type v} → (Fin n ↪ GlobalCollege) → Finset GlobalCollege

value:
fun {n : ℕ} {GlobalCollege : Type v} (embedding : Fin n ↪ GlobalCollege) => Finset.map embedding Finset.univ
```

PG24NoisyMatchingMarkets.theorem4ActualCoalition

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → {GlobalCollege : Type u} → (Fin n ↪ GlobalCollege) → Finset GlobalCollege

value:
fun {n : ℕ} {GlobalCollege : Type u} (embedding : Fin n ↪ GlobalCollege) =>
  PG24NoisyMatchingMarkets.theorem4EmbeddedCoalition embedding
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem4EmbeddedCoalition

PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset

Declaration location: paper-local

Lean declaration body

```
field actualSubset:
{n : ℕ} →
  {GlobalCollege : Type u} →
    {embedding : Fin n ↪ GlobalCollege} →
      PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset embedding → Finset GlobalCollege

field subset_actualCoalition:
∀ {n : ℕ} {GlobalCollege : Type u} {embedding : Fin n ↪ GlobalCollege}
  (self : PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset embedding),
  self.actualSubset ⊆ PG24NoisyMatchingMarkets.theorem4ActualCoalition embedding
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.theorem4ActualCoalition

PG24NoisyMatchingMarkets.theorem4EmbeddedCoalitionSubset

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → {GlobalCollege : Type v} → (Fin n ↪ GlobalCollege) → Finset (Fin n) → Finset GlobalCollege

value:
fun {n : ℕ} {GlobalCollege : Type v} (embedding : Fin n ↪ GlobalCollege) (active : Finset (Fin n)) =>
  Finset.map embedding active
```

PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset.ofIndexed

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} →
  {GlobalCollege : Type u} →
    (embedding : Fin n ↪ GlobalCollege) →
      Finset (Fin n) → PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset embedding

value:
fun {n : ℕ} {GlobalCollege : Type u} (embedding : Fin n ↪ GlobalCollege) (indexedSubset : Finset (Fin n)) =>
  { actualSubset := PG24NoisyMatchingMarkets.theorem4EmbeddedCoalitionSubset embedding indexedSubset,
    subset_actualCoalition :=
      PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset.ofIndexed._proof_1 embedding indexedSubset }
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset
- PG24NoisyMatchingMarkets.theorem4ActualCoalition
- PG24NoisyMatchingMarkets.theorem4EmbeddedCoalitionSubset

PG24NoisyMatchingMarkets.theorem4PullbackCoalitionSubset

Declaration location: paper-local

Lean declaration body

```
type:
{n : ℕ} → {GlobalCollege : Type u} → (Fin n ↪ GlobalCollege) → Finset GlobalCollege → Finset (Fin n)

value:
fun {n : ℕ} {GlobalCollege : Type u} (embedding : Fin n ↪ GlobalCollege) (actualSubset : Finset GlobalCollege) =>
  {index : Fin n | embedding index ∈ actualSubset}
```

PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.pMuOnActualCoalitionSubset

Declaration location: paper-local

Lean declaration body

```
type:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Outcome : Type v} →
            [inst_1 : MeasurableSpace Outcome] →
              {GlobalCollege : Type w} →
                [inst_2 : Fintype GlobalCollege] →
                  {Cutoff : Type x} →
                    (data :
                        PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType
                          Outcome GlobalCollege Cutoff) →
                      ℝ → PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding → ℝ

value:
fun {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply : ℝ} {StudentType : Type u}
    [MeasurableSpace StudentType] {Outcome : Type v} [MeasurableSpace Outcome] {GlobalCollege : Type w}
    [Fintype GlobalCollege] {Cutoff : Type x}
    (data :
      PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData C noiseLaw eta totalSupply StudentType Outcome
        GlobalCollege Cutoff)
    (value : ℝ) (actualSubset : PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding) =>
  data.toLiteralSourceStableData.toExtendedCoalitionSourceStableInstance.pMu value
    (PG24NoisyMatchingMarkets.theorem4PullbackCoalitionSubset data.coalitionEmbedding actualSubset.actualSubset)
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.pMu
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.toLiteralSourceStableData
- PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset
- PG24NoisyMatchingMarkets.theorem4PullbackCoalitionSubset

Paper-specific semantic prerequisites

PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling

Paper-local declaration:
PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling
Source locator: cited publication, lines 129-190

Verbatim paper-source connection

```
\subsection{Model and Definitions}

\paragraph{General noisy economy.}
There is a continuum of students of measure 1 to be matched with a finite set of colleges $C^* = \{1, 2, \cdots, C^*\}.$ Each student is identified by their \textit{true type} $\theta = (\bm{v}, \succ)$, where
\begin{equation}
  \bm{v} = (v_1, v_2, \cdots, v_{C^*}) \in \RR^C
\end{equation}
are the \textit{true values} of the student. Here, $v_c$ is the true value of the student at college $c$. In contrast to the basic model, students have multiple true values, which differ across colleges.

A student with true values $\bm{v}$ has \textit{approximate values} $\hat{\bm{v}},$ where
\begin{equation}
  \hat{\bm{v}} = (\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*})
\end{equation}
is a random variable over $\RR^{C^*}$. ($\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*}$ need not be independent.) Thus, a student $(\bm{v},\succ)$ has true value at college $c$ equal to $v_c$, and the college obtains an approximation of the student's value equal to $\hat{v}_c.$

An \textit{economy} is given by an ordered pair $(\rho, \vec{S}, \hat{\bm{V}}),$ where $\rho$ is a probability measure over the set of student types $\Theta = \RR^{C^*}\times \mathcal{R}$ and $\vec{S}$ is a vector of college capacities. As before, we hold the total capacity of colleges to be a fixed constant $S\in (0,1)$, and assume that all students prefer being matched to any college over being unmatched. $\hat{\bm{V}} = \{\hat{\bm{v}}\}_{\bm{v}\in \RR^{C^*}}$ is a set of independent random variables over $\RR^{C^*}$, where $\hat{\bm{v}}$ give the approximate values of a student with true values $\bm{v}$. For example, one could choose $\hat{\bm{V}}$ such that $\hat{\bm{v}}$ is equal to $\bm{v}$ perturbed by noise drawn from a specified multivariate normal distribution for all $\bm{v}\in \RR^C$. Here, $\hat{\bm{V}}$ generalizes the role of the noise distribution $\DD$ in the basic model, by allowing for more broader relationships between true values and approximate values.

(Formally, we require $\hat{\bm{V}}$ to be chosen in a way that induces a valid probability distribution over true values and approximate values, where $\rho$ specifies the marginal distribution of true values, and $\hat{\bm{v}}$ specifies the conditional distribution for each true value.)

\paragraph{Cutoff characterization of stable matching.}
Consider an economy $E=(\rho, \vec{S}, \hat{\bm{V}}).$ Given cutoffs $\vec{P} = (P_1, \cdots, P_{C^*}),$ we have that a student with true values $\bm{v}$ can afford college $c$ if and only if $\hat{v}_c > P_c$. A student's demand $D^{\theta}(\vec{P})$ at cutoffs $\vec{P}$ is equal to the college they most prefer among those they can afford; $D^{\theta}(\vec{P})=\emptyset$ if the student cannot afford any college. Note that $D^{\theta}(\vec{P})$ is a random variable. Then, as in the basic model, the cutoffs $\vec{P}$ are market-clearing if and only if
\begin{equation}
  \int_{\theta} \Pr[D^{\theta}(\vec{P})=c]\,d\rho(\theta) = S_c
\end{equation}
for all $c\in {C^*}$.\footnote{Capacities are exactly filled since the total capacity $S$ is less than the total measure of students $1$, and because every student prefers to be matched over being unmatched.} If $\vec{P}$ is market-clearing, then the function $\mu:\theta\mapsto D^{\theta}(\vec{P})$ is a stable matching.

\paragraph{College coalitions.}
We are interested in analyzing the behavior of subsets of colleges in an economy that share the same true preferences over students, but who each make noisy decisions. We now formally define these subsets.

In an economy $E=(\rho, \vec{S}, \hat{\bm{V}})$, a subset of colleges $C\subseteq {C^*}$ is a $(\DD,\eta)$\vocab{-coalition} if and only if the following conditions are met:
\begin{enumerate}
  \item The true value of a student is same at each college in $C$.\footnote{Formally, the $\rho$-measure of students who have different true values at two colleges in $C$ is equal to $0$:
  \begin{equation}
    \rho\left(\{(\bm{v}, \succ) \in \Theta: \text{exists }c,c'\in C\text{ such that }v_c\neq v_c'\}\right) = 0.
  \end{equation}
  }
  If this condition is satisfied, for a student with true values $\bm{v}$, we simply use $v$ to refer to the common true value of the student at colleges in $C$. In other words, $v_c=v$ for all $c\in C$. We call $v$ the student's \textit{true value at $C$}.
  \item The approximate values of a student at colleges in $C$ is equal (in distribution) to their true value at $C$ perturbed by independent noise drawn from $\DD$: For all $\bm{v}\in \RR^{C^*},$
  \begin{equation}
    \hat{v}_c \eqdist v + X_c
  \end{equation}
  for all $c\in C$,
  where $X_1, X_2, \cdots, X_C$ are drawn i.i.d. according to $\DD$.
  \item The distribution of true values at $C$ is given by the probability measure $\eta$. That is,
  \begin{equation}
    \rho((\bm{v},\succ): v\in C) = \eta(V)
  \end{equation}
  for all $V\subseteq \RR.$
\end{enumerate}

In essence, a subset of colleges form a coalition if they behave like the full set of colleges in the basic model---i.e., the colleges in the coalition have the same true value for each student and each obtain independent noisy estimates of this value. (But, as in the basic model, students may have arbitrary preferences over colleges in the coalition, and colleges may differ in their capacities.) Indeed, one may check that in the basic model, the entire set of colleges $C$ forms a $(\DD,\eta)$-coalition.\footnote{The reader might observe that the conditions for a coalition are somewhat simpler than in the basic model: the constants $\alpha$ and $\gamma$ governing regularity conditions are notably absent in the extended model. The reason is that these regularity conditions are not necessary for the somewhat weaker results we establish in the extended model. On the other hand, the regularity conditions are also not sufficient to establish stronger results in the extended model, as we explain below.} As we will see, coalitions satisfy very similar properties to the full set of colleges in the basic model.

\paragraph{Preliminaries.}
Consider an economy $E = (\rho, \vec{S}, \hat{\bm{V}})$ and a $(\DD, \eta)$-coalition $C$. Let $\mu$ be a stable matching in $E$ with corresponding market-clearing cutoffs $\vec{P}$. Our results focus on the probability that a student $\theta=(\bm{v},\succ)$ can \textit{afford} a college in $C'\subseteq C$. Recall that the student's true value at $C$ is denoted simply by $v$. Then the probability the student can afford a college in $C'$ is
```

```
\begin{equation}
    \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
which does not depend on the true values of the student at colleges outside of the coalition, nor the preferences of the student. We set
\begin{equation}
    p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}]
\end{equation}
to denote the probability that a student with true value $v$ at coalition $C$ can afford a college in $C'\subseteq C$. In this way, we can begin to see how the setup of the extended model essentially reduces to that of the basic model, since the quantity we are interested in depends only on the true value of a student at the coalition $C$.
```

Lean-elaborated paper signature

```
field studentLaw:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome →
              MeasureTheory.Measure StudentType

field studentLaw_isProbability:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  MeasureTheory.IsProbabilityMeasure self.studentLaw

field outcomeLaw:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome →
              MeasureTheory.Measure Outcome

field outcomeLaw_isProbability:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  MeasureTheory.IsProbabilityMeasure self.outcomeLaw

field commonValue:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome → StudentType → ℝ

field commonValue_measurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  Measurable self.commonValue

field commonValue_map:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  MeasureTheory.Measure.map self.commonValue self.studentLaw = eta

field sourceStudent:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome →
              Outcome → StudentType

field coalitionNoise:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {StudentType : Type v} →
      [inst : MeasurableSpace StudentType] →
        {Outcome : Type u} →
          [inst_1 : MeasurableSpace Outcome] →
            PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome →
              Outcome → Fin (C + 1) → ℝ

field sourceCoordinates_measurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  Measurable fun (outcome : Outcome) => (self.sourceStudent outcome, self.coalitionNoise outcome)

field sourceCoordinates_map:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {StudentType : Type v} [inst : MeasurableSpace StudentType]
  {Outcome : Type u} [inst_1 : MeasurableSpace Outcome]
  (self : PG24NoisyMatchingMarkets.PG24CoalitionSourceSampling C noiseLaw eta StudentType Outcome),
  MeasureTheory.Measure.map (fun (outcome : Outcome) => (self.sourceStudent outcome, self.coalitionNoise outcome))
      self.outcomeLaw =
```

```
self.studentLaw.prod (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
```

Recorded source-to-declaration screening **Verdict:** matches

Reason: The corrected extended-coalition source bundle supplies a probability law over full student types, a common coalition value with marginal eta, and iid coalition noises with law D conditional on the student. The Lean record packages exactly that sampling component as a student law and a joint student-by-iid-noise product law while leaving the rest of StudentType unconstrained.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PG24NoisyMatchingMarkets.PG24LiteralBasicMarket

Paper-local declaration:
PG24NoisyMatchingMarkets.PG24LiteralBasicMarket
Source locator: cited publication, lines 9-18

Verbatim paper-source connection

```
There is a continuum of students of measure $1$ and a finite set of colleges $C = \{1, 2, \cdots, C\}$. A standard model of stable matching between students and colleges has three ingredients: student preferences over colleges, college preferences over students, and college capacities. Given these three ingredients, we may analyze the corresponding set of \textit{stable matchings}, matchings between students and colleges so that no student-college pair would jointly prefer to defect from the matching.

The focal point of our model is on how colleges form preferences over students, so this is where we begin. Each student has a \vocab{value} $v\in \RR$, such that colleges all prefer students with higher value. Student values specify the true preferences of colleges. Colleges, however, only obtain noisy estimates of student values, dictated by a \textit{noise distribution} $\DD$: for a student with value $v$, each college $c\in C$ obtains a noisy estimate equal to $v + X_c$, where $X_c$ is drawn independently from $\DD$. Given this noise, we are interested in the probability that a student with value $v$ matches. For the sake of comparison, it is convenient to hold two parameters of our model fixed:
\begin{itemize}
  \item $\eta$, a probability measure over student values, and
  \item $S\in (0,1)$, the total capacity of colleges.
\end{itemize}
We assume that $\eta$ has connected support, which implies that there is a unique value $v_S$ such that $\eta((v_S, \infty))=S$; i.e., the measure of students with value at least $v_S$ is equal to $S$, the total capacity of colleges. Holding $\eta$ and $S$ fixed is convenient, because we can specify two extreme behaviors. On one extreme, only students with value at least $v_S$ are matched, reflecting a ``noise free'' matching; on the other extreme, all students match with probability $S$, reflecting a ``fully noisy'' matching. Note that overall capacity is less than the measure of students.

We further assume a (weak) regularity condition on $\eta$: that there is a constant $\gamma>0$ such that $\eta((v, v+\delta)) = O(\delta^\gamma)$ for all $v\in \RR$ and $\delta > 0.$ This is equivalent to the cumulative distribution function of student values satisfying a $\gamma$-Hölder condition. We also fix a constant $\alpha>0$ that we use to specify a regularity condition on college capacities.
```

Lean-elaborated paper signature

```
field studentLaw:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              MeasureTheory.Measure StudentType

field studentLaw_isProbability:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  MeasureTheory.IsProbabilityMeasure self.studentLaw

field value:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              StudentType → ℝ

field value_measurable:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  Measurable self.value

field value_marginal:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  MeasureTheory.Measure.map self.value self.studentLaw = eta

field demand:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              PG24NoisyMatchingMarkets.PG24FinitePreferredDemand (StudentType × (Fin (C + 1) → ℝ)) (Fin (C + 1)) Cutoff

field studentPreferenceRank:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              StudentType → Fin (C + 1) → ℕ

field studentPreferenceRank_injective:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
```

```
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (student : StudentType), Function.Injective (self.studentPreferenceRank student)

field demand_preferenceRank_eq_studentPreferenceRank:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (student : StudentType) (noise : Fin (C + 1) → ℝ) (college : Fin (C + 1)),
  self.demand.preferenceRank (student, noise) college = self.studentPreferenceRank student college

field demand_outcomeLaw_eq_iid:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  self.demand.outcomeLaw = self.studentLaw.prod (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)

field capacity:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              Fin (C + 1) → ℝ

field totalCapacity_eq:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  ∑ college : Fin (C + 1), self.capacity college = totalSupply

field approximateScore:
{C : ℕ} →
  {noiseLaw eta : MeasureTheory.Measure ℝ} →
    {totalSupply alpha : ℝ} →
      {StudentType : Type u} →
        [inst : MeasurableSpace StudentType] →
          {Cutoff : Type v} →
            PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff →
              StudentType × (Fin (C + 1) → ℝ) → Fin (C + 1) → ℝ

field demand_globalScore_eq_approximateScore:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (P : Cutoff) (outcome : StudentType × (Fin (C + 1) → ℝ)) (college : Fin (C + 1)),
  self.demand.globalScore P outcome college = self.approximateScore outcome college

field basic_score:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (outcome : StudentType × (Fin (C + 1) → ℝ)) (college : Fin (C + 1)),
  self.approximateScore outcome college = self.value outcome.1 + outcome.2 college

field capacity_regular:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (self : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff),
  PG24NoisyMatchingMarkets.capacityRegular self.capacity alpha (C + 1)
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.capacityRegular

Recorded source-to-declaration screening **Verdict:** matches

Reason: The basic-economy source and scoped record context specify a probability law of student types with value marginal eta, student-level strict preferences, iid D-noise across colleges, scores v+X_c, finite capacities summing to S, and the per-college regularity premise. The target carries exactly those fields, and the repaired studentPreferenceRank plus demand-preference equality now prevents realized noise from changing the student's ranking.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.beta_max_concentratingSpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.beta_max_concentratingSpec
Source locator: cited publication, lines 55-60

Verbatim paper-source connection

```
We define two regimes of noise distributions $\DD$. $\DD$ is \vocab{$\beta$-max-concentrating} if and only if
\begin{equation}
  \Var[X^{(n)}] = O(n^{-\beta})
\end{equation}
for some $\beta > 0,$ where
$X^{(n)}$ is the maximum order statistic of $n$ samples from $\DD$ (i.e., the maximum of $n$ independent draws from $\DD$). For example, all bounded
↪   distributions are max-concentrating, and the Gaussian distribution is $1$-max-concentrating.\footnote{See, e.g., Proposition 4.7 of
↪   \cite{boucheron2012concentration}.} On the other hand, $\DD$ is \vocab{long-tailed} if and only if for all $d > 0$,
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ (noiseLaw : MeasureTheory.Measure ℝ) (maxVariance : ℕ → ℝ) (beta : ℝ),
  PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
    PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
      0 < beta ∧
        ∃ (K : ℝ) (N : ℕ),
          0 ≤ K ∧
            ∀ (n : ℕ),
              N ≤ n →
                ProbabilityTheory.variance
                    (fun (sample : Fin (n + 1) → ℝ) =>
                      AppliedModelingLib.Probability.upperOrderStatistic sample
                        AppliedModelingLib.Probability.topSampleRank)
                    (MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) ≤
                  K * (↑(n + 1)).rpow (-beta)
```

Direct retained dependencies

- AppliedModelingLib.Probability.topSampleRank
- AppliedModelingLib.Probability.upperOrderStatistic
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared 'audit/cited publication:55-60''s maximum of n independent draws and eventual Var[X^(n)] = O(n^(-beta)) for beta > 0 with the expanded target. 'topSampleRank' followed by 'upperOrderStatistic' selects the largest element of the Fin (n + 1) iid product sample, and the displayed iid source assumption supplies its eventual variance bound. 'betaMaxConcentratingVariance' supplies beta > 0 and the nonnegative K, N polynomial rate; composing the two gives exactly the target's eventual variance bound for the maximum. The n+1 indexing is an eventual reindexing of the source sample-size condition and does not change the rate or source-admissible cases.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.capacity_regularitySpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.capacity_regularitySpec
Source locator: cited publication, lines 42-48

Verbatim paper-source connection

```
With $\eta$ and $S$ fixed, we may fully parameterize an \vocab{economy} by the ordered tuple $(\DD, C, \vec{S}, \rho),$ where $\DD$ is the noise
↪    distribution, $C$ the total number of colleges, $\vec{S}=(S_1, \cdots, S_C)$ a vector of college capacities, and $\rho$ a probability measure over
↪    student types (to be defined). We require $\vec{S}$ and $\rho$ to satisfy some basic conditions:

As the total capacity $S$ is fixed, we must have that $S_1 + S_2 + \cdots + S_C = S.$ We also assume that
\begin{equation}
    S_c < \frac{\alpha}{C}
\end{equation}
for all $c\in C$ for some fixed $\alpha$, meaning that no college has a disproportionately large capacity. This kind of assumption is necessary for our
↪    results, since otherwise it is possible for an economy to technically have many colleges, but such that a few colleges take up most of the capacity; this
↪    results in the economy behaving as if it only had a few colleges. It will also be useful to define $S(C')$ as the total capacity of colleges in $C'\subseteq
↪    C$, that is, $\sum_{c\in C'} S_c.$ By definition, $S = S(C).$
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ {College : Type u} [inst : Fintype College] (capacity : College → ℝ) (alpha totalSupply : ℝ),
  PG24NoisyMatchingMarkets.capacityRegular capacity alpha (Fintype.card College) ∧
      ∑ c : College, capacity c = totalSupply ↔
    (∀ (c : College), capacity c < alpha / ↑(Fintype.card College)) ∧ ∑ c : College, capacity c = totalSupply
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.capacityRegular

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared ‘audit/cited publication:42-48’’s two capacity requirements, sum_c S_c = S and S_c < alpha/C for every college, with the target and ‘capacityRegular‘. Expanding ‘capacityRegular capacity alpha (Fintype.card College)‘ is precisely forall c, capacity c < alpha / card College, and the other conjunct is exactly the displayed total-capacity equality. The target’s biconditional only makes this definitional equivalence explicit; it changes neither the quantifier over colleges nor either source requirement.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.holder_value_law_regularitySpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.holder_value_law_regularitySpec
Source locator: cited publication, lines 16-18

Verbatim paper-source connection

```
We assume that $\eta$ has connected support, which implies that there is a unique value $v_S$ such that $\eta((v_S, \infty))=S$; i.e., the measure of
↪    students with value at least $v_S$ is equal to $S$, the total capacity of colleges. Holding $\eta$ and $S$ fixed is convenient, because we can specify
↪    two extreme behaviors. On one extreme, only students with value at least $v_S$ are matched, reflecting a ``noise free'' matching; on the other
↪    extreme, all students match with probability $S$, reflecting a ``fully noisy'' matching. Note that overall capacity is less than the measure of students.

We further assume a (weak) regularity condition on $\eta$: that there is a constant $\gamma>0$ such that $\eta((v, v+\delta)) = O(\delta^\gamma)$ for
↪    all $v\in \RR$ and $\delta > 0.$ This is equivalent to the cumulative distribution function of student values satisfying a $\gamma$-Hölder condition. We
↪    also fix a constant $\alpha>0$ that we use to specify a regularity condition on college capacities.
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ (eta : MeasureTheory.Measure ℝ),
  IsPreconnected eta.support ∧ PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta ↔
    IsPreconnected eta.support ∧
      ∃ (holderConstant : ℝ) (exponent : ℝ),
        0 < exponent ∧
          0 ≤ holderConstant ∧
            ∀ (x delta : ℝ), 0 < delta → eta.real (Set.Ioo x (x + delta)) ≤ holderConstant * delta.rpow exponent
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24HolderIntervalRegular

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared 'audit/cited publication:16-18''s connected support and the uniform interval condition eta((v,v+delta)) = O(delta^gamma) for all real v and delta > 0, with the target and 'PG24HolderIntervalRegular'. The latter expands to one global nonnegative holderConstant, a positive exponent, and forall x delta with delta > 0, eta. real(Ioo x (x + delta)) <= holderConstant * delta^exponent. This is the displayed Holder/O bound with the same interval endpoints, domains, and exponent sign, conjoined with the source connected-support condition; the biconditional is an exact expansion of that regularity predicate.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.long_tailed_noiseSpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.long_tailed_noiseSpec
Source locator: cited publication, lines 60-64

Verbatim paper-source connection

```
$X^{(n)}$ is the maximum order statistic of $n$ samples from $\DD$ (i.e., the maximum of $n$ independent draws from $\DD$). For example, all bounded
↪    distributions are max-concentrating, and the Gaussian distribution is $1$-max-concentrating.\footnote{See, e.g., Proposition 4.7 of
↪    \cite{boucheron2012concentration}.} On the other hand, $\DD$ is \vocab{long-tailed} if and only if for all $d > 0$,
\begin{equation}
  \lim_{x\rightarrow \infty} \Pr_{X\sim \DD}[X > x + d\,|\,X > x] = 1.
\end{equation}
For example, the Pareto distribution is long-tailed, as are all subexponential distributions. These two regimes do not cover the set of all noise distributions;
↪    for example, the exponential and Gumbel distributions are neither max-concentrating nor long-tailed.
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ (noiseLaw : MeasureTheory.Measure ℝ),
  PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) ↔
    ∀ (d : ℝ),
      0 < d →
        Filter.Tendsto
          (fun (x : ℝ) =>
            AppliedModelingLib.Probability.upperTailMass noiseLaw (x + d) /
              AppliedModelingLib.Probability.upperTailMass noiseLaw x)
          Filter.atTop (nhds 1)
```

Direct retained dependencies

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared ‘audit/cited publication:60-64’’s forall d > 0 limit of Pr[X > x+d | X > x] to 1 as x tends to infinity with the expanded target. ‘upperTailMass noiseLaw x‘ is noiseLaw.real (Ioi x), so the target ratio is exactly the strict upper-tail conditional ratio in the source, with the same positive-shift binder, atTop limit, and limit value 1. ‘LongTailedSurvival‘ expands to that same formula, making the target’s biconditional a direct source-definition expansion.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.match_probability_formulaSpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.match_probability_formulaSpec
Source locator: cited publication, lines 96-111

Verbatim paper-source connection

```
\subsection{Preliminaries}
We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

Consider a matching $\mu\in M(\DD, C)$, characterized by a vector of market-clearing cutoffs $\vec{P}=\vec{P}(\mu)$. Our results focus on the probability that a student $\theta=(v,\succ)$ matches to a college. This probability is equal to
\begin{equation}
  \Pr[\mu(\theta)\in C] = \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
where $X_1, \cdots, X_C \simiid \DD.$ Notice that this probability does not depend on $\succ$, the student's preferences over colleges. Indeed, the probability that a student matches is exactly the probability that a student can afford some college; whether or not they can afford a college only depends on their value and the college's cutoff (not the student's preferences). Accordingly, we can drop $\theta$ and define
\begin{equation}
  p_\mu(v) := \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
the probability that a student with value $v$ is matched under $\mu$. More generally, for a subset of colleges $C'\subseteq C$, we define
\begin{equation}
  p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
the probability that a student with value $v$ can afford some college in $C'$. By definition, $p_\mu(v) = p_\mu(v, C).$

[Next verbatim source excerpt]

\paragraph{Preliminaries.}
Consider an economy $E = (\rho, \vec{S}, \hat{\bm{V}})$ and a $(\DD, \eta)$-coalition $C$. Let $\mu$ be a stable matching in $E$ with corresponding market-clearing cutoffs $\vec{P}$. Our results focus on the probability that a student $\theta=(\bm{v},\succ)$ can \textit{afford} a college in $C'\subseteq C$. Recall that the student's true value at $C$ is denoted simply by $v$. Then the probability the student can afford a college in $C'$ is
\begin{equation}
  \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
which does not depend on the true values of the student at colleges outside of the coalition, nor the preferences of the student. We set
\begin{equation}
  p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}]
\end{equation}
to denote the probability that a student with true value $v$ at coalition $C$ can afford a college in $C'\subseteq C$. In this way, we can begin to see how the setup of the extended model essentially reduces to that of the basic model, since the quantity we are interested in depends only on the true value of a student at the coalition $C$.
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ {College : Type u} [Fintype College] (sampleLaw : MeasureTheory.Measure (College → ℝ)) (active : Finset College)
  (value : ℝ) (cutoff : College → ℝ),
  PG24NoisyMatchingMarkets.cutoffAffordanceProbability sampleLaw active value cutoff =
    sampleLaw.real {noise : College → ℝ | ∃ college ∈ active, cutoff college < value + noise college}
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared both pinned source passages in 'audit/cited publication:96-111; audit/cited publication:181-190' with the target. Expanding 'cutoffAffordanceProbability', 'cutoffCrossingProbability', 'cutoffCrossedOn', and 'noisyScore' yields sampleLaw.real of the event exists college in active with cutoff college < value + noise college. That is precisely Pr[v + X_c > P_c for some c in C'] in both basic and coalition passages, including the strict inequality and existential active-college restriction. The target permits an arbitrary noise-vector law and cutoff, which validly includes every iid, market-clearing source instance without changing the event on that source domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PG24NoisyMatchingMarkets.PaperInterface.stable_matching_cutoff_definitionSpec

Paper-local declaration:
PG24NoisyMatchingMarkets.PaperInterface.stable_matching_cutoff_definitionSpec
Source locator: cited publication, lines 78-94

Verbatim paper-source connection

```
\subsection{Cutoff Characterization of Stable Matching}
We now specify the map between economies and stable matchings, using the cutoff characterization shown by \citet{azevedo2016supply}. Consider an
↪    economy $E = (\DD, C, \vec{S}, \rho) \in E(\DD,C).$

Let $\vec{P} = (P_1, P_2, \cdots, P_C)\in \RR^C$ be a vector of \vocab{cutoffs}, such that college $c$ has cutoff $P_c$. Student $\theta = (v, \succ)$ can
↪    \vocab{afford} college $c$ if and only if their estimated value at $c$ exceeds the cutoff $P_c$. Here, we recall that a student with value $v$ receives
↪    an estimated value $v + X_c$ at college $c$, where $X_1,\cdots,X_C$ are drawn i.i.d. according to $\DD$. The \vocab{demand} $D^\theta(\vec{P})$
↪    of student $\theta$ at $\vec{P}$ is their most preferred firm among those they can afford; if they cannot afford any firm, $D^\theta(\vec{P}) =
↪    \emptyset$. In our setup, $D^\theta(\vec{P})$ is a random variable. Cutoffs $\vec{P}$ are \vocab{market-clearing} in the economy $E$ if and only if
\begin{equation}
    \int_{\Theta} \Pr\left[D^\theta(\vec{P})=c\right] \,d\rho(\theta) = S_c
\end{equation}
for all $c\in C$. In other words, the measure of students who demand a college is exactly equal to that college's capacity.\footnote{Capacities are exactly
↪    filled since the total capacity $S$ is less than the total mass of students $1$, and because every student prefers to be matched over being unmatched
↪    (i.e., colleges are overdemanded).}

A random function $\mu:\Theta\rightarrow C\cup \{\emptyset\}$ is a \vocab{stable matching} of $E$ if and only if
\begin{equation}
    \mu(\theta) = D^\theta(\vec{P})
\end{equation}
for cutoffs $\vec{P}$ that are market-clearing in $E$. We refer the reader to Lemma 1 of \cite{azevedo2016supply} to see that this condition is in fact
↪    equivalent to the classical definition of stable matching, where stability here is with respect to college preferences based on the \textit{estimated}
↪    value of students.\footnote{This corresponds to a model in which colleges admit students based only on their estimated values (and do not strategize
↪    based on, for example, which students other colleges admit).}
\\
\\
Let $M(\DD, C)$ denote the set of all matchings that are stable for some economy $E\in E(\DD, C).$ Then any matching $\mu\in M(\DD, C)$ can be
↪    characterized by a set of market-clearing cutoffs $\vec{P}(\mu).$
```

Lean-expanded paper semantic target

```
type:
Prop

value:
∀ {C : ℕ} {noiseLaw eta : MeasureTheory.Measure ℝ} {totalSupply alpha : ℝ} {StudentType : Type u}
  [inst : MeasurableSpace StudentType] {Cutoff : Type v}
  (market : PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha StudentType Cutoff)
  (matching : StudentType × (Fin (C + 1) → ℝ) → Option (Fin (C + 1))),
  market.isStable matching ↔
    ∃ (cutoffPoint : Cutoff), market.marketClearing cutoffPoint ∧ matching = market.demand.demandAt cutoffPoint
```

Direct retained dependencies

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.isStable
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.marketClearing

Recorded source-to-declaration screening **Verdict:** matches

Reason: Compared 'audit/cited publication:78-94' with the selected approved source-model convention 'PG24-BASIC-MARKET-RECORD-01', the complete 'PG24LiteralBasicMarket' fields, and the expanded stable-matching dependencies. The record binds the source student law/value marginal, strict student preferences, iid college noise, score value + noise, capacities and their total, and capacity regularity; 'theorem4AffordableColleges' uses the source strict score-cutoff test, 'demandAt' chooses the most-preferred affordable college, and 'marketClearing' equates each demand-event mass to capacity. Consequently 'isStable' expands to exactly existence of a market-clearing cutoff whose demand map equals the matching, which is the source cutoff characterization. The approved convention is source-equivalent rather than a materially non-equivalent corrected target, so ordinary 'matches' applies.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 259-268

Verbatim source input

```
\begin{theorem}[Attenuation]\label{thm:attenuating}
  Let $\DD$ be max-concentrating and $v\in \RR$. Then for all $\epsilon > 0$, there exists $C(\epsilon)$ such that for all $C > C(\epsilon)$ and $\mu\in
  ↪   M(\DD, C)$,
  \begin{equation}
    p_\mu(v) < \epsilon \qquad \text{if $v < v_S$},
  \end{equation}
  and
  \begin{equation}
    p_\mu(v) > 1 - \epsilon \qquad \text{if $v > v_S$}.
  \end{equation}
\end{theorem}
```

Expanded PaperInterface specification

```
∀ {Admissible : ℕ → Type w} {StudentType : (C : ℕ) → Admissible C → Type u}
 [inst : (C : ℕ) → (a : Admissible C) → MeasurableSpace (StudentType C a)] {Cutoff : (C : ℕ) → Admissible C → Type v}
 (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
 [MeasureTheory.IsProbabilityMeasure eta] {maxVariance : ℕ → ℝ} {alpha beta totalSupply vS : ℝ}
 (market :
  (C : ℕ) →
   (a : Admissible C) →
    PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha (StudentType C a) (Cutoff C a))
 (selectedCutoff : (C : ℕ) → (a : Admissible C) → Cutoff C a),
 (∀ (C : ℕ) (a : Admissible C), (market C a).marketClearing (selectedCutoff C a)) →
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
   PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
    0 < alpha →
     PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta →
      IsPreconnected eta.support →
       0 < totalSupply →
        totalSupply < 1 →
         eta.real (Set.Ioi vS) = totalSupply →
          (∀ (value epsilon : ℝ),
            value < vS →
             0 < epsilon →
              ∀ᶠ (C : ℕ) in Filter.atTop,
               ∀ (a : Admissible C),
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                  ((market C a).demand.cutoffCoordinates (selectedCutoff C a)) <
                 epsilon) ∧
           ∀ (value epsilon : ℝ),
            vS < value →
             0 < epsilon →
              ∀ᶠ (C : ℕ) in Filter.atTop,
               ∀ (a : Admissible C),
                1 - epsilon <
                 PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                  ((market C a).demand.cutoffCoordinates (selectedCutoff C a))
```

Retained governing declarations

- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24HolderIntervalRegular
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.marketClearing
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound

Lean-elaborated claim roles

```
1. parameter: ℕ → Type w
2. parameter: (C : ℕ) → Admissible C → Type u
3. parameter: (C : ℕ) → (a : Admissible C) → MeasurableSpace (StudentType C a)
4. parameter: (C : ℕ) → Admissible C → Type v
5. parameter: MeasureTheory.Measure ℝ
6. parameter: MeasureTheory.Measure ℝ
7. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
8. assumption: MeasureTheory.IsProbabilityMeasure eta
9. parameter: ℕ → ℝ
10. parameter: ℝ
11. parameter: ℝ
```

```
12. parameter: ℝ
13. parameter: ℝ
14. parameter: (C : ℕ) →
  (a : Admissible C) →
    PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha (StudentType C a) (Cutoff C a)
15. parameter: (C : ℕ) → (a : Admissible C) → Cutoff C a
16. assumption: ∀ (C : ℕ) (a : Admissible C), (market C a).marketClearing (selectedCutoff C a)
17. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
18. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
19. assumption: 0 < alpha
20. assumption: PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta
21. assumption: IsPreconnected eta.support
22. assumption: 0 < totalSupply
23. assumption: totalSupply < 1
24. assumption: eta.real (Set.Ioi vS) = totalSupply
25. conclusion: (∀ (value epsilon : ℝ),
    value < vS →
      0 < epsilon →
        ∀ᶠ (C : ℕ) in Filter.atTop,
          ∀ (a : Admissible C),
            PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                ((market C a).demand.cutoffCoordinates (selectedCutoff C a)) <
              epsilon) ∧
  ∀ (value epsilon : ℝ),
    vS < value →
      0 < epsilon →
        ∀ᶠ (C : ℕ) in Filter.atTop,
          ∀ (a : Admissible C),
            1 - epsilon <
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                ((market C a).demand.cutoffCoordinates (selectedCutoff C a))
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.review_theorem1_attenuationSpec_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: For every fixed below-threshold or above-threshold value and positive tolerance, the Spec preserves the source eventual uniformity over admissible basic economies and selected clearing cutoffs, with low probability tending to zero and high probability tending to one.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 274-279

Verbatim source input

```
\begin{theorem}[Amplification]\label{thm:amplifying}
  Let $\DD$ be long-tailed and $v\in \RR$. Then for all $\epsilon > 0$, there exists $C(\epsilon)$ such that for all $C > C(\epsilon)$ and $\mu\in M(\DD, C)$,
  \begin{equation}
    |p_\mu(v) - S| < \epsilon.
  \end{equation}
\end{theorem}
```

Expanded PaperInterface specification

```
∀ {StudentTypeSeq : ℕ → Type u} [inst : (C : ℕ) → MeasurableSpace (StudentTypeSeq C)] {Admissible : ℕ → Type v}
  (CutoffSeq : ℕ → Type w) (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta] {totalSupply alpha : ℝ}
  (market :
    (C : ℕ) →
      Admissible C →
        PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha (StudentTypeSeq C)
          (CutoffSeq C))
  (selectedCutoff : (C : ℕ) → Admissible C → CutoffSeq C),
  (∀ (C : ℕ) (a : Admissible C), (market C a).marketClearing (selectedCutoff C a)) →
    PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta →
      IsPreconnected eta.support →
        PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
          0 < totalSupply →
            totalSupply < 1 →
              0 < alpha →
                ∀ (target epsilon : ℝ),
                  0 < epsilon →
                    ∀ᶠ (C : ℕ) in Filter.atTop,
                      ∀ (a : Admissible C),
                        |PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ target
                                ((market C a).demand.cutoffCoordinates (selectedCutoff C a)) -
                              totalSupply| <
                          epsilon
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand
- PG24NoisyMatchingMarkets.PG24HolderIntervalRegular
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket
- PG24NoisyMatchingMarkets.PG24LiteralBasicMarket.marketClearing
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Lean-elaborated claim roles

```
1. parameter: ℕ → Type u
2. parameter: (C : ℕ) → MeasurableSpace (StudentTypeSeq C)
3. parameter: ℕ → Type v
4. parameter: ℕ → Type w
5. parameter: MeasureTheory.Measure ℝ
6. parameter: MeasureTheory.Measure ℝ
7. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
8. assumption: MeasureTheory.IsProbabilityMeasure eta
9. parameter: ℝ
10. parameter: ℝ
11. parameter: (C : ℕ) →
  Admissible C →
    PG24NoisyMatchingMarkets.PG24LiteralBasicMarket C noiseLaw eta totalSupply alpha (StudentTypeSeq C) (CutoffSeq C)
12. parameter: (C : ℕ) → Admissible C → CutoffSeq C
13. assumption: ∀ (C : ℕ) (a : Admissible C), (market C a).marketClearing (selectedCutoff C a)
14. assumption: PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta
15. assumption: IsPreconnected eta.support
16. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
17. assumption: 0 < totalSupply
18. assumption: totalSupply < 1
19. assumption: 0 < alpha
20. parameter: ℝ
21. parameter: ℝ
22. assumption: 0 < epsilon
23. conclusion: ∀ᶠ (C : ℕ) in Filter.atTop,
  ∀ (a : Admissible C),
    |PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
        Finset.univ target ((market C a).demand.cutoffCoordinates (selectedCutoff C a)) -
      totalSupply| <
    epsilon
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.review_theorem2_amplificationSpec_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: The Spec preserves the source long-tailed conclusion: every fixed real target has eventual uniformly small distance between full-coalition affordability probability and total supply across the selected basic economies.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 196-206

Verbatim source input

```
\begin{theorem}[Attenuation in Coalitions]\label{thm:attenuating-extended}
    Let $\DD$ be max-concentrating. Then for any $\epsilon > 0$, there exists $C(\epsilon)$ such that the following holds.
    Let $C$ be a $(\DD, \eta)$-coalition in an economy $E$ with stable matching $\mu$ such that $|C|>C(\epsilon)$. Then there exists $v'\in \RR$ and $C'\subseteq C$ with $\frac{|C'|}{|C|} > 1 - \epsilon$, such that
    \begin{equation}
        p_\mu(v, C') < \epsilon \qquad\text{ for all $v<v'-\epsilon$},
    \end{equation}
    and
    \begin{equation}
        p_\mu(v, C) > 1 - \epsilon \qquad\text{ for all $v>v'+\epsilon$}.
    \end{equation}
\end{theorem}

[Next verbatim source excerpt]

\subsection{Model and Definitions}

\paragraph{General noisy economy.}
There is a continuum of students of measure 1 to be matched with a finite set of colleges $C^* = \{1, 2, \cdots, C^*\}.$ Each student is identified by their \textit{true type} $\theta = (\bm{v}, \succ)$, where
\begin{equation}
    \bm{v} = (v_1, v_2, \cdots, v_{C^*}) \in \RR^C
\end{equation}
are the \textit{true values} of the student. Here, $v_c$ is the true value of the student at college $c$. In contrast to the basic model, students have multiple true values, which differ across colleges.

A student with true values $\bm{v}$ has \textit{approximate values} $\hat{\bm{v}},$ where
\begin{equation}
    \hat{\bm{v}} = (\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*})
\end{equation}
is a random variable over $\RR^{C^*}$. ($\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*}$ need not be independent.) Thus, a student $(\bm{v},\succ)$ has true value at college $c$ equal to $v_c$, and the college obtains an approximation of the student's value equal to $\hat{v}_c.$

An \textit{economy} is given by an ordered pair $(\rho, \vec{S}, \hat{\bm{V}}),$ where $\rho$ is a probability measure over the set of student types $\Theta = \RR^{C^*}\times \mathcal{R}$ and $\vec{S}$ is a vector of college capacities. As before, we hold the total capacity of colleges to be a fixed constant $S\in (0,1)$, and assume that all students prefer being matched to any college over being unmatched. $\hat{\bm{V}} = \{\hat{\bm{v}}\}_{\bm{v}\in \RR^{C^*}}$ is a set of independent random variables over $\RR^{C^*}$, where $\hat{\bm{v}}$ give the approximate values of a student with true values $\bm{v}$. For example, one could choose $\hat{\bm{V}}$ such that $\hat{\bm{v}}$ is equal to $\bm{v}$ perturbed by noise drawn from a specified multivariate normal distribution for all $\bm{v}\in \RR^C$. Here, $\hat{\bm{V}}$ generalizes the role of the noise distribution $\DD$ in the basic model, by allowing for more broader relationships between true values and approximate values.

(Formally, we require $\hat{\bm{V}}$ to be chosen in a way that induces a valid probability distribution over true values and approximate values, where $\rho$ specifies the marginal distribution of true values, and $\hat{\bm{v}}$ specifies the conditional distribution for each true value.)

\paragraph{Cutoff characterization of stable matching.}
Consider an economy $E=(\rho, \vec{S}, \hat{\bm{V}}).$ Given cutoffs $\vec{P} = (P_1, \cdots, P_{C^*}),$ we have that a student with true values $\bm{v}$ can afford college $c$ if and only if $\hat{v}_c > P_c$. A student's demand $D^{\theta}(\vec{P})$ at cutoffs $\vec{P}$ is equal to the college they most prefer among those they can afford; $D^{\theta}(\vec{P})=\emptyset$ if the student cannot afford any college. Note that $D^{\theta}(\vec{P})$ is a random variable. Then, as in the basic model, the cutoffs $\vec{P}$ are market-clearing if and only if
\begin{equation}
    \int_{\theta} \Pr[D^{\theta}(\vec{P})=c]\,d\rho(\theta) = S_c
\end{equation}
for all $c\in {C^*}$.\footnote{Capacities are exactly filled since the total capacity $S$ is less than the total measure of students $1$, and because every student prefers to be matched over being unmatched.} If $\vec{P}$ is market-clearing, then the function $\mu:\theta\mapsto D^{\theta}(\vec{P})$ is a stable matching.

\paragraph{College coalitions.}
We are interested in analyzing the behavior of subsets of colleges in an economy that share the same true preferences over students, but who each make noisy decisions. We now formally define these subsets.

In an economy $E=(\rho, \vec{S}, \hat{\bm{V}})$, a subset of colleges $C\subseteq {C^*}$ is a $(\DD,\eta)$\vocab{-coalition} if and only if the following conditions are met:
\begin{enumerate}
    \item The true value of a student is same at each college in $C$.\footnote{Formally, the $\rho$-measure of students who have different true values at two colleges in $C$ is equal to $0$:
    \begin{equation}
        \rho\left(\{(\bm{v}, \succ) \in \Theta: \text{exists }c,c'\in C\text{ such that }v_c\neq v_c'\}\right) = 0.
    \end{equation}
    }
    If this condition is satisfied, for a student with true values $\bm{v}$, we simply use $v$ to refer to the common true value of the student at colleges in $C$. In other words, $v_c=v$ for all $c\in C$. We call $v$ the student's \textit{true value at $C$}.
    \item The approximate values of a student at colleges in $C$ is equal (in distribution) to their true value at $C$ perturbed by independent noise drawn from $\DD$: For all $\bm{v}\in \RR^{C^*},$
    \begin{equation}
        \hat{v}_c \eqdist v + X_c
    \end{equation}
    for all $c\in C$,
    where $X_1, X_2, \cdots, X_C$ are drawn i.i.d. according to $\DD$.
    \item The distribution of true values at $C$ is given by the probability measure $\eta$. That is,
    \begin{equation}
        \rho((\bm{v},\succ): v\in C) = \eta(V)
    \end{equation}
    for all $V\subseteq \RR.$
\end{enumerate}
```

```
In essence, a subset of colleges form a coalition if they behave like the full set of colleges in the basic model---i.e., the colleges in the coalition have the
↪   same true value for each student and each obtain independent noisy estimates of this value. (But, as in the basic model, students may have arbitrary
↪   preferences over colleges in the coalition, and colleges may differ in their capacities.) Indeed, one may check that in the basic model, the entire set of
↪   colleges $C$ forms a $(\DD,\eta)$-coalition.\footnote{The reader might observe that the conditions for a coalition are somewhat simpler than in the
↪   basic model: the constants $\alpha$ and $\gamma$ governing regularity conditions are notably absent in the extended model. The reason is that these
↪   regularity conditions are not necessary for the somewhat weaker results we establish in the extended model. On the other hand, the regularity
↪   conditions are also not sufficient to establish stronger results in the extended model, as we explain below.} As we will see, coalitions satisfy very
↪   similar properties to the full set of colleges in the basic model.

\paragraph{Preliminaries.}
Consider an economy $E = (\rho, \vec{S}, \hat{\bm{V}})$ and a $(\DD, \eta)$-coalition $C$. Let $\mu$ be a stable matching in $E$ with corresponding
↪   market-clearing cutoffs $\vec{P}$. Our results focus on the probability that a student $\theta=(\bm{v},\succ)$ can \textit{afford} a college in
↪   $C'\subseteq C$. Recall that the student's true value at $C$ is denoted simply by $v$. Then the probability the student can afford a college in $C'$ is
\begin{equation}
  \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
which does not depend on the true values of the student at colleges outside of the coalition, nor the preferences of the student. We set
\begin{equation}
  p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}]
\end{equation}
to denote the probability that a student with true value $v$ at coalition $C$ can afford a college in $C'\subseteq C$. In this way, we can begin to see how
↪   the setup of the extended model essentially reduces to that of the basic model, since the quantity we are interested in depends only on the true value
↪   of a student at the coalition $C$.
```

Expanded PaperInterface specification

```
∀ (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta] {maxVariance : ℕ → ℝ} {beta : ℝ},
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
      ∀ {totalSupply : ℝ},
        0 < totalSupply →
          totalSupply < 1 →
            ∀ (epsilon : ℝ),
              0 < epsilon →
                ∃ (N : ℕ),
                  ∀ (C : ℕ),
                    N ≤ C →
                      ∀ (StudentType : Type u) [inst : MeasurableSpace StudentType] (Outcome : Type v)
                        [inst_1 : MeasurableSpace Outcome] (GlobalCollege : Type w) [inst_2 : Fintype GlobalCollege]
                        (Cutoff : Type x)
                        (data :
                          PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                            Outcome GlobalCollege Cutoff),
                        ∃ (threshold : ℝ) (actualCoalition : Finset GlobalCollege) (actualLargeSubset :
                          PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding),
                          actualCoalition = PG24NoisyMatchingMarkets.theorem4ActualCoalition data.coalitionEmbedding ∧
                            PG24NoisyMatchingMarkets.CoalitionLargeSubset actualCoalition actualLargeSubset.actualSubset
                                epsilon ∧
                              N < actualCoalition.card ∧
                                (∀ value < threshold - epsilon,
                                    data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value
                                        actualLargeSubset <
                                      epsilon) ∧
                                  ∀ (value : ℝ),
                                    threshold + epsilon < value →
                                      1 - epsilon <
                                        data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value
                                          (PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset.ofIndexed
                                            data.coalitionEmbedding Finset.univ)
```

Retained governing declarations

- PG24NoisyMatchingMarkets.CoalitionLargeSubset
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.pMuOnActualCoalitionSubset
- PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData
- PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData.toStudentTypedSourceStableData
- PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset
- PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset.ofIndexed
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.theorem4ActualCoalition

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. parameter: MeasureTheory.Measure ℝ
3. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
4. assumption: MeasureTheory.IsProbabilityMeasure eta
5. parameter: ℕ → ℝ
6. parameter: ℝ
7. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
8. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
```

```
9. parameter: ℝ
10. assumption: 0 < totalSupply
11. assumption: totalSupply < 1
12. parameter: ℝ
13. assumption: 0 < epsilon
14. conclusion: ∃ (N : ℕ),
  ∀ (C : ℕ),
    N ≤ C →
      ∀ (StudentType : Type u) [inst : MeasurableSpace StudentType] (Outcome : Type v)
        [inst_1 : MeasurableSpace Outcome] (GlobalCollege : Type w) [inst_2 : Fintype GlobalCollege] (Cutoff : Type x)
        (data :
          PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome
            GlobalCollege Cutoff),
        ∃ (threshold : ℝ) (actualCoalition : Finset GlobalCollege) (actualLargeSubset :
          PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding),
          actualCoalition = PG24NoisyMatchingMarkets.theorem4ActualCoalition data.coalitionEmbedding ∧
            PG24NoisyMatchingMarkets.CoalitionLargeSubset actualCoalition actualLargeSubset.actualSubset epsilon ∧
              N < actualCoalition.card ∧
                (∀ value < threshold - epsilon,
                    data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value actualLargeSubset < epsilon) ∧
                  ∀ (value : ℝ),
                    threshold + epsilon < value →
                      1 - epsilon <
                        data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value
                          (PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset.ofIndexed data.coalitionEmbedding
                            Finset.univ)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.review_theorem3_attenuationInCoalitionsSpec_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: The literal extended primitive bundle preserves the source coalition inputs. For every tolerance it selects a size threshold before an arbitrary economy and returns a large embedded subcoalition with the source low/high affordability threshold conclusions.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 213-220

Verbatim source input

```
\begin{theorem}[Amplification in Coalitions]\label{thm:amplifying-extended}
    Let $\DD$ be long-tailed. Then for any $\epsilon > 0$, there exists $C(\epsilon)$ such that the following holds.
    Let $C$ be a $(\DD, \eta)$-coalition in an economy $E$ with stable matching $\mu$ such that $|C|>C(\epsilon)$. Then there exists $S'\in \RR$ and $C'\subseteq C$ with $\frac{|C'|}{|C|} > 1 - \epsilon$, such that
    \begin{equation}
        |p_\mu(v, C')-S'| < \epsilon
    \end{equation}
    for all $v$ except on a set of $\eta$-measure at most $\epsilon$.
\end{theorem}

[Next verbatim source excerpt]

\subsection{Model and Definitions}

\paragraph{General noisy economy.}
There is a continuum of students of measure 1 to be matched with a finite set of colleges $C^* = \{1, 2, \cdots, C^*\}.$ Each student is identified by their \textit{true type} $\theta = (\bm{v}, \succ)$, where
\begin{equation}
    \bm{v} = (v_1, v_2, \cdots, v_{C^*}) \in \RR^C
\end{equation}
are the \textit{true values} of the student. Here, $v_c$ is the true value of the student at college $c$. In contrast to the basic model, students have multiple true values, which differ across colleges.

A student with true values $\bm{v}$ has \textit{approximate values} $\hat{\bm{v}},$ where
\begin{equation}
    \hat{\bm{v}} = (\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*})
\end{equation}
is a random variable over $\RR^{C^*}$. ($\hat{v}_1, \hat{v}_2, \cdots, \hat{v}_{C^*}$ need not be independent.) Thus, a student $(\bm{v},\succ)$ has true value at college $c$ equal to $v_c$, and the college obtains an approximation of the student's value equal to $\hat{v}_c.$

An \textit{economy} is given by an ordered pair $(\rho, \vec{S}, \hat{\bm{V}}),$ where $\rho$ is a probability measure over the set of student types $\Theta = \RR^{C^*}\times \mathcal{R}$ and $\vec{S}$ is a vector of college capacities. As before, we hold the total capacity of colleges to be a fixed constant $S\in (0,1)$, and assume that all students prefer being matched to any college over being unmatched. $\hat{\bm{V}} = \{\hat{\bm{v}}\}_{\bm{v}\in \RR^{C^*}}$ is a set of independent random variables over $\RR^{C^*}$, where $\hat{\bm{v}}$ give the approximate values of a student with true values $\bm{v}$. For example, one could choose $\hat{\bm{V}}$ such that $\hat{\bm{v}}$ is equal to $\bm{v}$ perturbed by noise drawn from a specified multivariate normal distribution for all $\bm{v}\in \RR^C$. Here, $\hat{\bm{V}}$ generalizes the role of the noise distribution $\DD$ in the basic model, by allowing for more broader relationships between true values and approximate values.

(Formally, we require $\hat{\bm{V}}$ to be chosen in a way that induces a valid probability distribution over true values and approximate values, where $\rho$ specifies the marginal distribution of true values, and $\hat{\bm{v}}$ specifies the conditional distribution for each true value.)

\paragraph{Cutoff characterization of stable matching.}
Consider an economy $E=(\rho, \vec{S}, \hat{\bm{V}}).$ Given cutoffs $\vec{P} = (P_1, \cdots, P_{C^*}),$ we have that a student with true values $\bm{v}$ can afford college $c$ if and only if $\hat{v}_c > P_c$. A student's demand $D^{\theta}(\vec{P})$ at cutoffs $\vec{P}$ is equal to the college they most prefer among those they can afford; $D^{\theta}(\vec{P})=\emptyset$ if the student cannot afford any college. Note that $D^{\theta}(\vec{P})$ is a random variable. Then, as in the basic model, the cutoffs $\vec{P}$ are market-clearing if and only if
\begin{equation}
    \int_{\theta} \Pr[D^{\theta}(\vec{P})=c]\,d\rho(\theta) = S_c
\end{equation}
for all $c\in {C^*}$.\footnote{Capacities are exactly filled since the total capacity $S$ is less than the total measure of students $1$, and because every student prefers to be matched over being unmatched.} If $\vec{P}$ is market-clearing, then the function $\mu:\theta\mapsto D^{\theta}(\vec{P})$ is a stable matching.

\paragraph{College coalitions.}
We are interested in analyzing the behavior of subsets of colleges in an economy that share the same true preferences over students, but who each make noisy decisions. We now formally define these subsets.

In an economy $E=(\rho, \vec{S}, \hat{\bm{V}})$, a subset of colleges $C\subseteq {C^*}$ is a $(\DD,\eta)$\vocab{-coalition} if and only if the following conditions are met:
\begin{enumerate}
    \item The true value of a student is same at each college in $C$.\footnote{Formally, the $\rho$-measure of students who have different true values at two colleges in $C$ is equal to $0$:
    \begin{equation}
        \rho\left(\{(\bm{v}, \succ) \in \Theta: \text{exists }c,c'\in C\text{ such that }v_c\neq v_c'\}\right) = 0.
    \end{equation}
    }
    If this condition is satisfied, for a student with true values $\bm{v}$, we simply use $v$ to refer to the common true value of the student at colleges in $C$. In other words, $v_c=v$ for all $c\in C$. We call $v$ the student's \textit{true value at $C$}.
    \item The approximate values of a student at colleges in $C$ is equal (in distribution) to their true value at $C$ perturbed by independent noise drawn from $\DD$: For all $\bm{v}\in \RR^{C^*},$
    \begin{equation}
        \hat{v}_c \eqdist v + X_c
    \end{equation}
    for all $c\in C$,
    where $X_1, X_2, \cdots, X_C$ are drawn i.i.d. according to $\DD$.
    \item The distribution of true values at $C$ is given by the probability measure $\eta$. That is,
    \begin{equation}
        \rho((\bm{v},\succ): v\in C) = \eta(V)
    \end{equation}
    for all $V\subseteq \RR.$
\end{enumerate}
```

```
In essence, a subset of colleges form a coalition if they behave like the full set of colleges in the basic model---i.e., the colleges in the coalition have the same true value for each student and each obtain independent noisy estimates of this value. (But, as in the basic model, students may have arbitrary preferences over colleges in the coalition, and colleges may differ in their capacities.) Indeed, one may check that in the basic model, the entire set of colleges $C$ forms a $(\DD,\eta)$-coalition.\footnote{The reader might observe that the conditions for a coalition are somewhat simpler than in the basic model: the constants $\alpha$ and $\gamma$ governing regularity conditions are notably absent in the extended model. The reason is that these regularity conditions are not necessary for the somewhat weaker results we establish in the extended model. On the other hand, the regularity conditions are also not sufficient to establish stronger results in the extended model, as we explain below.} As we will see, coalitions satisfy very similar properties to the full set of colleges in the basic model.

\paragraph{Preliminaries.}
Consider an economy $E = (\rho, \vec{S}, \hat{\bm{V}})$ and a $(\DD, \eta)$-coalition $C$. Let $\mu$ be a stable matching in $E$ with corresponding market-clearing cutoffs $\vec{P}$. Our results focus on the probability that a student $\theta=(\bm{v},\succ)$ can \textit{afford} a college in $C'\subseteq C$. Recall that the student's true value at $C$ is denoted simply by $v$. Then the probability the student can afford a college in $C'$ is
\begin{equation}
  \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
which does not depend on the true values of the student at colleges outside of the coalition, nor the preferences of the student. We set
\begin{equation}
  p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}]
\end{equation}
to denote the probability that a student with true value $v$ at coalition $C$ can afford a college in $C'\subseteq C$. In this way, we can begin to see how the setup of the extended model essentially reduces to that of the basic model, since the quantity we are interested in depends only on the true value of a student at the coalition $C$.
```

Expanded PaperInterface specification

```
∀ (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta],
  PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
    ∀ {totalSupply : ℝ},
      0 < totalSupply →
        totalSupply < 1 →
          ∀ (epsilon : ℝ),
            0 < epsilon →
              ∃ (N : ℕ),
                ∀ (C : ℕ),
                  N ≤ C →
                    ∀ (StudentType : Type u) [inst : MeasurableSpace StudentType] (Outcome : Type v)
                      [inst_1 : MeasurableSpace Outcome] (GlobalCollege : Type w) [inst_2 : Fintype GlobalCollege]
                      (Cutoff : Type x)
                      (data :
                        PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType
                          Outcome GlobalCollege Cutoff),
                      ∃ (effectiveSupply : ℝ) (actualCoalition : Finset GlobalCollege) (actualLargeSubset :
                        PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding) (regularSet :
                        Set ℝ),
                        actualCoalition = PG24NoisyMatchingMarkets.theorem4ActualCoalition data.coalitionEmbedding ∧
                          PG24NoisyMatchingMarkets.CoalitionLargeSubset actualCoalition actualLargeSubset.actualSubset
                              epsilon ∧
                            N < actualCoalition.card ∧
                              eta.real regularSetᶜ ≤ epsilon ∧
                                ∀ value ∈ regularSet,
                                  |data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value
                                          actualLargeSubset -
                                        effectiveSupply| <
                                    epsilon
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.CoalitionLargeSubset
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24StudentTypedSourceStableData.pMuOnActualCoalitionSubset
- PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData
- PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData.toStudentTypedSourceStableData
- PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset
- PG24NoisyMatchingMarkets.theorem4ActualCoalition

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. parameter: MeasureTheory.Measure ℝ
3. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
4. assumption: MeasureTheory.IsProbabilityMeasure eta
5. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
6. parameter: ℝ
7. assumption: 0 < totalSupply
8. assumption: totalSupply < 1
9. parameter: ℝ
10. assumption: 0 < epsilon
11. conclusion: ∃ (N : ℕ),
  ∀ (C : ℕ),
    N ≤ C →
      ∀ (StudentType : Type u) [inst : MeasurableSpace StudentType] (Outcome : Type v)
```

```
[inst_1 : MeasurableSpace Outcome] (GlobalCollege : Type w) [inst_2 : Fintype GlobalCollege] (Cutoff : Type x)
(data :
  PG24NoisyMatchingMarkets.PG24TrueValueSourceStableData C noiseLaw eta totalSupply StudentType Outcome
    GlobalCollege Cutoff),
∃ (effectiveSupply : ℝ) (actualCoalition : Finset GlobalCollege) (actualLargeSubset :
  PG24NoisyMatchingMarkets.Theorem4ActualCoalitionSubset data.coalitionEmbedding) (regularSet : Set ℝ),
  actualCoalition = PG24NoisyMatchingMarkets.theorem4ActualCoalition data.coalitionEmbedding ∧
    PG24NoisyMatchingMarkets.CoalitionLargeSubset actualCoalition actualLargeSubset.actualSubset epsilon ∧
      N < actualCoalition.card ∧
        eta.real regularSetᶜ ≤ epsilon ∧
          ∀ value ∈ regularSet,
            |data.toStudentTypedSourceStableData.pMuOnActualCoalitionSubset value actualLargeSubset -
                effectiveSupply| <
              epsilon
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.review_theorem4_amplificationInCoalitionsSpec_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: The Spec preserves the extended source quantifier order and returns an effective supply, large embedded subcoalition, and eta-exceptional regular set on which affordability is uniformly epsilon-close under long-tailed noise.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 312-321

Verbatim source input

```
\begin{proposition}\label{thm1v2}
    Let $\DD$ be $\beta$-max-concentrating and $v\in \RR$. Let $\mu \in M(\DD, C)$. Then
    \begin{equation}
        \int_{v_S}^\infty p_\mu(v)\,d\eta(v) = O\left(C^{-K(\beta,\gamma)}\right),
    \end{equation}
    where
    \begin{equation}
        K(\beta,\gamma) := \frac{2\beta\gamma}{3\beta\gamma + 2\beta + 5\gamma + 6}.
    \end{equation}
\end{proposition}
```

Additional assumptions The iid one-draw noise law satisfies MemLp (fun x : ℝ => x) 2 noiseLaw; this finite second moment is used only for the Case-2 one-draw Chebyshev step.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

```
Under the source beta-max and value-law conditions plus a finite iid one-draw second moment, uniformly over literal selected stable instances the matched
↪    mass at values at or below vS is O(C^(-K(beta,gamma))).
```

Archival source anchor: cited publication, lines 312-321

Expanded PaperInterface specification

```
∀ {Admissible : ℕ → Type w} {StudentType : (C : ℕ) → Admissible C → Type u}
  [inst : (C : ℕ) → (a : Admissible C) → MeasurableSpace (StudentType C a)] {Cutoff : (C : ℕ) → Admissible C → Type v}
  (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta] {maxVariance : ℕ → ℝ} {alpha beta totalSupply vS : ℝ}
  (inst_3 :
    (C : ℕ) →
      (a : Admissible C) →
        PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha (StudentType C a)
          (Cutoff C a)),
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
      PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw →
        0 ≤ alpha →
          PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta →
            eta.real (Set.Ioi vS) = totalSupply →
              ∃ (holderConstant : ℝ) (gamma : ℝ) (A : ℝ),
                0 < gamma ∧
                  0 ≤ holderConstant ∧
                    (∀ (x delta : ℝ),
                        0 < delta → eta.real (Set.Ioo x (x + delta)) ≤ holderConstant * delta.rpow gamma) ∧
                      0 ≤ A ∧
                        ∀ᶠ (C : ℕ) in Filter.atTop,
                          ∀ (a : Admissible C),
                            (AppliedModelingLib.Matching.eventMass
                                ((inst_3 C a).studentLaw.prod
                                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw))
                                fun (outcome : StudentType C a × (Fin (C + 1) → ℝ)) =>
                                (inst_3 C a).value outcome.1 ∈ Set.Iic vS ∧
                                  AppliedModelingLib.Matching.chosenInActive
                                    ((inst_3 C a).literal.demand.demandAt (inst_3 C a).literal.selectedCutoff)
                                    Finset.univ outcome) ≤
                              A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Retained governing declarations

- AppliedModelingLib.Matching.chosenInActive
- AppliedModelingLib.Matching.eventMass
- PG24NoisyMatchingMarkets.PG24FinitePreferredDemand.demandAt
- PG24NoisyMatchingMarkets.PG24HolderIntervalRegular
- PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment
- PG24NoisyMatchingMarkets.theorem1TailK

Lean-elaborated claim roles

```
1. parameter: ℕ → Type w
2. parameter: (C : ℕ) → Admissible C → Type u
3. parameter: (C : ℕ) → (a : Admissible C) → MeasurableSpace (StudentType C a)
4. parameter: (C : ℕ) → Admissible C → Type v
5. parameter: MeasureTheory.Measure ℝ
6. parameter: MeasureTheory.Measure ℝ
7. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
8. assumption: MeasureTheory.IsProbabilityMeasure eta
9. parameter: ℕ → ℝ
10. parameter: ℝ
11. parameter: ℝ
12. parameter: ℝ
13. parameter: ℝ
14. parameter: (C : ℕ) →
  (a : Admissible C) →
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha (StudentType C a)
      (Cutoff C a)
15. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
16. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
17. assumption: PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw
18. assumption: 0 ≤ alpha
19. assumption: PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta
20. assumption: eta.real (Set.Ioi vS) = totalSupply
21. conclusion: ∃ (holderConstant : ℝ) (gamma : ℝ) (A : ℝ),
  0 < gamma ∧
    0 ≤ holderConstant ∧
      (∀ (x delta : ℝ), 0 < delta → eta.real (Set.Ioo x (x + delta)) ≤ holderConstant * delta.rpow gamma) ∧
        0 ≤ A ∧
          ∀ᶠ (C : ℕ) in Filter.atTop,
            ∀ (a : Admissible C),
              (AppliedModelingLib.Matching.eventMass
                  ((inst C a).studentLaw.prod (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw))
                  fun (outcome : StudentType C a × (Fin (C + 1) → ℝ)) =>
                  (inst C a).value outcome.1 ∈ Set.Iic vS ∧
                    AppliedModelingLib.Matching.chosenInActive
                      ((inst C a).literal.demand.demandAt (inst C a).literal.selectedCutoff) Finset.univ outcome) ≤
                A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_thm1_corrected_lower_tailSpec_proof

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The correction bundle is complete: cited publication:312-321 displays the archival upper-tail integral, 8c8afe63cf028d729de91564a303f0de9bcc06685eef7305ac115fca8ff0b9f4 binds the exact corrected target, the two governing defect IDs and scope note are present, and the maintainer approval excerpt is displayed. The expanded target deliberately formalizes the approved non-equivalent lower-tail result: uniformly eventually in C and across admissible selected stable instances, eventMass under the stated product law of value at or below vS jointly with chosenInActive over the full coalition is bounded by A*C^(-K(beta,gamma)). The displayed eventMass and chosenInActive definitions make this exactly lower-value matched mass, and the Holder, iid-max, and approved one-draw-second-moment prerequisites are displayed. It therefore matches the approved corrected lower-tail target, not the archival upper-tail wording.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 391-395

Verbatim source input

```
\begin{proposition}\label{prop:duck-1}
\begin{equation}
  S(\CC_1) \le \alpha C^{\phi_3 - 1} = O\left(C^{-K(\beta,\gamma)}\right).
\end{equation}
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {College : Type u} (active : Finset College) (capacity : College → ℝ) {alpha beta gamma : ℝ} {C : ℕ},
  0 < ↑C →
    0 < alpha →
      ↑active.card < (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma) →
        (∀ c ∈ active, capacity c < alpha / ↑C) →
          AppliedModelingLib.Matching.activeCapacity active capacity ≤
            alpha * (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma - 1)
```

Retained governing declarations

- AppliedModelingLib.Matching.activeCapacity
- PG24NoisyMatchingMarkets.theorem1TailPhi3

Lean-elaborated claim roles

```
1. parameter: Type u
2. parameter: Finset College
3. parameter: College → ℝ
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: ℕ
8. assumption: 0 < ↑C
9. assumption: 0 < alpha
10. assumption: ↑active.card < (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)
11. assumption: ∀ c ∈ active, capacity c < alpha / ↑C
12. conclusion: AppliedModelingLib.Matching.activeCapacity active capacity ≤
  alpha * (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma - 1)
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_duck_1Spec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: At source_duck_1, audit/cited publication:391-395, S(CC_1) is bounded by α C^(φ_3-1). The Lean conclusion is the corresponding total active capacity, and activeCapacity is exactly the finite capacity sum. theorem1TailPhi3 = 1 - 2βγ/denom and theorem1TailK = 2βγ/denom, so φ_3-1 = -K; the displayed O(C^-K) restatement follows from the target's stronger concrete bound under its displayed capacity/cardinality prerequisites.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 404-416

Verbatim source input

```
\begin{proposition}\label{prop:tomato}
\quad
    \begin{itemize}
        \item[(i)] If $v > P^* - \EE[X^{(C^{\phi_2})}]  + 2\delta,$
        \begin{equation}
            p_\mu(v, \CC_2) \ge 1 - O\left(C^{-2\phi_1 - \beta\phi_2}\right).
        \end{equation}
        \item[(ii)] If $v < P^* - \EE[X^{(C^{\phi_2})}]  - \delta,$
        \begin{equation}
            p_\mu(v, \CC_2) \le O\left(C^{1-2\phi_1 - (1 + \beta)\phi_2}\right).
        \end{equation}
    \end{itemize}
\end{proposition}

[Next verbatim source excerpt]

\subsection{Preliminaries}
We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

Consider a matching $\mu\in M(\DD, C)$, characterized by a vector of market-clearing cutoffs $\vec{P}=\vec{P}(\mu)$. Our results focus on the probability that a student $\theta=(v,\succ)$ matches to a college. This probability is equal to
\begin{equation}
    \Pr[\mu(\theta)\in C] = \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
where $X_1, \cdots, X_C \simiid \DD.$ Notice that this probability does not depend on $\succ$, the student's preferences over colleges. Indeed, the probability that a student matches is exactly the probability that a student can afford some college; whether or not they can afford a college only depends on their value and the college's cutoff (not the student's preferences). Accordingly, we can drop $\theta$ and define
\begin{equation}
    p_\mu(v) := \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
the probability that a student with value $v$ is matched under $\mu$. More generally, for a subset of colleges $C'\subseteq C$, we define
\begin{equation}
    p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
the probability that a student with value $v$ can afford some college in $C'$. By definition, $p_\mu(v) = p_\mu(v, C).$

[Next verbatim source excerpt]

\end{itemize}
We choose these two cases for the following reason. The first case guarantees the existence of a dense set of cutoffs ``early on'' in the sequence of colleges. The second case, does not have a dense set of cutoffs early on. By the pigeonhole principle, this instead implies that there must exist a large gap between cutoffs early on. As we will see, both the existence of an ``early'' dense set or large gap is sufficient to obtain the desired result.

\subsubsection*{Case 1: Few Cutoffs Below $P^*$}

At a high level, our analysis in this case proceeds as follows. Set
\begin{align}
    \CC_1 &:= C((-\infty, P^*))\\
    \CC_2 &:= C((P^*, \infty)).
\end{align}
First, because there are few cutoffs below $P^*$, the mass of students who match to colleges in $\CC_1$ is small---i.e., going to $0$ as $C\rightarrow \infty$. Therefore, we can focus on the probability a student can afford a college in $\CC_2.$ This probability is approximately a step function as for $C$ large, since the dense cluster of colleges in $[P^*, P^*+\delta]$ contains a large number of colleges, serving as an ``arbitrarily strong filter.''
\\
\\
In accordance with this plan, observe that
\begin{equation}
    \int_{v_S}^\infty p_\mu(v)\,d\eta(v) \le S(\CC_1) + \int_{v_S}^\infty p_\mu(v,\CC_2)\,d\eta(v),
\end{equation}
since the total measure of students with value below $v_S$ who match to a college is at most the total measure $S(\CC_1)$ of students (of any value) who match with a college in $\CC_1$ plus the total measure of students with value below $v_S$ who can afford a college in $\CC_2.$

We proceed by bounding the two components of the RHS. Bounding $S(\CC_1)$ is straightforward: there are at most $C^{\phi_3}$ colleges in this set, each of which has capacity at most $\frac{\alpha}{C}.$ This gives the following bound.
\begin{proposition}\label{prop:duck-1}
\begin{equation}
    S(\CC_1) \le \alpha C^{\phi_3 - 1} = O\left(C^{-K(\beta,\gamma)}\right).
\end{equation}
\end{proposition}
\begin{proof}
    The capacity of each college is at most $\frac{\alpha}{C}$. Since there are fewer than $C^{\phi_3}$ colleges in $C((-\infty, P^*))$, the mass of students who match to these colleges is less than $C^{\phi_3}\cdot \frac{\alpha}{C} = \alpha C^{\phi_3 - 1}.$ The result follows since
    \begin{equation}
        \phi_3 - 1 = -\frac{2\beta\gamma}{3\beta\gamma + 2\beta + 5\gamma + 6} = K(\beta,\gamma).
    \end{equation}
\end{proof}
Bounding the second component---the measure of students with value below $v_S$ who match with a college in $C([P^*,\infty))$---is more involved. We start by showing that $p_\mu(v, \CC_2)$, the probability that a student with value $v$ can afford a college in $\CC_2$, resembles a step function in $v$ that jumps at $P^* - \EE[X^{(C^{\phi_2})}],$ where we recall that $\phi_2$ is the number of colleges in the dense cluster. We can use this result to analyze the second integral, since if $p_\mu(v, \CC_2)$ resembles a step function, then this step must occur at roughly $v_S$ or otherwise too many (more than $S$) or too few (at most $S$) students will be able to afford a college. In turn, $p_\mu(v, \CC_2)$ must be small for almost all students with value at most $v_S$.
```


```
\begin{proposition}\label{prop:tomato}
\quad
    \begin{itemize}
        \item[(i)] If $v > P^* - \EE[X^{(C^{\phi_2})}]  + 2\delta,$
        \begin{equation}
```

```
        p_\mu(v, \CC_2) \ge 1 - O\left(C^{-2\phi_1 - \beta\phi_2}\right).
    \end{equation}
    \item[(ii)] If $v < P^* - \EE[X^{(C^{\phi_2})}]  - \delta,$
    \begin{equation}
        p_\mu(v, \CC_2) \le O\left(C^{1-2\phi_1 - (1 + \beta)\phi_2}\right).
    \end{equation}
  \end{itemize}
\end{proposition}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
In the source Case-1 closed high-cutoff block C([P*, infinity)), the two displayed dense-cluster cutoff-affordance rates hold under the strict source value
↪    separations.
```

Archival source anchor: cited publication, lines 371-416

Expanded PaperInterface specification

```
∀ (noiseLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw] (maxVariance : ℕ → ℝ)
  {beta gamma : ℝ},
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    0 < gamma →
      PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
        ∀ (dense upper : (C : ℕ) → Finset (Fin C)) (cutoff : (C : ℕ) → Fin C → ℝ)
          (highValue ceiling lowValue pivot : ℕ → ℝ),
          (∀ᶠ (C : ℕ) in Filter.atTop, upper C = {c : Fin C | pivot C ≤ cutoff C c}) →
            (∀ᶠ (C : ℕ) in Filter.atTop, dense C ⊆ upper C) →
              (∀ᶠ (C : ℕ) in Filter.atTop,
                  (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma) ≤ ↑(dense C).card) →
                (∀ᶠ (C : ℕ) in Filter.atTop, ∀ c ∈ dense C, cutoff C c ≤ ceiling C) →
                  (∀ᶠ (C : ℕ) in Filter.atTop,
                      ceiling C - highValue C <
                        PG24NoisyMatchingMarkets.theorem1DenseGroupCenter noiseLaw C beta gamma -
                          PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius C beta gamma) →
                    (∀ᶠ (C : ℕ) in Filter.atTop, ∀ c ∈ upper C, pivot C ≤ cutoff C c) →
                      (∀ᶠ (C : ℕ) in Filter.atTop,
                          PG24NoisyMatchingMarkets.theorem1DenseGroupCenter noiseLaw C beta gamma +
                              PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius C beta gamma <
                            pivot C - lowValue C) →
                        (∃ (A : ℝ),
                            0 ≤ A ∧
                              ∀ᶠ (C : ℕ) in Filter.atTop,
                                1 -
                                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                      (MeasureTheory.Measure.pi fun (x : Fin C) => noiseLaw) (upper C) (highValue C)
                                      (cutoff C) ≤
                                  A *
                                    (↑C).rpow
                                      (-2 * PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma -
                                        beta * PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)) ∧
                          ∃ (B : ℝ),
                            0 ≤ B ∧
                              ∀ᶠ (C : ℕ) in Filter.atTop,
                                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                    (MeasureTheory.Measure.pi fun (x : Fin C) => noiseLaw) (upper C) (lowValue C)
                                    (cutoff C) ≤
                                  B *
                                    (↑C).rpow
                                      (1 - 2 * PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma -
                                        (1 + beta) * PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
```

Retained governing declarations

- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius
- PG24NoisyMatchingMarkets.theorem1DenseGroupCenter
- PG24NoisyMatchingMarkets.theorem1TailPhi1
- PG24NoisyMatchingMarkets.theorem1TailPhi2

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
3. parameter: ℕ → ℝ
4. parameter: ℝ
5. parameter: ℝ
6. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
7. assumption: 0 < gamma
8. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
9. parameter: (C : ℕ) → Finset (Fin C)
10. parameter: (C : ℕ) → Finset (Fin C)
```

```
11. parameter: (C : ℕ) → Fin C → ℝ
12. parameter: ℕ → ℝ
13. parameter: ℕ → ℝ
14. parameter: ℕ → ℝ
15. parameter: ℕ → ℝ
16. assumption: ∀ᶠ (C : ℕ) in Filter.atTop, upper C = {c : Fin C | pivot C ≤ cutoff C c}
17. assumption: ∀ᶠ (C : ℕ) in Filter.atTop, dense C ⊆ upper C
18. assumption: ∀ᶠ (C : ℕ) in Filter.atTop, (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma) ≤ ↑(dense C).card
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    pivot C - lowValue C
23. conclusion: (∃ (A : ℝ),
    0 ≤ A ∧
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        1 -
            PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin C) => noiseLaw)
              (upper C) (highValue C) (cutoff C) ≤
          A *
            (↑C).rpow
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            (upper C) (lowValue C) (cutoff C) ≤
          B *
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              (1 - 2 * PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma -
                (1 + beta) * PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_tomatoSpec_proof

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The target implements the approved Case-1 closed high-cutoff block and the strict high- and low-value separations. Its two eventual conclusions are the source rates for 1-p_mu at the high value and p_mu at the low value, with the dense-subblock geometry explicit.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 469-473

Verbatim source input

```
\begin{proposition}\label{prop:duck-2}
\begin{equation}
    \int_{-\infty}^{v_S} p_\mu(v, \CC_2)\,d\eta(v) = O\left(C^{-K(\beta,\gamma)}\right).
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[Next verbatim source excerpt]

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We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

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\end{equation}
since the total measure of students with value below $v_S$ who match to a college is at most the total measure $S(\CC_1)$ of students (of any value) who match with a college in $\CC_1$ plus the total measure of students with value below $v_S$ who can afford a college in $\CC_2.$

We proceed by bounding the two components of the RHS. Bounding $S(\CC_1)$ is straightforward: there are at most $C^{\phi_3}$ colleges in this set, each of which has capacity at most $\frac{\alpha}{C}.$ This gives the following bound.
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    The capacity of each college is at most $\frac{\alpha}{C}$. Since there are fewer than $C^{\phi_3}$ colleges in $C((-\infty, P^*))$, the mass of students who match to these colleges is less than $C^{\phi_3}\cdot \frac{\alpha}{C} = \alpha C^{\phi_3 - 1}.$ The result follows since
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\begin{proposition}\label{prop:tomato}
\quad
    \begin{itemize}
        \item[(i)] If $v > P^* - \EE[X^{(C^{\phi_2})}]  + 2\delta,$
        \begin{equation}
            p_\mu(v, \CC_2) \ge 1 - O\left(C^{-2\phi_1 - \beta\phi_2}\right).
        \end{equation}
        \item[(ii)] If $v < P^* - \EE[X^{(C^{\phi_2})}]  - \delta,$
        \begin{equation}
            p_\mu(v, \CC_2) \le O\left(C^{1-2\phi_1 - (1 + \beta)\phi_2}\right).
        \end{equation}
    \end{itemize}
\end{proposition}
```

```latex
We first sketch the proof strategy and intuition for \Cref{prop:tomato}. For part (i), we will lower bound $p_\mu(v, \CC_2)$ by the probability that a student with value $v$ can afford a college in the dense cluster $C([P^*, P^* + \delta]).$ In turn, this can be lower bounded by the probability that among the at least $C^{\phi_2}$ scores $\theta$ receives from the colleges in the dense cluster, at least one exceeds $P^* + \delta$, i.e., $\Pr[v + X^{(C^{\phi_2})} > P^* + \delta$]. This probability is close to $1$, since $v + \EE[X^{(C^{\phi_2})}] > P^* + 2\delta$ and $X^{(C^{\phi_2})}$ concentrates as $C$ grows large (due to the $\beta$-max-concentrating property).

Our approach for part (ii) is similar, but uses an additional observation. We can upper bound $p_\mu(v, \CC_2)$ by the probability that among the at most $C$ scores $v$ receives from the colleges in $C([P^*,\infty)),$ at least one exceeds $P^*$---i.e., $\Pr[v + X^{(C)} > P^*].$ The key observation is that this probability is at most $C^{1 - \phi_2}\Pr[v + X^{(C^{\phi_2})} > P^*]$ by a union bound. $\Pr[v + X^{(C^{\phi_2})} > P^*]$ is small since $v + \EE[X^{(C^{\phi_2})}] < P^* - \delta$ and $X^{(C^{\phi_2})}$ concentrates as $C$ grows large; in fact, we can show that it is so small that it overcomes the factor of $C^{1 - \phi_2}$.

We now provide this analysis formally.

\begin{proof}[Proof of (i)]
  We bound $p_\mu(v, \CC_2)$ from below by the probability that $v$ can afford a college in $C([P^*, P^* + \delta])\subseteq C([P^*, \infty)).$
  Observe that if $\max_{c\in C(P^*, P^* + \delta)} \{v + X_c\} > P^* + \delta$, then $v$ can afford at least one college in $C([P^*, \infty))$.
  Therefore,
  \begin{equation}
    p_\mu(v, \CC_2) \ge \Pr\left[\max_{c\in C([P^*, P^* + \delta])} \{v + X_c\} > P^* + \delta\right] \ge \Pr\left[v + X^{(C^{\phi_2})} > P^* + \delta\right],
  \end{equation}
  where the last inequality follows since there are more than $C^{\phi_2}$ colleges in $C([P^*, P^* + \delta])$.

  Now observe that
  \begin{align}
    \Pr\left[v + X^{(C^{\phi_2})} > P^* + \delta\right] &= \Pr\left[X^{(C^{\phi_2})} > P^* + \delta - v\right]\\
    &\ge \Pr\left[X^{(C^{\phi_2})} > \EE\left[X^{(C^{\phi_2})}\right] - \delta \right] \label{eq:potato-1} \\
    &\ge \Pr\left[\left|X^{(C^{\phi_2})} - \EE\left[X^{(C^{\phi_2})}\right]\right| < \delta \right] \\
    &> 1 - \frac{\Var\left[X^{(C^{\phi_2})}\right]}{\delta^2} \label{eq:potato-2} \\
    &= 1 - O\left(\frac{C^{-\beta\phi_2}}{\delta^2}\right) \label{eq:potato-3} \\
    &= 1 - O\left(C^{-2\phi_1 - \beta\phi_2}\right),
  \end{align}
  where \eqref{eq:potato-1} follows because $v > P^* - \EE\left[X^{(C^{\phi_2})}\right]  + 2\delta,$ \eqref{eq:potato-2} follows from Chebyshev's inequality, and \eqref{eq:potato-3} follows from the fact that $\Var[X^{(n)}] = O\left(n^{-\beta}\right).$
\end{proof}

\begin{proof}[Proof of (ii).]
Note that $v$ can afford at least one college in $C(P^*, \infty)$ only if $v + X_c > P^*$ for some $c\in C(P^*, \infty).$ Thus,
  \begin{align}
    p_\mu(v, \CC_2) &\le \Pr[v + X^{(C)} > P^*]\\
    &\le C^{1 - \phi_2} \cdot \Pr[v + X^{(C^{\phi_2})} > P^*],
  \end{align}
  where the last inequality holds by a union bound (split the $C$ colleges into $C^{1-\phi_2}$ groups each of size $C^{\phi_2}$, and then bound the probability that the student is admitted to a college in any of these groups).

  Then, proceeding as in part (i),
  \begin{align}
    \Pr\left[v + X^{(C^{\phi_2})} > P^* \right] &= \Pr\left[X^{(C^{\phi_2})} > P^* - v\right]\\
    &\ge \Pr\left[X^{(C^{\phi_2})} > \EE\left[X^{(C^{\phi_2})}\right] + \delta \right] \label{eq:yam-1} \\
    &\ge \Pr\left[\left|X^{(C^{\phi_2})} - \EE\left[X^{(C^{\phi_2})}\right]\right| > \delta \right] \\
    &< \frac{\Var\left[X^{(C^{\phi_2})}\right]}{\delta^2} \label{eq:yam-2} \\
    &= O\left(\frac{C^{-\beta \phi_2}}{\delta^2}\right), \label{eq:yam-3}
  \end{align}
  where \eqref{eq:yam-1} follows because $v < P^* - \EE\left[X^{(C^{\phi_2})}\right]  - \delta,$ \eqref{eq:yam-2} follows from Chebyshev's inequality, and \eqref{eq:yam-3} follows from the fact that $\Var[X^{(n)}] = O\left(\frac{1}{n}\right).$ It follows that
  \begin{equation}
    p_\mu(v, \CC_2) \le C^{1 - \phi_2}\cdot O\left(\frac{C^{-\beta \phi_2}}{\delta^2}\right) = O\left(C^{1-2\phi_1-(1+\beta)\phi_2}\right).
  \end{equation}
\end{proof}

We may now use the preceding result (\Cref{prop:tomato}) to bound the second integral, the mass of students with value at most $v_S$ who can afford a college in $C([P^*,\infty))$.

\begin{proposition}\label{prop:duck-2}
\begin{equation}
  \int_{-\infty}^{v_S} p_\mu(v, \CC_2)\,d\eta(v) = O\left(C^{-K(\beta,\gamma)}\right).
\end{equation}
\end{proposition}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

In the source Case-1 closed high-cutoff block C([P*, infinity)), the below-vS cutoff-affordance integral has the displayed O(C^(-K(beta,gamma))) rate.

Archival source anchor: cited publication, lines 371-473

Expanded PaperInterface specification

```
∀ (noiseLaw valueLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure valueLaw] (maxVariance : ℕ → ℝ) {beta gamma vS : ℝ},
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    0 < gamma →
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        ∀ (upper : (C : ℕ) → Finset (Fin C)) (cutoff : (C : ℕ) → Fin C → ℝ) (pivot : ℕ → ℝ),
          (∀ᶠ (C : ℕ) in Filter.atTop, upper C = {c : Fin C | pivot C ≤ cutoff C c}) →
            (∀ᶠ (C : ℕ) in Filter.atTop, ∀ c ∈ upper C, pivot C ≤ cutoff C c) →
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                  0 ≤ A ∧
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```

```
∫ (v : ℝ),
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      (fun (v : ℝ) =>
        PG24NoisyMatchingMarkets.cutoffAffordanceProbability
          (MeasureTheory.Measure.pi fun (x : Fin C) => noiseLaw) (upper C) v (cutoff C))
      v ∂valueLaw ≤
  A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Retained governing declarations

- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius
- PG24NoisyMatchingMarkets.theorem1DenseGroupCenter
- PG24NoisyMatchingMarkets.theorem1TailK

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. parameter: MeasureTheory.Measure ℝ
3. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
4. assumption: MeasureTheory.IsProbabilityMeasure valueLaw
5. parameter: ℕ → ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
10. assumption: 0 < gamma
11. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
12. parameter: (C : ℕ) → Finset (Fin C)
13. parameter: (C : ℕ) → Fin C → ℝ
14. parameter: ℕ → ℝ
15. assumption: ∀ᶠ (C : ℕ) in Filter.atTop, upper C = {c : Fin C | pivot C ≤ cutoff C c}
16. assumption: ∀ᶠ (C : ℕ) in Filter.atTop, ∀ c ∈ upper C, pivot C ≤ cutoff C c
17. assumption: ∀ᶠ (C : ℕ) in Filter.atTop,
  PG24NoisyMatchingMarkets.theorem1DenseGroupCenter noiseLaw C beta gamma +
      PG24NoisyMatchingMarkets.theorem1DenseDeviationRadius C beta gamma <
    pivot C - vS
18. conclusion: ∃ (A : ℝ),
  0 ≤ A ∧
    ∀ᶠ (C : ℕ) in Filter.atTop,
      ∫ (v : ℝ),
          (Set.Iic vS).indicator
            (fun (v : ℝ) =>
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin C) => noiseLaw) (upper C) v (cutoff C))
            v ∂valueLaw ≤
        A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_duck_2Spec_proof
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Against the approved closed-block correction, upper C = {c | pivot C <= cutoff C c} realizes C([P*,infinity)). The Iic vS integral and eventual nonnegative C^(-K(beta,gamma)) bound retain the source endpoint and rate under the supplied iid variance and separation context.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 553-557

Verbatim source input

```
\begin{proposition}\label{prop:goose-1}
\begin{equation}
  S(\CC_1) < \alpha C^{-\frac{2\beta\gamma}{3\beta\gamma + 2\beta + 5\gamma + 6}}.
\end{equation}
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {College : Type u} (active : Finset College) (capacity : College → ℝ) {alpha beta gamma : ℝ} {C : ℕ},
  0 < ↑C →
    0 < alpha →
      ↑active.card < (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma) →
        (∀ c ∈ active, capacity c < alpha / ↑C) →
          AppliedModelingLib.Matching.activeCapacity active capacity <
            alpha * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Retained governing declarations

- AppliedModelingLib.Matching.activeCapacity
- PG24NoisyMatchingMarkets.theorem1TailK
- PG24NoisyMatchingMarkets.theorem1TailPhi3

Lean-elaborated claim roles

```
1. parameter: Type u
2. parameter: Finset College
3. parameter: College → ℝ
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: ℕ
8. assumption: 0 < ↑C
9. assumption: 0 < alpha
10. assumption: ↑active.card < (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)
11. assumption: ∀ c ∈ active, capacity c < alpha / ↑C
12. conclusion: AppliedModelingLib.Matching.activeCapacity active capacity <
  alpha * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_goose_1Spec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: The strict source bound at source_goose_1, audit/cited publication:553-557, is α C^(-2βγ/(3βγ+2β+5γ+6)). The Lean conclusion is the same strict active-capacity bound α C^(-theorem1TailK β γ), and theorem1TailK expands to exactly that exponent; the displayed finite-set capacity prerequisites supply the source S(CC_1) quantity.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 565-570

Verbatim source input

```
\begin{lemma}\label{lem:log-bound}
  If $\DD$ is $\beta$-max-concentrating for some $\beta>0$, then
  \begin{equation}
    \EE[X^{(n)}] = o(\log n).
  \end{equation}
\end{lemma}
```

Expanded PaperInterface specification

```
∀ (noiseLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw] (maxVariance : ℕ → ℝ) (beta : ℝ),
 PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
  PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
   Filter.Tendsto
    (fun (n : ℕ) =>
     AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
       (fun (k : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (k + 1)) => noiseLaw) n /
      Real.log ↑n)
    Filter.atTop (nhds 0)
```

Retained governing declarations

- AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
3. parameter: ℕ → ℝ
4. parameter: ℝ
5. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
6. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
7. conclusion: Filter.Tendsto
 (fun (n : ℕ) =>
  AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
    (fun (k : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (k + 1)) => noiseLaw) n /
   Real.log ↑n)
 Filter.atTop (nhds 0)
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_log_boundSpec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: At source_log_bound, audit/cited publication:565-570, β-max-concentration with β>0 yields E[X^(n)]=o(log n). betaMaxConcentratingVariance explicitly entails 0<β and the iid variance prerequisite concerns the top order statistic. The Lean Tendsto conclusion is precisely the little-o quotient for the expected maximum; expectedTopOrderStatisticSeq uses n+1 iid draws, the stated nonempty-product indexing shift, which leaves the asymptotic rate unchanged.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 599-611

Verbatim source input

```
\begin{proposition}\label{prop:beet}
\quad
\begin{itemize}
  \item[(i)] If $v < P_{C^{\phi_3}} - \EE[X^{(C)}] - C^{\phi_4},$
  \begin{equation}
    p_\mu(v, \CC_2) \le O\left(C^{-\beta-2\phi_4}\right).
  \end{equation}
  \item[(ii)] If $v > P_{C^{\phi_3}} - \EE[X^{(C)}] - C^{\phi_4},$
  \begin{equation}
    p_\mu(v) \ge 1 - O\left(C^{-2\phi_1 - 2\phi_3 + 2\phi_2}\right).
  \end{equation}
\end{itemize}
\end{proposition}
```

Additional assumptions

The iid one-draw noise law satisfies MemLp (fun x : ℝ => x) 2 noiseLaw; this finite second moment is used only for the Appendix Proposition 7(ii) one-draw Chebyshev step.

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
Appendix Proposition 7(i) has its literal full-maximum polynomial rate, and Appendix Proposition 7(ii) has its displayed near-one polynomial rate under the
↪   source beta-max condition plus a finite iid one-draw second moment, with an atom-safe lower-tail complement event.
```

Archival source anchor: cited publication, lines 599-611

Expanded PaperInterface specification

```
∀ (noiseLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw] (maxVariance : ℕ → ℝ)
  {beta gamma : ℝ},
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    0 < gamma →
      PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
        PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw →
          ∀ (cutoff : (C : ℕ) → Fin (C + 1) → ℝ),
            (∀ᶠ (C : ℕ) in Filter.atTop,
                PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C) 0 +
                    ↑(PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount C
                        (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
                        (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) *
                      (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma) <
                  PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                    (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                      (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))) →
              (∃ (A : ℝ),
                  0 ≤ A ∧
                    ∀ᶠ (C : ℕ) in Filter.atTop,
                      PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                          (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
                          (PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock Finset.univ (cutoff C)
                            (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                              (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                                (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))))
                          (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                                (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                                  (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) -
                              AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
                                (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) C -
                            (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma))
                          (cutoff C) ≤
                        A * (↑C).rpow (-beta - 2 * PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma)) ∧
                ∃ (B : ℝ),
                  0 ≤ B ∧
                    ∀ (value : ℕ → ℝ),
                      (∀ᶠ (C : ℕ) in Filter.atTop,
                          PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                                  (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                                    (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) -
                                AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
                                  (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) C -
                              (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma) <
                            value C) →
                        ∀ᶠ (C : ℕ) in Filter.atTop,
                          1 -
                              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ (value C)
                                (cutoff C) ≤
                            B * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Retained governing declarations

- AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance

- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment
- PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock
- PG24NoisyMatchingMarkets.theorem1TailK
- PG24NoisyMatchingMarkets.theorem1TailPhi1
- PG24NoisyMatchingMarkets.theorem1TailPhi2
- PG24NoisyMatchingMarkets.theorem1TailPhi3
- PG24NoisyMatchingMarkets.theorem1TailPhi4
- PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount
- PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank
- PG24NoisyMatchingMarkets.theorem3RankedCutoffNat

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
3. parameter: ℕ → ℝ
4. parameter: ℝ
5. parameter: ℝ
6. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
7. assumption: 0 < gamma
8. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
9. assumption: PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw
10. parameter: (C : ℕ) → Fin (C + 1) → ℝ
11. assumption: ∀ᶠ (C : ℕ) in Filter.atTop,
  PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C) 0 +
      ↑(PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount C (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
          (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) *
        (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma) <
    PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
      (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))
12. conclusion: (∃ (A : ℝ),
    0 ≤ A ∧
      ∀ᶠ (C : ℕ) in Filter.atTop,
        PG24NoisyMatchingMarkets.cutoffAffordanceProbability
            (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
            (PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock Finset.univ (cutoff C)
              (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                  (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))))
            (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                  (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                    (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) -
                AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
                  (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) C -
              (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma))
            (cutoff C) ≤
          A * (↑C).rpow (-beta - 2 * PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma)) ∧
  ∃ (B : ℝ),
    0 ≤ B ∧
      ∀ (value : ℕ → ℝ),
        (∀ᶠ (C : ℕ) in Filter.atTop,
            PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                    (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                      (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) -
                  AppliedModelingLib.Probability.expectedTopOrderStatisticSeq
                    (fun (n : ℕ) => MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) C -
                (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi4 beta gamma) <
              value C) →
          ∀ᶠ (C : ℕ) in Filter.atTop,
            1 -
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ (value C) (cutoff C) ≤
              B * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_beetSpec_proof

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The correction bundle is complete: its archival display is cited publication:599-611, its exact corrected-target review binding is 9583dda1015f19d203baee53e623751eb80e41c84c5ef8424e0e633bb6d9a722, it identifies PG24-APP-PROP7-ONE-DRAW-MOMENT-01, states the scope, and includes the maintainer approval excerpt. The expanded target has the two displayed Proposition 7 cases: the cutoff-crossing probability for the high-cutoff block at the printed pivot has rate C^(-beta-2*phi4), and, for every eventually strictly-above-pivot value sequence, its full-block failure probability has rate C^(-K(beta,gamma)). The displayed definitions make cutoffAffordanceProbability the relevant crossing probability and make K=2*beta*gamma/(3*beta*gamma+2*beta+5*gamma+6); the latter equals 2*phi1+2*phi3-2*phi2, so the second exponent is the printed one. The added iid one-draw MemLp-2 premise is exactly the approved correction; the materially corrected target therefore matches rather than the archival wording alone.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 647-651

Verbatim source input

```
\begin{proposition}\label{prop:goose-2}
  \begin{equation}
  \int_{-\infty}^{v_S} p_\mu(v, \CC_2)\,d\eta(v) = O\left(C^{-K(\beta,\gamma)}\right).
\end{equation}
\end{proposition}
```

Additional assumptions

The iid one-draw noise law satisfies MemLp (fun x : ℝ => x) 2 noiseLaw; this finite second moment supplies the inherited Appendix Proposition 8 Case-2 rate.

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

```
Under the source beta-max and value-law conditions plus a finite iid one-draw second moment, Appendix Proposition 8's below-vS high-cutoff-affordance
↪   integral is O(C^(-K(beta,gamma))).
```

Archival source anchor: cited publication, lines 647-651

Expanded PaperInterface specification

```
∀ (noiseLaw valueLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure valueLaw] (maxVariance : ℕ → ℝ) {beta gamma vS totalSupply : ℝ},
  PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta →
    0 < gamma →
      PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance →
        PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw →
          ∀ (cutoff : (C : ℕ) → Fin (C + 1) → ℝ),
            (∀ (C : ℕ),
                ∫ (value : ℝ),
                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                      (cutoff C) ∂valueLaw =
                  totalSupply) →
              valueLaw.real (Set.Ioi vS) = totalSupply →
                (∀ᶠ (C : ℕ) in Filter.atTop,
                    PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C) 0 +
                        ↑(PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount C
                            (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
                            (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) *
                          (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma) <
                      PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                        (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                          (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))) →
                  ∃ (A : ℝ),
                    0 ≤ A ∧
                      ∀ᶠ (C : ℕ) in Filter.atTop,
                        ∫ (value : ℝ),
                            (Set.Iic vS).indicator
                              (fun (value : ℝ) =>
                                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
                                  (PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock Finset.univ (cutoff C)
                                    (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                                      (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                                        (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))))
                                  value (cutoff C))
                              value ∂valueLaw ≤
                          A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Retained governing declarations

- PG24NoisyMatchingMarkets.betaMaxConcentratingVariance
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound
- PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment
- PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock
- PG24NoisyMatchingMarkets.theorem1TailK
- PG24NoisyMatchingMarkets.theorem1TailPhi1
- PG24NoisyMatchingMarkets.theorem1TailPhi2
- PG24NoisyMatchingMarkets.theorem1TailPhi3
- PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount
- PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank
- PG24NoisyMatchingMarkets.theorem3RankedCutoffNat

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. parameter: MeasureTheory.Measure ℝ
3. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
4. assumption: MeasureTheory.IsProbabilityMeasure valueLaw
5. parameter: ℕ → ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. assumption: PG24NoisyMatchingMarkets.betaMaxConcentratingVariance maxVariance beta
11. assumption: 0 < gamma
12. assumption: PG24NoisyMatchingMarkets.source_assumption_iid_beta_max_variance_bound noiseLaw maxVariance
13. assumption: PG24NoisyMatchingMarkets.source_clarification_one_draw_finite_second_moment noiseLaw
14. parameter: (C : ℕ) → Fin (C + 1) → ℝ
15. assumption: ∀ (C : ℕ),
  ∫ (value : ℝ),
      PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
        Finset.univ value (cutoff C) ∂valueLaw =
    totalSupply
16. assumption: valueLaw.real (Set.Ioi vS) = totalSupply
17. assumption: ∀ᶠ (C : ℕ) in Filter.atTop,
  PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C) 0 +
      ↑(PG24NoisyMatchingMarkets.theorem3DenseGapBlockCount C (PG24NoisyMatchingMarkets.theorem1TailPhi2 beta gamma)
          (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma)) *
        (↑C).rpow (PG24NoisyMatchingMarkets.theorem1TailPhi1 beta gamma) <
    PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
      (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))
18. conclusion: ∃ (A : ℝ),
  0 ≤ A ∧
    ∀ᶠ (C : ℕ) in Filter.atTop,
      ∫ (value : ℝ),
          (Set.Iic vS).indicator
            (fun (value : ℝ) =>
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
                (PG24NoisyMatchingMarkets.theorem1CutoffAtOrAboveBlock Finset.univ (cutoff C)
                  (PG24NoisyMatchingMarkets.theorem3RankedCutoffNat C (cutoff C)
                    (PG24NoisyMatchingMarkets.theorem3EarlyPrefixRank C
                      (PG24NoisyMatchingMarkets.theorem1TailPhi3 beta gamma))))
                value (cutoff C))
            value ∂valueLaw ≤
        A * (↑C).rpow (-PG24NoisyMatchingMarkets.theorem1TailK beta gamma)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_goose_2Spec_proof

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The correction bundle is complete: cited publication:647-651 supplies the archival display, e161f44e117d228fb252f3a1cff12284f87187c8d018c321009c8884a020432e binds the exact corrected target, PG24-APP-PROP8-INHERITED-ONE-DRAW-MOMENT-01 and its scope note are present, and the maintainer approval excerpt is displayed. The expanded conclusion is precisely an eventual O(C^(-K(beta,gamma))) bound on the valueLaw integral of the cutoff-crossing probability over Set.Iic vS, with the active set defined as the high-cutoff block. That is the displayed lower-tail integral in the archival proposition, while the explicit iid one-draw MemLp-2 premise is the approved inherited Case-2 correction. All displayed prerequisite domains, quantifiers, and rate arguments agree with that corrected target.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 703-709

Verbatim source input

```
\begin{proposition}\label{thm2v2}
Let $\DD$ be long-tailed. Then, for all $C$ sufficiently large, for all $\epsilon>0$ and any $\mu\in M(\DD, C)$,
\begin{equation}
    \Pr[\mu(v)\in C]\in (S-\epsilon, S+\epsilon).
\end{equation}
for all $v\in (v_\epsilon^-, v_\epsilon^+)$ for $v_\epsilon^-, v_\epsilon^+$ chosen such that $\eta(-\infty, v_\epsilon^-) = \eta((v_\epsilon, \infty)) =
↪    \epsilon$.
\end{proposition}
```

Expanded PaperInterface specification

```
∀ {StudentTypeSeq : ℕ → Type u} [inst : (C : ℕ) → MeasurableSpace (StudentTypeSeq C)] {Admissible : ℕ → Type v}
  (CutoffSeq : ℕ → Type w) (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta] {totalSupply alpha : ℝ}
  (inst_3 :
    (C : ℕ) →
      Admissible C →
        PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha (StudentTypeSeq C)
          (CutoffSeq C)),
  PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta →
    IsPreconnected eta.support →
      PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
        0 < totalSupply →
          totalSupply < 1 →
            0 < alpha →
              ∀ (epsilon vLow vHigh : ℝ),
                0 < epsilon →
                  eta.real (Set.Iio vLow) = epsilon →
                    eta.real (Set.Ioi vHigh) = epsilon →
                      vLow < vHigh →
                        ∀ᶠ (C : ℕ) in Filter.atTop,
                          ∀ (a : Admissible C) (value : ℝ),
                            vLow < value →
                              value < vHigh →
                                totalSupply - epsilon <
                                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                                      (inst_3 C a).selectedCutoffVector ∧
                                  PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
                                      (inst_3 C a).selectedCutoffVector <
                                    totalSupply + epsilon
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24HolderIntervalRegular
- PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance
- PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance.selectedCutoffVector
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Lean-elaborated claim roles

```
1. parameter: ℕ → Type u
2. parameter: (C : ℕ) → MeasurableSpace (StudentTypeSeq C)
3. parameter: ℕ → Type v
4. parameter: ℕ → Type w
5. parameter: MeasureTheory.Measure ℝ
6. parameter: MeasureTheory.Measure ℝ
7. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
8. assumption: MeasureTheory.IsProbabilityMeasure eta
9. parameter: ℝ
10. parameter: ℝ
11. parameter: (C : ℕ) →
  Admissible C →
    PG24NoisyMatchingMarkets.PG24LiteralBasicTwoScaleInstance C noiseLaw eta totalSupply alpha (StudentTypeSeq C)
      (CutoffSeq C)
12. assumption: PG24NoisyMatchingMarkets.PG24HolderIntervalRegular eta
13. assumption: IsPreconnected eta.support
14. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
15. assumption: 0 < totalSupply
16. assumption: totalSupply < 1
17. assumption: 0 < alpha
18. parameter: ℝ
19. parameter: ℝ
20. parameter: ℝ
21. assumption: 0 < epsilon
```

```
22. assumption: eta.real (Set.Iio vLow) = epsilon
23. assumption: eta.real (Set.Ioi vHigh) = epsilon
24. assumption: vLow < vHigh
25. conclusion: ∀ᶠ (C : ℕ) in Filter.atTop,
  ∀ (a : Admissible C) (value : ℝ),
    vLow < value →
      value < vHigh →
        totalSupply - epsilon <
            PG24NoisyMatchingMarkets.cutoffAffordanceProbability
              (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
              (inst C a).selectedCutoffVector ∧
          PG24NoisyMatchingMarkets.cutoffAffordanceProbability
              (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) Finset.univ value
              (inst C a).selectedCutoffVector <
            totalSupply + epsilon
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_thm2_equivalent_boundSpec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: The target retains the long-tailed basic-market and selected-clearing setting, 0 < epsilon, the exact lower and upper eta tail choices, an eventual C threshold uniform over source instances, the open value window, and the strict (S-epsilon,S+epsilon) affordance-probability conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 744-750

Verbatim source input

```
\begin{proposition}\label{prop:lt-large-firms}
There exists a constant $\sigma > 0$ such that for $C$ sufficiently large,
    \begin{equation}\label{eq:lt-large-firms}
    p_\mu(v, \CC_2) \in (S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})).
    \end{equation}
for all $v\in (v_-, v_+).$
\end{proposition}

[Next verbatim source excerpt]

\subsection{Preliminaries}
We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

Consider a matching $\mu\in M(\DD, C)$, characterized by a vector of market-clearing cutoffs $\vec{P}=\vec{P}(\mu)$. Our results focus on the probability that a student $\theta=(v,\succ)$ matches to a college. This probability is equal to
\begin{equation}
    \Pr[\mu(\theta)\in C] = \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
where $X_1, \cdots, X_C \simiid \DD.$ Notice that this probability does not depend on $\succ$, the student's preferences over colleges. Indeed, the probability that a student matches is exactly the probability that a student can afford some college; whether or not they can afford a college only depends on their value and the college's cutoff (not the student's preferences). Accordingly, we can drop $\theta$ and define
\begin{equation}
    p_\mu(v) := \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
the probability that a student with value $v$ is matched under $\mu$. More generally, for a subset of colleges $C'\subseteq C$, we define
\begin{equation}
    p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
the probability that a student with value $v$ can afford some college in $C'$. By definition, $p_\mu(v) = p_\mu(v, C).$

[Next verbatim source excerpt]

\end{equation}
This tells us that if a college's cutoff is large, two students (with different values) can afford the college with approximately equal probability. To leverage this fact, we will show that almost all colleges must have large cutoffs. Roughly speaking, this is true because if there were a large number of colleges with small cutoffs, then more students would be able to afford these firms than there is total capacity.

\subsection*{Proof of \Cref{thm2v2}}
We consider a long-tailed probability distribution $\DD$. To prove \Cref{thm2v2}, we show the following equivalent statement: There exists a constant $\sigma > 0$ such that for all $C$ sufficiently large, if $\mu \in M(\DD, C),$
\begin{equation}\label{eq:equiv-lt}
    p_\mu(v) \in \left(S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma}) + \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}\right)
\end{equation}
for all $v\in (v_-,v^*)$, where $v_-, v_+$ are chosen such that $\eta(-\infty,v_-) = \eta(v_+,\infty) = \frac{\epsilon}{2}$ and $v^*$ is chosen such that $\eta(v^*,v_+)=\sqrt{\epsilon}.$ Indeed, the result follows since the interval in \eqref{eq:equiv-lt} is a vanishingly small interval around $S$ as $\epsilon\rightarrow 0$, and since every $v$ is contained in $(v_-, v^*)$ for $\epsilon$ sufficiently small.
\\
\\
Consider $\mu \in M(\DD, C)$, with cutoffs $P_1\le P_2 \le \cdots \le P_C.$ Define the following two sets of colleges:
\begin{align}
    \FF_1 &:= \{1, 2, \cdots, \epsilon C\}\\
    \FF_2 &:= \{\epsilon C + 1, \epsilon C + 2, \cdots, C\}.
\end{align}
Then observe that
\begin{equation}\label{eq:lt-bounds}
    p_\mu(v, \CC_2) \le p_\mu(v) \le p_\mu(v, \CC_1) + p_\mu(v, \CC_2).
\end{equation}
We use \eqref{eq:lt-bounds} to show the desired result. To briefly outline our plan: First, we show that $\Pr[\mu(v)\in \FF_1]$ must be small for all $v$ except on a set of small $\eta$-measure. Since $\FF_1$ is a small fraction of the firms in the market, $p_\mu(v, \CC_1)$ can be thought of as an ``error term.'' Second, and more substantially, we show that $p_\mu(v, \CC_2) \approx S$ for all $v$ except on a set of small $\eta$-measure. Then the result will follow by applying the bounds in \eqref{eq:lt-bounds}.
\\
\\
We focus first on $p_\mu(v, \CC_2)$. We will show the following result.
\begin{proposition}\label{prop:lt-large-firms}
There exists a constant $\sigma > 0$ such that for $C$ sufficiently large,
    \begin{equation}\label{eq:lt-large-firms}
    p_\mu(v, \CC_2) \in (S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})).
    \end{equation}
for all $v\in (v_-, v_+).$
\end{proposition}

Observe that by using the bounds in \Cref{prop:lt-small-firms} and \Cref{prop:lt-large-firms} in combination with \eqref{eq:lt-bounds}, we obtain the desired result \eqref{eq:equiv-lt}. Therefore, it suffices to show \Cref{prop:lt-large-firms}, which we devote the remainder of this section towards.

\paragraph{Proof of \Cref{prop:lt-large-firms}.}
We first show a few useful lemmas. The first lemma demonstrates that as $C$ is taken large, almost all college cutoffs are must also be large.

\begin{lemma}\label{lem:unbounded-cutoffs}
    For any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large.
\end{lemma}

\begin{proof}
    Suppose otherwise. Then there exists $P$ such that there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le P_C$ such that $P_{\epsilon C} \le P$.
    Then we show that the number of applicants who can afford a firm in $\FF_1$ exceeds the total capacity $S$. Indeed,
```

```latex
    \begin{align}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{-\infty}^\infty \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P_{\epsilon C}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P]\,d\eta(v).
    \end{align}
    Since $\DD$ is unbounded from above, the final integral approaches $1$ as $C$ grows large. So for $C$ sufficiently large,
    \begin{equation}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v) > S,
    \end{equation}
    which contradicts the fact that the total capacity of colleges is $S$.
\end{proof}

We now use \Cref{lem:unbounded-cutoffs} to show that the probability that applicants with values $v_-$ and $v_+$ have approximately equal chances of affording a college in $\FF_2$ when $C$ is sufficiently large.

\begin{lemma}\label{lem:apple}
    For $C$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P_c]}{\Pr[v_+ + X > P_c]} > 1 - \epsilon
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    For $P$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P]}{\Pr[v_+ + X > P]} > 1 - \epsilon
    \end{equation}
    since $\DD$ is long-tailed. By \Cref{lem:unbounded-cutoffs}, for any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large. Observing that $P_c\ge P_{\epsilon C}$ for all $c\in \FF_2,$ the result follows.
\end{proof}

\Cref{lem:apple} shows that the probability of affording a firm in $\FF_2$ is approximately the same for $v_-$ and $v_+$ We require a slightly stronger result: that the probability of affording \textit{at least one} college in $\FF_2$ is approximately the same. To show this, we also require the following bound.

\begin{lemma}
    There exists a constant $\sigma>0$ such that for $C$ sufficiently large,
    \begin{equation}
        \Pr[v_+ + X > P_c] \le \frac{\sigma}{C}
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    Assume otherwise, such that such that for all $\sigma > 0$, there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le P_C$ such that
    \begin{equation}
        \Pr[v_+ + X > P_{\epsilon C + 1}] > \frac{\sigma}{C}.
    \end{equation}

    Then if we take $C$ and $\sigma$ sufficiently large, and choose such a $\mu,$ we have that
    \begin{align}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{v_-}^{v_+} \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge (1 - \epsilon) \Pr[v_- + X_c > P_{\epsilon c}\text{ for some $c\in \FF_1$}] \label{eq:building-1}\\
        &= (1-\epsilon)\left( 1 - \left(1 - \Pr[v_- + X > P_{\epsilon m}]\right)^{|\FF_1|}\right) \label{eq:building-2}\\
        &> (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right) \label{eq:building-3}
    \end{align}
    where \eqref{eq:building-1} follows by noting that $\eta(v_-, v_+)=1-\epsilon$ and that $\Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}] \ge \Pr[v_- + X_c > P_c\text{ for some $c\in \FF_1$}]$ for $v\in (v_-,v_+).$ \eqref{eq:building-3} follows from taking $C$ sufficiently large and applying \Cref{lem:apple}. Continuing,
    \begin{align}
        (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right)
        &> (1-\epsilon)\left(1 - \left(1 - \frac{(1-\epsilon)\sigma}{C}\right)^{|\FF_1|}\right)\\
        &> (1-\epsilon)\left(1 - \exp\left[- \frac{|\FF_1|(1-\epsilon)\sigma}{C}\right]\right)\\
        &= (1-\epsilon)\left(1 - \exp\left[-\epsilon(1-\epsilon)\sigma\right]\right)\\
        &> S
    \end{align}
    for $\sigma$ sufficiently large when $\epsilon < 1 - S$. This provides the desired contradiction, since the number of applicants who can afford a college in $\FF_2$ cannot exceed the total capacity of colleges.
\end{proof}

We may now show that the probability $v_-$ and $v_+$ can afford at least one college in $\FF_2$ is approximately the same when $C$ is large.
\begin{proposition}\label{prop:lt-approx-F2}
For $C$ sufficiently large,
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}.
\end{equation}
\end{proposition}
\begin{proof}
We have that
\begin{align}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2)
    &= \prod_{c\in \FF_2}\Pr[v_- + X < P_c] - \prod_{c\in \FF_2}\Pr[v_+ + X < P_c]\\
    &= \left(1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}\right) \prod_{c\in \FF_2}\Pr[v_- + X < P_c]\\
    &< 1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
\end{align}
From \Cref{lem:apple},
\begin{equation}
    \Pr[v_- + X > P_c] > (1-\epsilon)\Pr[v_+ + X > P_c]
\end{equation}
for all $c\in \FF_2$. Therefore,
\begin{align}
    \frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]} &= \frac{1 - \Pr[v_+ + X > P_c]}{1 - \Pr[v_- + X > P_c]}\\
    &> \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]}.
\end{align}
Continuing,
```

```
\begin{align}
    \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]} &= 1 - \frac{\epsilon \Pr[v_+ + X > P_c]}{1 - (1-\epsilon) \Pr[v_+ + X > P_c]}\\
    &\ge 1 - \frac{\epsilon\frac{\sigma}{C}}{1 - (1-\epsilon)\frac{\sigma}{C}}\\
    &= 1 - \frac{\epsilon\sigma}{C - (1-\epsilon)\sigma}\\
    &> \exp\left[-\frac{2\epsilon\sigma}{C}\right]
\end{align}
where the last inequality holds when $C$ is sufficiently large. Then we have that
\begin{align}
    1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
    &< 1 - \left(\exp\left[-\frac{2\epsilon\sigma}{C}\right]\right)^{|\FF_2|}\\
    &< 1 - e^{-2\epsilon\sigma}.
\end{align}
\end{proof}

We now use \Cref{prop:lt-approx-F2} to show \Cref{prop:lt-large-firms}, where primary remaining observation is that the average probability a student with value in $(v_-,v_+)$ can afford a college in $\FF_2$ must be approximately $S.$ Then, since the probabilities for $v$ in this interval must be about the same (by \Cref{prop:lt-approx-F2}), all the probabilities in this interval must be about $S$.

Formally, we observe that
\begin{align}
    (1 - \epsilon) p_\mu(v_-, \CC_2) \le \int_{v_-}^{v_+} p_\mu(v, \CC_2)\,d\eta(v) \le S,
\end{align}
so
\begin{equation}\label{eq:bbq-1}
    p_\mu(v_-, \CC_2) \le \frac{S}{1-\epsilon} < S + \epsilon,
\end{equation}
where the last inequality holds for $\epsilon$ sufficiently small. We also have that
\begin{align}
    p_\mu(v_+, \CC_2) &\ge \int_{v_-}^{v_+} \Pr[B(v)\cap \FF_2\neq \emptyset]\,d\eta(v)\\
    &\ge \int_{-\infty}^\infty \Pr[B(v)\cap \FF_2\neq \emptyset]\,d\eta(v) - \epsilon \\
    &\ge \int_{-\infty}^\infty \Pr[\mu(v)\cap \FF_2\neq \emptyset]\,d\eta(v) - \epsilon \label{eq:ginger}\\
    &= \left(S - \int_{-\infty}^\infty \Pr[\mu(v)\in \FF_1\neq \emptyset]\,d\eta(v)\right) - \epsilon\\
    &\ge S - \alpha\epsilon - \epsilon,
\end{align}
so
\begin{equation}\label{eq:bbq-2}
    p_\mu(v_+, \CC_2) \ge S - \alpha\epsilon - \epsilon.
\end{equation}
Then, since
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}
\end{equation}
(by \Cref{prop:lt-approx-F2}), we have that
\begin{align}
    p_\mu(v_+, \CC_2) &< p_\mu(v_-, \CC_2) + (1 - e^{-2\epsilon\sigma})\\
    &< S + \epsilon + (1 - e^{-2\epsilon\sigma})\label{eq:cucumber-1},
\end{align}
where the last line follows from \eqref{eq:bbq-1}, and that
\begin{align}
    \Pr[B(v_-)\cap \FF_2\neq \emptyset] &> \Pr[B(v_+)\cap \FF_2\neq \emptyset] - (1 - e^{-2\epsilon\sigma})\\
    &\ge S - \alpha\epsilon - \epsilon - (1 - e^{-2\epsilon\sigma}),\label{eq:cucumber-2}
\end{align}
where the last line follows from \eqref{eq:bbq-2}.
Finally, since
\begin{equation}
    p_\mu(v_-, \CC_2) < p_\mu(v, \CC_2) < p_\mu(v_+, \CC_2),
\end{equation}
it follows from \eqref{eq:cucumber-1} and \eqref{eq:cucumber-2} that
\begin{equation}
    p_\mu(v, \CC_2) \in (S - \alpha\epsilon - \epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})),
\end{equation}
as desired.
\\
\\
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

The large-firm interval proposition is the source p_mu(v,F2) strict open interval with its positive sigma witness, under the preceding endpoint and product-gap estimates.

Archival source anchor: cited publication, lines 744-916

Expanded PaperInterface specification

```
∀ (noiseLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  {active : (C : ℕ) → Finset (Fin (C + 1))} {cutoff : (C : ℕ) → Fin (C + 1) → ℝ}
  {vLow vHigh totalSupply alpha epsilon : ℝ},
  0 < epsilon →
    (∃ (sigma : ℝ),
        0 < sigma ∧
          (∀ᶠ (C : ℕ) in Filter.atTop,
              Monotone fun (v : ℝ) =>
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C)) ∧
            (∀ᶠ (C : ℕ) in Filter.atTop,
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                    (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vLow (cutoff C) ≤
                  totalSupply + epsilon) ∧
              (∀ᶠ (C : ℕ) in Filter.atTop,
                  totalSupply - (1 + alpha) * epsilon ≤
                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vHigh (cutoff C)) ∧
                ∀ᶠ (C : ℕ) in Filter.atTop,
                  PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                        (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vHigh (cutoff C) -
```

```
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vLow (cutoff C) <
                1 - Real.exp (-(2 * epsilon * sigma))) →
    ∃ (sigma : ℝ),
      0 < sigma ∧
        ∀ᶠ (C : ℕ) in Filter.atTop,
          ∀ (v : ℝ),
            vLow < v →
              v < vHigh →
                totalSupply - (1 + alpha) * epsilon - (1 - Real.exp (-(2 * epsilon * sigma))) <
                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C) ∧
                  PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C) <
                    totalSupply + epsilon + (1 - Real.exp (-(2 * epsilon * sigma)))
```

Retained governing declarations

- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
3. parameter: (C : ℕ) → Finset (Fin (C + 1))
4. parameter: (C : ℕ) → Fin (C + 1) → ℝ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. assumption: 0 < epsilon
11. assumption: ∃ (sigma : ℝ),
  0 < sigma ∧
    (∀ᶠ (C : ℕ) in Filter.atTop,
        Monotone fun (v : ℝ) =>
          PG24NoisyMatchingMarkets.cutoffAffordanceProbability
            (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C)) ∧
      (∀ᶠ (C : ℕ) in Filter.atTop,
          PG24NoisyMatchingMarkets.cutoffAffordanceProbability
              (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vLow (cutoff C) ≤
            totalSupply + epsilon) ∧
        (∀ᶠ (C : ℕ) in Filter.atTop,
            totalSupply - (1 + alpha) * epsilon ≤
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vHigh (cutoff C)) ∧
          ∀ᶠ (C : ℕ) in Filter.atTop,
            PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vHigh (cutoff C) -
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vLow (cutoff C) <
              1 - Real.exp (-(2 * epsilon * sigma))
12. conclusion: ∃ (sigma : ℝ),
  0 < sigma ∧
    ∀ᶠ (C : ℕ) in Filter.atTop,
      ∀ (v : ℝ),
        vLow < v →
          v < vHigh →
            totalSupply - (1 + alpha) * epsilon - (1 - Real.exp (-(2 * epsilon * sigma))) <
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C) ∧
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                  (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C) <
                totalSupply + epsilon + (1 - Real.exp (-(2 * epsilon * sigma)))
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_lt_large_firmsSpec_proof
```

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The expanded target now includes the source-context premise 0 < epsilon. It preserves the positive sigma witness, iid p_mu(v,F2) object, preceding monotonicity/endpoint/product-gap inputs, eventual uniformity, and both strict endpoints of the approved open interval.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 757-759

Verbatim source input

```
\begin{lemma}\label{lem:unbounded-cutoffs}
  For any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large.
\end{lemma}
```

Expanded PaperInterface specification

```
∀ {StudentTypeSeq : ℕ → Type u} [inst : (C : ℕ) → MeasurableSpace (StudentTypeSeq C)] {OutcomeSeq : ℕ → Type v}
  [inst_1 : (C : ℕ) → MeasurableSpace (OutcomeSeq C)] {GlobalCollegeSeq : ℕ → Type w}
  [inst_2 : (C : ℕ) → Fintype (GlobalCollegeSeq C)] (CutoffSeq : ℕ → Type x) (noiseLaw eta : MeasureTheory.Measure ℝ)
  [MeasureTheory.IsProbabilityMeasure noiseLaw] [MeasureTheory.IsProbabilityMeasure eta],
  PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
    ∀ {totalSupply epsilon : ℝ},
      0 < epsilon →
        totalSupply < 1 →
          ∀
            (data :
              (C : ℕ) →
                PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C)
                  (OutcomeSeq C) (GlobalCollegeSeq C) (CutoffSeq C)),
            (∀ (C : ℕ) (i j : Fin (C + 1)),
                ↑i ≤ ↑j →
                  (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
                    (data C).toExtendedCoalitionSourceStableInstance.localCutoff j) →
              ∀ (P : ℝ),
                ∀ᶠ (C : ℕ) in Filter.atTop,
                  ∀
                    c ∈
                      PG24NoisyMatchingMarkets.indexSuffixLarge
                        (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
                    P < (data C).toExtendedCoalitionSourceStableInstance.localCutoff c
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.localCutoff
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.epsilonFloorSplitIndex
- PG24NoisyMatchingMarkets.indexSuffixLarge

Lean-elaborated claim roles

```
1. parameter: ℕ → Type u
2. parameter: (C : ℕ) → MeasurableSpace (StudentTypeSeq C)
3. parameter: ℕ → Type v
4. parameter: (C : ℕ) → MeasurableSpace (OutcomeSeq C)
5. parameter: ℕ → Type w
6. parameter: (C : ℕ) → Fintype (GlobalCollegeSeq C)
7. parameter: ℕ → Type x
8. parameter: MeasureTheory.Measure ℝ
9. parameter: MeasureTheory.Measure ℝ
10. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
11. assumption: MeasureTheory.IsProbabilityMeasure eta
12. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
13. parameter: ℝ
14. parameter: ℝ
15. assumption: 0 < epsilon
16. assumption: totalSupply < 1
17. parameter: (C : ℕ) →
  PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C) (OutcomeSeq C)
    (GlobalCollegeSeq C) (CutoffSeq C)
18. assumption: ∀ (C : ℕ) (i j : Fin (C + 1)),
  ↑i ≤ ↑j →
    (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
      (data C).toExtendedCoalitionSourceStableInstance.localCutoff j
19. parameter: ℝ
20. conclusion: ∀ᶠ (C : ℕ) in Filter.atTop,
  ∀ c ∈ PG24NoisyMatchingMarkets.indexSuffixLarge (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
    P < (data C).toExtendedCoalitionSourceStableInstance.localCutoff c
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_unbounded_cutoffsSpec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: The source conclusion at source_unbounded_cutoffs, audit/cited publication:757-759, says every fixed P is eventually below P_{εC}. The Lean target has the same ∀P/eventually quantifier order and strengthens the sorted-index reading to every c in the suffix indexSuffixLarge (epsilonFloorSplitIndex epsilon C) C. The displayed localCutoff and suffix definitions identify these as the source sorted cutoffs past the εC prefix; its C+1/floor convention is asymptotically the same nonempty-product indexing convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 779-785

Verbatim source input

```
\begin{lemma}\label{lem:apple}
  For $C$ sufficiently large,
  \begin{equation}
    \frac{\Pr[v_- + X > P_c]}{\Pr[v_+ + X > P_c]} > 1 - \epsilon
  \end{equation}
  for all $c\in \FF_2.$
\end{lemma}
```

Expanded PaperInterface specification

```
∀ {StudentTypeSeq : ℕ → Type u} [inst : (C : ℕ) → MeasurableSpace (StudentTypeSeq C)] {OutcomeSeq : ℕ → Type v}
  [inst_1 : (C : ℕ) → MeasurableSpace (OutcomeSeq C)] {GlobalCollegeSeq : ℕ → Type w}
  [inst_2 : (C : ℕ) → Fintype (GlobalCollegeSeq C)] (CutoffSeq : ℕ → Type x) (noiseLaw eta : MeasureTheory.Measure ℝ)
  [MeasureTheory.IsProbabilityMeasure noiseLaw] [MeasureTheory.IsProbabilityMeasure eta],
  PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
    ∀ {totalSupply epsilon vLow vHigh : ℝ},
      0 < epsilon →
        vLow < vHigh →
          totalSupply < 1 →
            ∀
              (data :
                (C : ℕ) →
                  PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C)
                    (OutcomeSeq C) (GlobalCollegeSeq C) (CutoffSeq C)),
              (∀ (C : ℕ) (i j : Fin (C + 1)),
                  ↑i ≤ ↑j →
                    (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
                      (data C).toExtendedCoalitionSourceStableInstance.localCutoff j) →
                ∀ᶠ (C : ℕ) in Filter.atTop,
                  ∀
                    c ∈
                      PG24NoisyMatchingMarkets.indexSuffixLarge
                        (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
                    0 <
                        AppliedModelingLib.Probability.upperTailMass noiseLaw
                          ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh) ∧
                      1 - epsilon <
                        AppliedModelingLib.Probability.upperTailMass noiseLaw
                            ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vLow) /
                          AppliedModelingLib.Probability.upperTailMass noiseLaw
                            ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh)
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.localCutoff
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.epsilonFloorSplitIndex
- PG24NoisyMatchingMarkets.indexSuffixLarge

Lean-elaborated claim roles

```
1. parameter: ℕ → Type u
2. parameter: (C : ℕ) → MeasurableSpace (StudentTypeSeq C)
3. parameter: ℕ → Type v
4. parameter: (C : ℕ) → MeasurableSpace (OutcomeSeq C)
5. parameter: ℕ → Type w
6. parameter: (C : ℕ) → Fintype (GlobalCollegeSeq C)
7. parameter: ℕ → Type x
8. parameter: MeasureTheory.Measure ℝ
9. parameter: MeasureTheory.Measure ℝ
10. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
11. assumption: MeasureTheory.IsProbabilityMeasure eta
12. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
13. parameter: ℝ
14. parameter: ℝ
15. parameter: ℝ
16. parameter: ℝ
17. assumption: 0 < epsilon
18. assumption: vLow < vHigh
19. assumption: totalSupply < 1
20. parameter: (C : ℕ) →
  PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C) (OutcomeSeq C)
    (GlobalCollegeSeq C) (CutoffSeq C)
21. assumption: ∀ (C : ℕ) (i j : Fin (C + 1)),
```

```
  ↑i ≤ ↑j →
    (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
      (data C).toExtendedCoalitionSourceStableInstance.localCutoff j
22. conclusion: ∀ᶠ (C : ℕ) in Filter.atTop,
  ∀ c ∈ PG24NoisyMatchingMarkets.indexSuffixLarge (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
    0 <
        AppliedModelingLib.Probability.upperTailMass noiseLaw
          ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh) ∧
      1 - epsilon <
        AppliedModelingLib.Probability.upperTailMass noiseLaw
            ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vLow) /
          AppliedModelingLib.Probability.upperTailMass noiseLaw
            ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh)
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_appleSpec_proof

Recorded source-to-Spec screening **Verdict:** matches

Reason: The conclusion at source_apple, audit/cited publication:779-785, is the eventual all-c∈FF_2 tail ratio. The Lean target conclusion is the same strict ratio after rewriting Pr[v± + X > P_c] as upperTailMass noiseLaw (localCutoff c - v±); upperTailMass is the strict Ioi tail. indexSuffixLarge (epsilonFloorSplitIndex epsilon C) is the displayed large-firm suffix, and the explicit positive denominator makes the displayed quotient well-defined without changing its rate or domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 797-803

Verbatim source input

```
\begin{lemma}
  There exists a constant $\sigma>0$ such that for $C$ sufficiently large,
  \begin{equation}
    \Pr[v_+ + X > P_c] \le \frac{\sigma}{C}
  \end{equation}
  for all $c\in \FF_2.$
\end{lemma}
```

Expanded PaperInterface specification

```
∀ {StudentTypeSeq : ℕ → Type u} [inst : (C : ℕ) → MeasurableSpace (StudentTypeSeq C)] {OutcomeSeq : ℕ → Type v}
  [inst_1 : (C : ℕ) → MeasurableSpace (OutcomeSeq C)] {GlobalCollegeSeq : ℕ → Type w}
  [inst_2 : (C : ℕ) → Fintype (GlobalCollegeSeq C)] (CutoffSeq : ℕ → Type x) (noiseLaw eta : MeasureTheory.Measure ℝ)
  [MeasureTheory.IsProbabilityMeasure noiseLaw] [MeasureTheory.IsProbabilityMeasure eta],
  PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw) →
    ∀ {totalSupply epsilon vHigh : ℝ},
      0 < epsilon →
        totalSupply < 1 →
          ∀
            (data :
              (C : ℕ) →
                PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C)
                  (OutcomeSeq C) (GlobalCollegeSeq C) (CutoffSeq C)),
            (∀ (C : ℕ) (i j : Fin (C + 1)),
                ↑i ≤ ↑j →
                  (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
                    (data C).toExtendedCoalitionSourceStableInstance.localCutoff j) →
              ∃ (sigma : ℝ),
                0 < sigma ∧
                  ∀ᶠ (C : ℕ) in Filter.atTop,
                    ∀
                      c ∈
                        PG24NoisyMatchingMarkets.indexSuffixLarge
                          (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
                      AppliedModelingLib.Probability.upperTailMass noiseLaw
                          ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh) ≤
                        sigma / ↑C
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.LongTailedSurvival
- PG24NoisyMatchingMarkets.PG24ExtendedCoalitionSourceStableInstance.localCutoff
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData
- PG24NoisyMatchingMarkets.PG24LiteralSourceStableData.toExtendedCoalitionSourceStableInstance
- PG24NoisyMatchingMarkets.epsilonFloorSplitIndex
- PG24NoisyMatchingMarkets.indexSuffixLarge

Lean-elaborated claim roles

```
1. parameter: ℕ → Type u
2. parameter: (C : ℕ) → MeasurableSpace (StudentTypeSeq C)
3. parameter: ℕ → Type v
4. parameter: (C : ℕ) → MeasurableSpace (OutcomeSeq C)
5. parameter: ℕ → Type w
6. parameter: (C : ℕ) → Fintype (GlobalCollegeSeq C)
7. parameter: ℕ → Type x
8. parameter: MeasureTheory.Measure ℝ
9. parameter: MeasureTheory.Measure ℝ
10. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
11. assumption: MeasureTheory.IsProbabilityMeasure eta
12. assumption: PG24NoisyMatchingMarkets.LongTailedSurvival (AppliedModelingLib.Probability.upperTailMass noiseLaw)
13. parameter: ℝ
14. parameter: ℝ
15. parameter: ℝ
16. assumption: 0 < epsilon
17. assumption: totalSupply < 1
18. parameter: (C : ℕ) →
  PG24NoisyMatchingMarkets.PG24LiteralSourceStableData C noiseLaw eta totalSupply (StudentTypeSeq C) (OutcomeSeq C)
    (GlobalCollegeSeq C) (CutoffSeq C)
19. assumption: ∀ (C : ℕ) (i j : Fin (C + 1)),
  ↑i ≤ ↑j →
    (data C).toExtendedCoalitionSourceStableInstance.localCutoff i ≤
      (data C).toExtendedCoalitionSourceStableInstance.localCutoff j
20. conclusion: ∃ (sigma : ℝ),
  0 < sigma ∧
    ∀ᶠ (C : ℕ) in Filter.atTop,
```

```
∀ c ∈ PG24NoisyMatchingMarkets.indexSuffixLarge (PG24NoisyMatchingMarkets.epsilonFloorSplitIndex epsilon C) C,
  AppliedModelingLib.Probability.upperTailMass noiseLaw
      ((data C).toExtendedCoalitionSourceStableInstance.localCutoff c - vHigh) ≤
    sigma / ↑C
```

Lean proof endpoint (built separately)

```
PG24NoisyMatchingMarkets.ProofInterface.source_large_firm_tail_boundSpec_proof
```

Recorded source-to-Spec screening **Verdict:** matches

Reason: The existential/eventual/all-large-firm statement at source_large_firm_tail_bound, audit/cited publication:797-803, is preserved: Lean concludes ∃σ>0, eventually for every c in the large suffix, upperTailMass noiseLaw (localCutoff c-vHigh) ≤ σ/C. By upperTailMass = Pr[X > ·], this is exactly Pr[v_+ + X > P_c] ≤ σ/C, with the displayed source cutoff restriction supplying P_c.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 831-836

Verbatim source input

```
\begin{proposition}\label{prop:lt-approx-F2}
For $C$ sufficiently large,
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}.
\end{equation}
\end{proposition}

[Next verbatim source excerpt]

\subsection{Preliminaries}
We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

Consider a matching $\mu\in M(\DD, C)$, characterized by a vector of market-clearing cutoffs $\vec{P}=\vec{P}(\mu)$. Our results focus on the probability that a student $\theta=(v,\succ)$ matches to a college. This probability is equal to
\begin{equation}
    \Pr[\mu(\theta)\in C] = \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
where $X_1, \cdots, X_C \simiid \DD.$ Notice that this probability does not depend on $\succ$, the student's preferences over colleges. Indeed, the probability that a student matches is exactly the probability that a student can afford some college; whether or not they can afford a college only depends on their value and the college's cutoff (not the student's preferences). Accordingly, we can drop $\theta$ and define
\begin{equation}
    p_\mu(v) := \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
the probability that a student with value $v$ is matched under $\mu$. More generally, for a subset of colleges $C'\subseteq C$, we define
\begin{equation}
    p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
the probability that a student with value $v$ can afford some college in $C'$. By definition, $p_\mu(v) = p_\mu(v, C).$

[Next verbatim source excerpt]

\end{equation}
This tells us that if a college's cutoff is large, two students (with different values) can afford the college with approximately equal probability. To leverage this fact, we will show that almost all colleges must have large cutoffs. Roughly speaking, this is true because if there were a large number of colleges with small cutoffs, then more students would be able to afford these firms than there is total capacity.

\subsection*{Proof of \Cref{thm2v2}}
We consider a long-tailed probability distribution $\DD$. To prove \Cref{thm2v2}, we show the following equivalent statement: There exists a constant $\sigma > 0$ such that for all $C$ sufficiently large, if $\mu \in M(\DD, C),$
\begin{equation}\label{eq:equiv-lt}
    p_\mu(v) \in \left(S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma}) + \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}\right)
\end{equation}
for all $v\in (v_-,v^*)$, where $v_-, v_+$ are chosen such that $\eta(-\infty,v_-) = \eta(v_+,\infty) = \frac{\epsilon}{2}$ and $v^*$ is chosen such that $\eta(v^*,v_+)=\sqrt{\epsilon}.$ Indeed, the result follows since the interval in \eqref{eq:equiv-lt} is a vanishingly small interval around $S$ as $\epsilon\rightarrow 0$, and since every $v$ is contained in $(v_-, v^*)$ for $\epsilon$ sufficiently small.
\\
\\
Consider $\mu \in M(\DD, C)$, with cutoffs $P_1\le P_2 \le \cdots \le P_C.$ Define the following two sets of colleges:
\begin{align}
    \FF_1 &:= \{1, 2, \cdots, \epsilon C\}\\
    \FF_2 &:= \{\epsilon C + 1, \epsilon C + 2, \cdots, C\}.
\end{align}
Then observe that
\begin{equation}\label{eq:lt-bounds}
    p_\mu(v, \CC_2) \le p_\mu(v) \le p_\mu(v, \CC_1) + p_\mu(v, \CC_2).
\end{equation}
We use \eqref{eq:lt-bounds} to show the desired result. To briefly outline our plan: First, we show that $\Pr[\mu(v)\in \FF_1]$ must be small for all $v$ except on a set of small $\eta$-measure. Since $\FF_1$ is a small fraction of the firms in the market, $p_\mu(v, \CC_1)$ can be thought of as an ``error term.'' Second, and more substantially, we show that $p_\mu(v, \CC_2) \approx S$ for all $v$ except on a set of small $\eta$-measure. Then the result will follow by applying the bounds in \eqref{eq:lt-bounds}.
\\
\\
We focus first on $p_\mu(v, \CC_2)$. We will show the following result.
\begin{proposition}\label{prop:lt-large-firms}
There exists a constant $\sigma > 0$ such that for $C$ sufficiently large,
    \begin{equation}\label{eq:lt-large-firms}
    p_\mu(v, \CC_2) \in (S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})).
    \end{equation}
for all $v\in (v_-, v_+).$
\end{proposition}

Observe that by using the bounds in \Cref{prop:lt-small-firms} and \Cref{prop:lt-large-firms} in combination with \eqref{eq:lt-bounds}, we obtain the desired result \eqref{eq:equiv-lt}. Therefore, it suffices to show \Cref{prop:lt-large-firms}, which we devote the remainder of this section towards.

\paragraph{Proof of \Cref{prop:lt-large-firms}.}
We first show a few useful lemmas. The first lemma demonstrates that as $C$ is taken large, almost all college cutoffs are must also be large.

\begin{lemma}\label{lem:unbounded-cutoffs}
    For any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large.
\end{lemma}

\begin{proof}
    Suppose otherwise. Then there exists $P$ such that there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le P_C$ such that $P_{\epsilon C} \le P$.
    Then we show that the number of applicants who can afford a firm in $\FF_1$ exceeds the total capacity $S$. Indeed,
    \begin{align}
```

```latex
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{-\infty}^\infty \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P_{\epsilon C}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P]\,d\eta(v).
    \end{align}
    Since $\DD$ is unbounded from above, the final integral approaches $1$ as $C$ grows large. So for $C$ sufficiently large,
    \begin{equation}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v) > S,
    \end{equation}
    which contradicts the fact that the total capacity of colleges is $S$.
\end{proof}

We now use \Cref{lem:unbounded-cutoffs} to show that the probability that applicants with values $v_-$ and $v_+$ have approximately equal chances of
↪   affording a college in $\FF_2$ when $C$ is sufficiently large.

\begin{lemma}\label{lem:apple}
    For $C$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P_c]}{\Pr[v_+ + X > P_c]} > 1 - \epsilon
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    For $P$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P]}{\Pr[v_+ + X > P]} > 1 - \epsilon
    \end{equation}
    since $\DD$ is long-tailed. By \Cref{lem:unbounded-cutoffs}, for any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large. Observing that
    ↪   $P_c\ge P_{\epsilon C}$ for all $c\in \FF_2,$ the result follows.
\end{proof}

\Cref{lem:apple} shows that the probability of affording a firm in $\FF_2$ is approximately the same for $v_-$ and $v_+$ We require a slightly stronger
↪   result: that the probability of affording \textit{at least one} college in $\FF_2$ is approximately the same. To show this, we also require the following
↪   bound.

\begin{lemma}
    There exists a constant $\sigma>0$ such that for $C$ sufficiently large,
    \begin{equation}
        \Pr[v_+ + X > P_c] \le \frac{\sigma}{C}
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    Assume otherwise, such that such that for all $\sigma > 0$, there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le
    ↪   P_C$ such that
    \begin{equation}
        \Pr[v_+ + X > P_{\epsilon C + 1}] > \frac{\sigma}{C}.
    \end{equation}

    Then if we take $C$ and $\sigma$ sufficiently large, and choose such a $\mu,$ we have that
    \begin{align}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{v_-}^{v_+} \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge (1 - \epsilon) \Pr[v_- + X_c > P_{\epsilon c}\text{ for some $c\in \FF_1$}] \label{eq:building-1}\\
        &= (1-\epsilon)\left( 1 - \left(1 - \Pr[v_- + X > P_{\epsilon m}]\right)^{|\FF_1|}\right) \label{eq:building-2}\\
        &> (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right) \label{eq:building-3}
    \end{align}
    where \eqref{eq:building-1} follows by noting that $\eta(v_-, v_+)=1-\epsilon$ and that $\Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}] \ge \Pr[v_- + X_c
    ↪   > P_c\text{ for some $c\in \FF_1$}]$ for $v\in (v_-,v_+).$ \eqref{eq:building-3} follows from taking $C$ sufficiently large and applying
    ↪   \Cref{lem:apple}. Continuing,
    \begin{align}
        (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right)
        &> (1-\epsilon)\left(1 - \left(1 - \frac{(1-\epsilon)\sigma}{C}\right)^{|\FF_1|}\right)\\
        &> (1-\epsilon)\left(1 - \exp\left[- \frac{|\FF_1|(1-\epsilon)\sigma}{C}\right]\right)\\
        &= (1-\epsilon)\left(1 - \exp\left[-\epsilon(1-\epsilon)\sigma\right]\right)\\
        &> S
    \end{align}
    for $\sigma$ sufficiently large when $\epsilon < 1 - S$. This provides the desired contradiction, since the number of applicants who can afford a college in
    ↪   $\FF_2$ cannot exceed the total capacity of colleges.
\end{proof}

We may now show that the probability $v_-$ and $v_+$ can afford at least one college in $\FF_2$ is approximately the same when $C$ is large.
\begin{proposition}\label{prop:lt-approx-F2}
For $C$ sufficiently large,
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}.
\end{equation}
\end{proposition}
\begin{proof}
We have that
\begin{align}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2)
    &= \prod_{c\in \FF_2}\Pr[v_- + X < P_c] - \prod_{c\in \FF_2}\Pr[v_+ + X < P_c]\\
    &= \left(1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}\right) \prod_{c\in \FF_2}\Pr[v_- + X < P_c]\\
    &< 1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
\end{align}
From \Cref{lem:apple},
\begin{equation}
    \Pr[v_- + X > P_c] > (1-\epsilon)\Pr[v_+ + X > P_c]
\end{equation}
for all $c\in \FF_2$. Therefore,
\begin{align}
    \frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]} &= \frac{1 - \Pr[v_+ + X > P_c]}{1 - \Pr[v_- + X > P_c]}\\
    &> \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]}.
\end{align}
Continuing,
\begin{align}
```

```
    \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]} &= 1 - \frac{\epsilon \Pr[v_+ + X > P_c]}{1 - (1-\epsilon) \Pr[v_+ + X > P_c]}\\
    &\ge 1 - \frac{\epsilon\frac{\sigma}{C}}{1 - (1-\epsilon)\frac{\sigma}{C}}\\
    &= 1 - \frac{\epsilon\sigma}{C - (1-\epsilon)\sigma}\\
    &> \exp\left[-\frac{2\epsilon\sigma}{C}\right]
\end{align}
where the last inequality holds when $C$ is sufficiently large. Then we have that
\begin{align}
    1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
    &< 1 - \left(\exp\left[-\frac{2\epsilon\sigma}{C}\right]\right)^{|\FF_2|}\\
    &< 1 - e^{-2\epsilon\sigma}.
\end{align}
\end{proof}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

The large-firm affordance-difference proposition is the strict source p_mu(v_+,F2)-p_mu(v_-,F2) bound under its preceding tail-ratio and nonempty-block ↪ estimates.

Archival source anchor: cited publication, lines 831-867

Expanded PaperInterface specification

```
∀ (noiseLaw : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  {active : (n : ℕ) → Finset (Fin (n + 1))} {cutoff : (n : ℕ) → Fin (n + 1) → ℝ} {vLow vHigh epsilon sigma : ℝ},
  0 < epsilon →
    0 < sigma →
      (∀ᶠ (n : ℕ) in Filter.atTop, (active n).Nonempty) →
        (∀ᶠ (n : ℕ) in Filter.atTop,
            ∀ c ∈ active n,
              Real.exp (-(2 * epsilon * sigma / ↑(n + 1))) <
                (1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vHigh)) /
                  (1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow))) →
          (∀ᶠ (n : ℕ) in Filter.atTop,
              ∀ c ∈ active n, 0 < 1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow)) →
            (∀ᶠ (n : ℕ) in Filter.atTop,
                ∀ c ∈ active n, 1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow) ≤ 1) →
              ∀ᶠ (n : ℕ) in Filter.atTop,
                PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) (active n) vHigh (cutoff n) -
                    PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                      (MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw) (active n) vLow (cutoff n) <
                  1 - Real.exp (-(2 * epsilon * sigma))
```

Retained governing declarations

- AppliedModelingLib.Probability.upperTailMass
- PG24NoisyMatchingMarkets.cutoffAffordanceProbability

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
3. parameter: (n : ℕ) → Finset (Fin (n + 1))
4. parameter: (n : ℕ) → Fin (n + 1) → ℝ
5. parameter: ℝ
6. parameter: ℝ
7. parameter: ℝ
8. parameter: ℝ
9. assumption: 0 < epsilon
10. assumption: 0 < sigma
11. assumption: ∀ᶠ (n : ℕ) in Filter.atTop, (active n).Nonempty
12. assumption: ∀ᶠ (n : ℕ) in Filter.atTop,
  ∀ c ∈ active n,
    Real.exp (-(2 * epsilon * sigma / ↑(n + 1))) <
      (1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vHigh)) /
        (1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow))
13. assumption: ∀ᶠ (n : ℕ) in Filter.atTop,
  ∀ c ∈ active n, 0 < 1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow)
14. assumption: ∀ᶠ (n : ℕ) in Filter.atTop,
  ∀ c ∈ active n, 1 - AppliedModelingLib.Probability.upperTailMass noiseLaw (cutoff n c - vLow) ≤ 1
15. conclusion: ∀ᶠ (n : ℕ) in Filter.atTop,
  PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw)
      (active n) vHigh (cutoff n) -
    PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (n + 1)) => noiseLaw)
      (active n) vLow (cutoff n) <
  1 - Real.exp (-(2 * epsilon * sigma))
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_lt_approx_f2Spec_proof

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: The conclusion is exactly the strict source p_mu(v_+,F2)-p_mu(v_-,F2) bound, with p_mu represented by iid cutoff-affordance probability and the same 1-exp(-2 epsilon sigma) endpoint. The approved nonempty-block, tail-ratio, and complement-probability context is explicit.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 918-923

Verbatim source input

```
\begin{proposition}\label{prop:lt-small-firms}
\begin{equation}
    p_\mu(v, \CC_1) \le \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}
\end{equation}
for all $v < v^*$ where $\eta((v^*, v_+))=\sqrt{\epsilon}.$
\end{proposition}

[Next verbatim source excerpt]

\subsection{Preliminaries}
We now make some basic observations that will be useful for our results. We show that key properties of stable matchings operate at the level of student values rather than student types (which also encode student preferences). This means that in our analysis, we can focus on the fixed student value distribution $\eta$ while mostly ignoring the varying student type distribution $\rho$.

Consider a matching $\mu\in M(\DD, C)$, characterized by a vector of market-clearing cutoffs $\vec{P}=\vec{P}(\mu)$. Our results focus on the probability that a student $\theta=(v,\succ)$ matches to a college. This probability is equal to
\begin{equation}
    \Pr[\mu(\theta)\in C] = \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
where $X_1, \cdots, X_C \simiid \DD.$ Notice that this probability does not depend on $\succ$, the student's preferences over colleges. Indeed, the probability that a student matches is exactly the probability that a student can afford some college; whether or not they can afford a college only depends on their value and the college's cutoff (not the student's preferences). Accordingly, we can drop $\theta$ and define
\begin{equation}
    p_\mu(v) := \Pr[v + X_c > P_c\text{ for some $c\in C$}],
\end{equation}
the probability that a student with value $v$ is matched under $\mu$. More generally, for a subset of colleges $C'\subseteq C$, we define
\begin{equation}
    p_\mu(v, C') := \Pr[v + X_c > P_c\text{ for some $c\in C'$}],
\end{equation}
the probability that a student with value $v$ can afford some college in $C'$. By definition, $p_\mu(v) = p_\mu(v, C).$

[Next verbatim source excerpt]

\end{equation}
This tells us that if a college's cutoff is large, two students (with different values) can afford the college with approximately equal probability. To leverage this fact, we will show that almost all colleges must have large cutoffs. Roughly speaking, this is true because if there were a large number of colleges with small cutoffs, then more students would be able to afford these firms than there is total capacity.

\subsection*{Proof of \Cref{thm2v2}}
We consider a long-tailed probability distribution $\DD$. To prove \Cref{thm2v2}, we show the following equivalent statement: There exists a constant $\sigma > 0$ such that for all $C$ sufficiently large, if $\mu \in M(\DD, C),$
\begin{equation}\label{eq:equiv-lt}
    p_\mu(v) \in \left(S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma}) + \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}\right)
\end{equation}
for all $v\in (v_-,v^*)$, where $v_-, v_+$ are chosen such that $\eta(-\infty,v_-) = \eta(v_+,\infty) = \frac{\epsilon}{2}$ and $v^*$ is chosen such that $\eta(v^*,v_+)=\sqrt{\epsilon}.$ Indeed, the result follows since the interval in \eqref{eq:equiv-lt} is a vanishingly small interval around $S$ as $\epsilon\rightarrow 0$, and since every $v$ is contained in $(v_-, v^*)$ for $\epsilon$ sufficiently small.
\\
\\
Consider $\mu \in M(\DD, C)$, with cutoffs $P_1\le P_2 \le \cdots \le P_C.$ Define the following two sets of colleges:
\begin{align}
    \FF_1 &:= \{1, 2, \cdots, \epsilon C\}\\
    \FF_2 &:= \{\epsilon C + 1, \epsilon C + 2, \cdots, C\}.
\end{align}
Then observe that
\begin{equation}\label{eq:lt-bounds}
    p_\mu(v, \CC_2) \le p_\mu(v) \le p_\mu(v, \CC_1) + p_\mu(v, \CC_2).
\end{equation}
We use \eqref{eq:lt-bounds} to show the desired result. To briefly outline our plan: First, we show that $\Pr[\mu(v)\in \FF_1]$ must be small for all $v$ except on a set of small $\eta$-measure. Since $\FF_1$ is a small fraction of the firms in the market, $p_\mu(v, \CC_1)$ can be thought of as an ``error term.'' Second, and more substantially, we show that $p_\mu(v, \CC_2) \approx S$ for all $v$ except on a set of small $\eta$-measure. Then the result will follow by applying the bounds in \eqref{eq:lt-bounds}.
\\
\\
We focus first on $p_\mu(v, \CC_2)$. We will show the following result.
\begin{proposition}\label{prop:lt-large-firms}
There exists a constant $\sigma > 0$ such that for $C$ sufficiently large,
    \begin{equation}\label{eq:lt-large-firms}
    p_\mu(v, \CC_2) \in (S - (1 + \alpha)\epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})).
    \end{equation}
for all $v\in (v_-, v_+).$
\end{proposition}

Observe that by using the bounds in \Cref{prop:lt-small-firms} and \Cref{prop:lt-large-firms} in combination with \eqref{eq:lt-bounds}, we obtain the desired result \eqref{eq:equiv-lt}. Therefore, it suffices to show \Cref{prop:lt-large-firms}, which we devote the remainder of this section towards.

\paragraph{Proof of \Cref{prop:lt-large-firms}.}
We first show a few useful lemmas. The first lemma demonstrates that as $C$ is taken large, almost all college cutoffs are must also be large.

\begin{lemma}\label{lem:unbounded-cutoffs}
    For any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large.
\end{lemma}

\begin{proof}
    Suppose otherwise. Then there exists $P$ such that there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le P_C$ such that $P_{\epsilon C} \le P$.
    Then we show that the number of applicants who can afford a firm in $\FF_1$ exceeds the total capacity $S$. Indeed,
    \begin{align}
```

```
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{-\infty}^\infty \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P_{\epsilon C}]\,d\eta(v)\\
        &\ge \int_{-\infty}^\infty \Pr[v + X^{(\epsilon C)} > P]\,d\eta(v).
    \end{align}
    Since $\DD$ is unbounded from above, the final integral approaches $1$ as $C$ grows large. So for $C$ sufficiently large,
    \begin{equation}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v) > S,
    \end{equation}
    which contradicts the fact that the total capacity of colleges is $S$.
\end{proof}

We now use \Cref{lem:unbounded-cutoffs} to show that the probability that applicants with values $v_-$ and $v_+$ have approximately equal chances of affording a college in $\FF_2$ when $C$ is sufficiently large.

\begin{lemma}\label{lem:apple}
    For $C$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P_c]}{\Pr[v_+ + X > P_c]} > 1 - \epsilon
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    For $P$ sufficiently large,
    \begin{equation}
        \frac{\Pr[v_- + X > P]}{\Pr[v_+ + X > P]} > 1 - \epsilon
    \end{equation}
    since $\DD$ is long-tailed. By \Cref{lem:unbounded-cutoffs}, for any $P\in \RR$, $P_{\epsilon C} > P$ for all $C$ sufficiently large. Observing that $P_c\ge P_{\epsilon C}$ for all $c\in \FF_2,$ the result follows.
\end{proof}

\Cref{lem:apple} shows that the probability of affording a firm in $\FF_2$ is approximately the same for $v_-$ and $v_+$ We require a slightly stronger result: that the probability of affording \textit{at least one} college in $\FF_2$ is approximately the same. To show this, we also require the following bound.

\begin{lemma}
    There exists a constant $\sigma>0$ such that for $C$ sufficiently large,
    \begin{equation}
        \Pr[v_+ + X > P_c] \le \frac{\sigma}{C}
    \end{equation}
    for all $c\in \FF_2.$
\end{lemma}

\begin{proof}
    Assume otherwise, such that such that for all $\sigma > 0$, there exists $C$ arbitrarily large and $\mu \in M(\DD, C)$ with cutoffs $P_1\le P_2\le \cdots \le P_C$ such that
    \begin{equation}
        \Pr[v_+ + X > P_{\epsilon C + 1}] > \frac{\sigma}{C}.
    \end{equation}

    Then if we take $C$ and $\sigma$ sufficiently large, and choose such a $\mu,$ we have that
    \begin{align}
        \int_{-\infty}^\infty p_\mu(v, \CC_1)\,d\eta(v)
        &= \int_{v_-}^{v_+} \Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}]\,d\eta(v)\\
        &\ge (1 - \epsilon) \Pr[v_- + X_c > P_{\epsilon c}\text{ for some $c\in \FF_1$}] \label{eq:building-1}\\
        &= (1-\epsilon)\left( 1 - \left(1 - \Pr[v_- + X > P_{\epsilon m}]\right)^{|\FF_1|}\right) \label{eq:building-2}\\
        &> (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right) \label{eq:building-3}
    \end{align}
    where \eqref{eq:building-1} follows by noting that $\eta(v_-, v_+)=1-\epsilon$ and that $\Pr[v + X_c > P_c\text{ for some $c\in \FF_1$}] \ge \Pr[v_- + X_c > P_c\text{ for some $c\in \FF_1$}]$ for $v\in (v_-,v_+).$ \eqref{eq:building-3} follows from taking $C$ sufficiently large and applying \Cref{lem:apple}. Continuing,
    \begin{align}
        (1-\epsilon)\left(1 - \left(1 - (1-\epsilon)\Pr[v_+ + X > P_{\epsilon C}]\right)^{|\FF_1|}\right)
        &> (1-\epsilon)\left(1 - \left(1 - \frac{(1-\epsilon)\sigma}{C}\right)^{|\FF_1|}\right)\\
        &> (1-\epsilon)\left(1 - \exp\left[- \frac{|\FF_1|(1-\epsilon)\sigma}{C}\right]\right)\\
        &= (1-\epsilon)\left(1 - \exp\left[-\epsilon(1-\epsilon)\sigma\right]\right)\\
        &> S
    \end{align}
    for $\sigma$ sufficiently large when $\epsilon < 1 - S$. This provides the desired contradiction, since the number of applicants who can afford a college in $\FF_2$ cannot exceed the total capacity of colleges.
\end{proof}

We may now show that the probability $v_-$ and $v_+$ can afford at least one college in $\FF_2$ is approximately the same when $C$ is large.
\begin{proposition}\label{prop:lt-approx-F2}
For $C$ sufficiently large,
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}.
\end{equation}
\end{proposition}
\begin{proof}
We have that
\begin{align}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2)
    &= \prod_{c\in \FF_2}\Pr[v_- + X < P_c] - \prod_{c\in \FF_2}\Pr[v_+ + X < P_c]\\
    &= \left(1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}\right) \prod_{c\in \FF_2}\Pr[v_- + X < P_c]\\
    &< 1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
\end{align}
From \Cref{lem:apple},
\begin{equation}
    \Pr[v_- + X > P_c] > (1-\epsilon)\Pr[v_+ + X > P_c]
\end{equation}
for all $c\in \FF_2$. Therefore,
\begin{align}
    \frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]} &= \frac{1 - \Pr[v_+ + X > P_c]}{1 - \Pr[v_- + X > P_c]}\\
    &> \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]}.
\end{align}
Continuing,
\begin{align}
```

```
    \frac{1 - \Pr[v_+ + X > P_c]}{1 - (1-\epsilon)\Pr[v_+ + X > P_c]} &= 1 - \frac{\epsilon \Pr[v_+ + X > P_c]}{1 - (1-\epsilon) \Pr[v_+ + X > P_c]}\\
    &\ge 1 - \frac{\epsilon\frac{\sigma}{C}}{1 - (1-\epsilon)\frac{\sigma}{C}}\\
    &= 1 - \frac{\epsilon\sigma}{C - (1-\epsilon)\sigma}\\
    &> \exp\left[-\frac{2\epsilon\sigma}{C}\right]
\end{align}
where the last inequality holds when $C$ is sufficiently large. Then we have that
\begin{align}
    1 - \prod_{c\in \FF_2}\frac{\Pr[v_+ + X < P_c]}{\Pr[v_- + X < P_c]}
    &< 1 - \left(\exp\left[-\frac{2\epsilon\sigma}{C}\right]\right)^{|\FF_2|}\\
    &< 1 - e^{-2\epsilon\sigma}.
\end{align}
\end{proof}

We now use \Cref{prop:lt-approx-F2} to show \Cref{prop:lt-large-firms}, where primary remaining observation is that the average probability a student with value in $(v_-,v_+)$ can afford a college in $\FF_2$ must be approximately $S.$ Then, since the probabilities for $v$ in this interval must be about the same (by \Cref{prop:lt-approx-F2}), all the probabilities in this interval must be about $S$.

Formally, we observe that
\begin{align}
    (1 - \epsilon) p_\mu(v_-, \CC_2) \le \int_{v_-}^{v_+} p_\mu(v, \CC_2)\,d\eta(v) \le S,
\end{align}
so
\begin{equation}\label{eq:bbq-1}
    p_\mu(v_-, \CC_2) \le \frac{S}{1-\epsilon} < S + \epsilon,
\end{equation}
where the last inequality holds for $\epsilon$ sufficiently small. We also have that
\begin{align}
    p_\mu(v_+, \CC_2) &\ge \int_{v_-}^{v_+} \Pr[B(v)\cap \FF_2\neq \emptyset]\,d\eta(v)\\
    &\ge \int_{-\infty}^\infty \Pr[B(v)\cap \FF_2\neq \emptyset]\,d\eta(v) - \epsilon \\
    &\ge \int_{-\infty}^\infty \Pr[\mu(v)\cap \FF_2\neq \emptyset]\,d\eta(v) - \epsilon \label{eq:ginger}\\
    &= \left(S - \int_{-\infty}^\infty \Pr[\mu(v)\in \FF_1\neq \emptyset]\,d\eta(v)\right) - \epsilon\\
    &\ge S - \alpha\epsilon - \epsilon,
\end{align}
so
\begin{equation}\label{eq:bbq-2}
    p_\mu(v_+, \CC_2) \ge S - \alpha\epsilon - \epsilon.
\end{equation}
Then, since
\begin{equation}
    p_\mu(v_+, \CC_2) - p_\mu(v_-, \CC_2) < 1 - e^{-2\epsilon\sigma}
\end{equation}
(by \Cref{prop:lt-approx-F2}), we have that
\begin{align}
    p_\mu(v_+, \CC_2) &< p_\mu(v_-, \CC_2) + (1 - e^{-2\epsilon\sigma})\\
    &< S + \epsilon + (1 - e^{-2\epsilon\sigma})\label{eq:cucumber-1},
\end{align}
where the last line follows from \eqref{eq:bbq-1}, and that
\begin{align}
    \Pr[B(v_-)\cap \FF_2\neq \emptyset] &> \Pr[B(v_+)\cap \FF_2\neq \emptyset] - (1 - e^{-2\epsilon\sigma})\\
    &\ge S - \alpha\epsilon - \epsilon - (1 - e^{-2\epsilon\sigma}),\label{eq:cucumber-2}
\end{align}
where the last line follows from \eqref{eq:bbq-2}.
Finally, since
\begin{equation}
    p_\mu(v_-, \CC_2) < p_\mu(v, \CC_2) < p_\mu(v_+, \CC_2),
\end{equation}
it follows from \eqref{eq:cucumber-1} and \eqref{eq:cucumber-2} that
\begin{equation}
    p_\mu(v, \CC_2) \in (S - \alpha\epsilon - \epsilon - (1 - e^{-2\epsilon\sigma}), S + \epsilon + (1 - e^{-2\epsilon\sigma})),
\end{equation}
as desired.
\\
\\
We now analyze $p_\mu(v, \CC_1)$.
\begin{proposition}\label{prop:lt-small-firms}
\begin{equation}
    p_\mu(v, \CC_1) \le \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}
\end{equation}
for all $v < v^*$ where $\eta((v^*, v_+))=\sqrt{\epsilon}.$
\end{proposition}
\begin{proof}
Note that any student who cannot afford any college in $\CC_2$ but can afford a college in $\CC_1$ will be matched to a college in $\CC_1.$ Therefore, the measure of students with value in $(v_*, v_+)$ who match with a college in $\CC_1$ is at least
\begin{equation}\label{eq:monkey}
    \int_{v^*}^{v_+} (1 - p_\mu(v, \CC_2)) \cdot p_\mu(v, \CC_1)\,d\eta(v).
\end{equation}
We have from \eqref{eq:cucumber-1} that
\begin{equation}
    1 - p_\mu(v, \CC_2) \ge 1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})
\end{equation}
for all $v\in (v^*, v_+)\subseteq (v_-, v_+).$

Also $S(\CC_1)\le \epsilon C\cdot \frac{\alpha}{C} = \epsilon\alpha.$ \eqref{eq:monkey} can be at most $S(\CC_1),$ so we have that
\begin{align}
    \epsilon\alpha &\ge \int_{v^*}^{v_+} (1 - p_\mu(v, \CC_2)) \cdot p_\mu(v, \CC_1)\,d\eta(v)\\
    &\ge \int_{v^*}^{v_+} \left(1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})\right) \cdot p_\mu(v, \CC_1)\,d\eta(v)\\
    &\ge \eta((v^*,v_+)) \left(1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})\right) \cdot p_\mu(v^*, \CC_1)\\
    &= \sqrt{\epsilon} \left(1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})\right) \cdot p_\mu(v^*, \CC_1).
\end{align}
\end{proof}
It follows that
\begin{equation}
    p_\mu(v^*, \CC_1) \le \frac{\alpha\sqrt{\epsilon}}{1 - S - \epsilon - (1 - e^{-2\epsilon\sigma})}.
\end{equation}
The result follows since $p_\mu(v^*, \CC_1)$ is monotonically increasing in $v$.
% === end archive member: proof-amplifying.tex ===
% === archive member: proofs-extended.tex ===
\section{Proofs of Extended Results}\label{sec:proofs-extended}
```

Formalized review target

The archival source above is not asserted equivalent to this formalized target.

The small-firm proposition is the source p_mu(v,F1) bound below v*, with eta((v*,v_+)) = sqrt(epsilon), under its preceding capacity-mass estimate.

Archival source anchor: cited publication, lines 918-950

Expanded PaperInterface specification

```
∀ (noiseLaw eta : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure noiseLaw]
  [MeasureTheory.IsProbabilityMeasure eta] {active : (C : ℕ) → Finset (Fin (C + 1))}
  {cutoff : (C : ℕ) → Fin (C + 1) → ℝ} {vStar vHigh totalSupply alpha epsilon sigma : ℝ},
  0 < epsilon →
    eta.real (Set.Ioo vStar vHigh) = √epsilon →
      0 < 1 - totalSupply - epsilon - (1 - Real.exp (-(2 * epsilon * sigma))) →
        (∀ᶠ (C : ℕ) in Filter.atTop,
            Monotone fun (v : ℝ) =>
              PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C)) →
          (∀ᶠ (C : ℕ) in Filter.atTop,
              √epsilon * (1 - totalSupply - epsilon - (1 - Real.exp (-(2 * epsilon * sigma)))) *
                  PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                    (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) vStar (cutoff C) ≤
                epsilon * alpha) →
            ∀ᶠ (C : ℕ) in Filter.atTop,
              ∀ v < vStar,
                PG24NoisyMatchingMarkets.theorem2_smallFirmSourceBound totalSupply alpha epsilon sigma
                  (PG24NoisyMatchingMarkets.cutoffAffordanceProbability
                    (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw) (active C) v (cutoff C))
```

Retained governing declarations

- PG24NoisyMatchingMarkets.cutoffAffordanceProbability
- PG24NoisyMatchingMarkets.theorem2_smallFirmSourceBound

Lean-elaborated claim roles

```
1. parameter: MeasureTheory.Measure ℝ
2. parameter: MeasureTheory.Measure ℝ
3. assumption: MeasureTheory.IsProbabilityMeasure noiseLaw
4. assumption: MeasureTheory.IsProbabilityMeasure eta
5. parameter: (C : ℕ) → Finset (Fin (C + 1))
6. parameter: (C : ℕ) → Fin (C + 1) → ℝ
7. parameter: ℝ
8. parameter: ℝ
9. parameter: ℝ
10. parameter: ℝ
11. parameter: ℝ
12. parameter: ℝ
13. assumption: 0 < epsilon
14. assumption: eta.real (Set.Ioo vStar vHigh) = √epsilon
15. assumption: 0 < 1 - totalSupply - epsilon - (1 - Real.exp (-(2 * epsilon * sigma)))
16. assumption: ∀ᶠ (C : ℕ) in Filter.atTop,
  Monotone fun (v : ℝ) =>
    PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
      (active C) v (cutoff C)
17. assumption: ∀ᶠ (C : ℕ) in Filter.atTop,
  √epsilon * (1 - totalSupply - epsilon - (1 - Real.exp (-(2 * epsilon * sigma)))) *
      PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
        (active C) vStar (cutoff C) ≤
    epsilon * alpha
18. conclusion: ∀ᶠ (C : ℕ) in Filter.atTop,
  ∀ v < vStar,
    PG24NoisyMatchingMarkets.theorem2_smallFirmSourceBound totalSupply alpha epsilon sigma
      (PG24NoisyMatchingMarkets.cutoffAffordanceProbability (MeasureTheory.Measure.pi fun (x : Fin (C + 1)) => noiseLaw)
        (active C) v (cutoff C))
```

Lean proof endpoint (built separately)

PG24NoisyMatchingMarkets.ProofInterface.source_lt_small_firmsSpec_proof

Recorded source-to-Spec screening

Verdict: matches_approved_corrected_target

Reason: The target retains eta((v*,v_+)) = sqrt(epsilon), the source p_mu(v,F1) as iid cutoff-affordance probability, and its alpha sqrt(epsilon) denominator bound below v*. The supplied positive-denominator, v*-level capacity-mass, and monotonicity premises are the approved immediate context.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation